COMP2711 / DISCRETE MATHEMATICAL TOOLS

Number Theory

Past Paper Solutions

143	3	2013-25
verified exercises	topic groups	source papers

English solutions aligned with the supplied official answers.

Each answer has two parts: the original Official Answer and the existing Supplementary Explanation. Source PDF pages are retained for verification.

Generated 2 October 2026

All verified, in-scope questions linked to Number Theory in the knowledge base.

Each question is included once, including cross-topic applications.

READING GUIDE

Contents

Modular Arithmetic
37 exercises — N001-N037

GCDs and Congruences
69 exercises — N038-N106

Cryptography
37 exercises — N107-N143

How to use this booklet

Questions are grouped by their first Number Theory subtopic, then ordered by difficulty and year (newest first). Each question appears once as N001-N143. Difficulty ranges from 1 (introductory) to 4 (challenging).

Official Answer reproduces the answer in the source paper, with its physical PDF page reference.

Supplementary Explanation retains the existing independently authored reasoning, including the stated result and derivation. The two parts remain separate throughout the booklet.

Source references use physical PDF pages. File links work with the local source collection at its existing location.

The answer provenance shown for each exercise retains its knowledge-base classification; attaching an official quotation does not reclassify a derived correction as official.

N001

Modular Arithmetic

2019 Midterm | Question 6(b) | Source page 8 | Difficulty 1 | 3 marks
Q-2019-MID-P06-b | Answer provenance: official

> (b) For all integers a, b, if $a \mid b$ and $b \mid a$, then $a = b$.

Official Answer

Source: physical PDF page 8 (2019-midterm.pdf).

False. Set $a = 1$ and $b = -1$.

Supplementary Explanation

Answer. False; $a = 1, b = -1$.

Derivation/proof. Here $-1 = 1 \cdot (-1)$, so $1 \mid -1$. Also $1 = (-1)(-1)$, so $-1 \mid 1$. Yet $1 \neq -1$. The hypotheses determine absolute values for nonzero inputs, not their signs, and the integer domain allows this counterexample.

Mutual divisibility forces equal absolute values, not necessarily equal signs.

The example $1 = (-1) \cdot (-1)$ shows why a product equal to one need not have both integer factors equal to positive one; the negative unit is the missing case.

N002

Modular Arithmetic

2019 Midterm | Question 9(a) | Source page 11 | Difficulty 1 | 2 marks
Q-2019-MID-P09-a | Answer provenance: official

> Suppose that a and b are integers, $a \equiv 11 \pmod{19}$, and $b \equiv 3 \pmod{19}$. Find the integer c with $0 \leq c \leq 18$ such that
>
> (a) $c \equiv a - b \pmod{19}$

Official Answer

Source: physical PDF page 11 (2019-midterm.pdf).

8.

Supplementary Explanation

Answer. $c = 8$.

Derivation/proof. Congruence respects subtraction, so $a - b \equiv 11 - 3 = 8 \pmod{19}$. Since 8 lies between 0 and 18, no further adjustment is necessary. Any other solution differs by a multiple of 19, so only this representative lies in the requested interval.

N003

Modular Arithmetic

2019 Midterm | Question 9(b) | Source page 11 | Difficulty 1 | 2 marks
Q-2019-MID-P09-b | Answer provenance: official

> Suppose that a and b are integers, $a \equiv 11 \pmod{19}$, and $b \equiv 3 \pmod{19}$. Find the integer c with $0 \leq c \leq 18$ such that
>
> (b) $c \equiv 7a + 3b \pmod{19}$

Official Answer

Source: physical PDF page 11 (2019-midterm.pdf).

10.

Supplementary Explanation

Answer. $c = 10$.

Derivation/proof. Use the given congruences to obtain $7a + 3b \equiv 7 \cdot 11 + 3 \cdot 3 = 77 + 9 = 86$. Since $86 = 4 \cdot 19 + 10$, the remainder is 10. It belongs to the specified interval, so it is the unique requested value.

Substituting the known residues in the linear combination gives $7 \cdot 11 + 3 \cdot 3 = 86$. Reducing modulo nineteen gives $86 = 4 \cdot 19 + 10$, so the required least nonnegative residue is ten.

N004

Modular Arithmetic

2019 Midterm | Question 9(c) | Source page 11 | Difficulty 1 | 2 marks
Q-2019-MID-P09-c | Answer provenance: official

> Suppose that a and b are integers, $a \equiv 11 \pmod{19}$, and $b \equiv 3 \pmod{19}$. Find the integer c with $0 \leq c \leq 18$ such that
>
> (c) $c \equiv ab^2 \pmod{19}$

Official Answer

Source: physical PDF page 11 (2019-midterm.pdf).

4.

Supplementary Explanation

Answer. $c = 4$.

Derivation/proof. First $b^2 \equiv 3^2 = 9 \pmod{19}$. Then $ab^2 \equiv 11 \cdot 9 = 99 = 5 \cdot 19 + 4$. Thus $c = 4$, already in the range 0 through 18. The expression is a times b^2, not $(ab)^2$.

Direct substitution gives $11 \cdot 3^2 = 99$. Reduce this value modulo the stated modulus to obtain the required residue.

N005

Modular Arithmetic

2017 Midterm | Question 4(a) | Source page 6 | Difficulty 1
Q-2017-MID-P04-a | Answer provenance: official

> Prove or disprove the following statements. Assume that a, b, c, d, and m are integers and $m > 1$.
>
> (a) If $a \equiv b \pmod{m}$ and $a \equiv c \pmod{m}$, then $a \equiv b + c \pmod{m}$.

Official Answer

Source: physical PDF page 6 (2017-midterm.pdf).

False. Set $a = 1, b = 3, c = 5, m = 2$ as a counter example. We have $1 \equiv 3 \pmod{2}$, $1 \equiv 5 \pmod{2}$, but $1 \not\equiv 5 \pmod{2}$.

Supplementary Explanation

Answer. False; $a = 1, b = 3, c = 5, m = 2$ is a counterexample.

Derivation/proof. Here $1 \equiv 3 \equiv 5 \pmod{2}$, so both hypotheses hold. But $b + c = 8 \equiv 0 \pmod{2}$, while $a \equiv 1 \pmod{2}$. The two residues differ, so the conclusion fails. In general the hypotheses give $b + c \equiv 2a$, not a.

N006

Modular Arithmetic

2017 Midterm | Question 4(b) | Source page 6 | Difficulty 1
Q-2017-MID-P04-b | Answer provenance: official

> Prove or disprove the following statements. Assume that a, b, c, d, and m are integers and $m > 1$.
>
> (b) If $a \equiv b \pmod{m}$, then $2a \equiv 2b \pmod{2m}$.

Official Answer

Source: physical PDF page 6 (2017-midterm.pdf).

True. $a \equiv b \pmod{m}$ implies $m \mid a - b$, i.e., $a - b = km$ for some integer k. Multiplying both sides by 2 gives $2a - 2b = 2mk$. Therefore $2a \equiv 2b \pmod{2m}$.

Supplementary Explanation

Answer. True.

Derivation/proof. The hypothesis means $a - b = km$ for some integer k. Multiplying the equality by two gives $2a - 2b = k(2m)$. Thus $2m$ divides $2a - 2b$, exactly the definition of the required congruence.

Starting from $m \mid (a - b)$, multiplication by two yields $2m \mid (2a - 2b)$, which is precisely the congruence modulo $2m$. Conversely, cancel two from the integer multiple identity to recover the original divisibility.

N007

Modular Arithmetic

2015 Final | Question 5(b) | Source page 10 | Difficulty 1 | 3 marks
Q-2015-FINAL-P05-b | Answer provenance: official

> Let p be a prime number. For any two integers x and y from $\mathbb{Z}_p$, if
>
> $$x = y^2 \bmod p,$$
>
> then we say that x is a quadratic residue mod p, and that y is a square root of x mod p.
>
> (b) Show that $p-1$ is a square root of 1 mod p.

Official Answer

Source: PDF physical page 10 (2015-final.pdf).

(b) $(p-1)^2 \bmod p = (p^2 - 2p + 1) \bmod p = 1 \bmod p = 1$

Supplementary Explanation

Answer. $(p-1)^2 \equiv 1 \pmod{p}$.

Derivation/proof. Expand $(p-1)^2 = p^2 - 2p + 1 = p(p-2) + 1$. The first term is divisible by p, leaving residue 1; therefore $p-1$ satisfies the definition of a square root of 1.

As $p \geq 2$, $p-1$ is an allowed representative in $\{0, \ldots, p-1\}$.

N008

Modular Arithmetic

2025 Spring Final | Question 2(a) | Source page 3 | Difficulty 2 | 3 marks
Q-2025S-FINAL-P02-a | Answer provenance: official

> (a) [3 pts] Calculate $2^{62} \pmod{63}$.

Official Answer

Source: physical PDF page 3 (COMP2711_S2025_Final.pdf).

(a) $2^6 \equiv 1 \pmod{63} \implies 2^{60} \equiv 1 \implies 2^{62} \equiv 4 \pmod{63}$

Supplementary Explanation

Answer. 4.

Derivation/proof. Directly, $2^6 = 64 \equiv 1 \pmod{63}$. Decompose the exponent as $62 = 6 \cdot 10 + 2$. Then $2^{62} = (2^6)^{10} 2^2 \equiv 1^{10} \cdot 4 = 4 \pmod{63}$. Four lies between zero and sixty-two and is the reduced remainder. The calculation uses explicit integer congruences and does not rely on sixty-three being prime.

N009

Modular Arithmetic

2025 Spring Final | Question 3(a) | Source page 4 | Difficulty 2 | 2 marks
Q-2025S-FINAL-P03-a | Answer provenance: official

> For a positive integer p, define the positive integer n to be p-safe if n differs in absolute value by more than 2 from all multiples of p. For example, the set of 10-safe numbers are $3, 4, 5, 6, 7, 13, 14, 15, 16, 17, \cdots$.
>
> (a) [2 pts] Let x be a 7-safe number. What are the possible values of $x \bmod 7$?

Official Answer

Source: physical PDF page 4 (COMP2711_S2025_Final.pdf).

(a) $3, 4$

Supplementary Explanation

Answer. $3, 4$.

Derivation/proof. Write $x = 7q + r$, $0 \le r < 7$. The distance to $7q$ is r and the distance to $7(q+1)$ is $7 - r$. Both must exceed two: $r > 2$ and $7 - r > 2$. Thus $2 < r < 5$, allowing exactly the integer residues three and four. Every more distant multiple is farther away, so both residues satisfy the full condition.

N010

Modular Arithmetic

2025 Spring Midterm | Question 7 | Source page 11 | Difficulty 2 | 8 marks
Q-2025S-MID-P07 | Answer provenance: official

> Prove that for any odd natural number a, $8 \mid a^2 - 1$.

Official Answer

Source: physical PDF page 11 (COMP2711_S2025_Midterm.pdf).

We need to prove $(a-1)(a+1)$ is divisible by 8.

Note that $a = 2k + 1$ So we know that $a - 1 = 2k$ and $a + 1 = 2k + 2$.

$(a-1)(a+1) = 2k \cdot (2k+2) = 4k^2 + 4k = 4 \cdot k \cdot (k+1)$. Note that $2 \mid k \cdot (k+1)$ as k and $k+1$ are two consecutive numbers. So $(a-1)(a+1) = 4 \cdot 2k' = 8k'$ and this completes the proof.

Supplementary Explanation

Answer. $8 \mid a^2 - 1$.

Derivation/proof. An odd natural number has the form $a = 2k + 1$. Thus $a^2 - 1 = (a-1)(a+1) = 2k(2k+2) = 4k(k+1)$. One of the consecutive integers $k, k+1$ is even, so $k(k+1) = 2t$ for some integer t. Substitution gives $a^2 - 1 = 8t$, exactly the divisibility claim. At $a = 1$, $k = 0$ and the difference is zero; eight divides zero, so there is no boundary exception.

N011

Modular Arithmetic

2024 Fall Final | Question 5(a) | Source page 15 | Difficulty 2 | 3 marks
Q-2024F-FINAL-P05-a | Answer provenance: official

> (a) Assume a to be an even integer. What are the possible values for $a^2 \pmod 8$? Hint: Assume $a = 4k$ or $a = 4k + 2$.

Official Answer

Source: physical PDF page 15 (COMP2711_F2024_Final.pdf).

> Answer: (a) If $a = 4k$ then $a^2 = 16k^2 \implies a^2 \equiv 0 \pmod 8$ If $a = 4k + 2$ then $a^2 = 16k^2 + 16k + 4 \implies a^2 \equiv 4 \pmod 8$. So if a is even then the possible remainder of $a^2 \pmod 8$ is either $\{0, 4\}$.

Supplementary Explanation

Answer. $\{0, 4\}$.

Derivation/proof. Every even integer has residue zero or two modulo four. If $a = 4k$, then $a^2 = 16k^2 \equiv 0 \pmod 8$. If $a = 4k + 2$, then $a^2 = 16k^2 + 16k + 4 \equiv 4 \pmod 8$. The cases cover positive and negative even integers. Taking $a = 0$ and $a = 2$ realizes zero and four respectively, proving these are exactly the possible residues.

N012

Modular Arithmetic

2024 Fall Final | Question 5(b) | Source page 15 | Difficulty 2 | 3 marks
Q-2024F-FINAL-P05-b | Answer provenance: official

> (b) Assume a to be an odd integer. What are the possible values for $a^2 \pmod 8$?

Official Answer

Source: physical PDF page 15 (COMP2711_F2024_Final.pdf).

> (b) $a = 2k + 1 \implies a^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$. Note that $k(k + 1)$ is always even. So $4k(k + 1) \equiv 0 \pmod 8$. We can conclude that $a^2 \equiv 1 \pmod 8$.

Supplementary Explanation

Answer. Only 1.

Derivation/proof. Every odd integer is $a = 2k + 1$ for an integer k. Squaring gives $a^2 = 4k^2 + 4k + 1 = 4k(k + 1) + 1$. Among consecutive integers $k, k + 1$, one is even, so $k(k + 1)$ is divisible by two and $4k(k + 1)$ by eight. Thus $a^2 \equiv 1 \pmod 8$. Taking $a = 1$ realizes the residue; negative odd integers are covered by the same form.

N013

Modular Arithmetic

2023 Fall Final | Question 6(b) | Source page 10 | Difficulty 2 | 3 marks
Q-2023F-FINAL-P06-b | Answer provenance: official

> What is the remainder of $17^{2023} \bmod 14$?

Official Answer

Source: physical PDF page 10 (COMP2711_F2023_Final.pdf).

> (b) $17 \equiv 3 \pmod{14}$ So we have: $3^3 \equiv 27 \equiv -1 \pmod{14}$. $(3^3)^{674} \equiv 3^{2022} \equiv (-1)^{674} \equiv 1 \pmod{14}$ $3^{2023} \equiv 3 \cdot 3^{2022} \equiv 3 \cdot 1 \equiv 3$.

Supplementary Explanation

Answer. 3.

Derivation/proof. Since $17 \equiv 3 \pmod{14}$ and $3^3 = 27 \equiv -1 \pmod{14}$, write $2023 = 3 \cdot 674 + 1$. Then

$$17^{2023} \equiv 3^{2023} = (3^3)^{674} \cdot 3 \equiv (-1)^{674} \cdot 3 = 3 \pmod{14}.$$

The residue 3 is already between 0 and 13, so it is the requested remainder. Equivalently, $3^6 \equiv 1$ and $2023 \equiv 1 \pmod{6}$, confirming the same cycle calculation.

Because $2023 = 3 \cdot 674 + 1$, the remaining expression is $(-1)^{674} \cdot 3 = 3$. This is already in the standard remainder range $0 \le r < 14$.

N014

Modular Arithmetic

2022 Final Part 1 | Question 4 | Source page 3 | Difficulty 2 | 8 marks
Q-2022-FINAL1-P04 | Answer provenance: official

> Use induction to prove that, for any integer $n \geq 1$, $5^n + 2 \cdot 11^n$ is divisible by 3.

Official Answer

Source: physical PDF page 3 (2022-final-part-1.pdf).

> Base case: When $n = 1$, we have $5^1 + 2 \cdot 11^1 = 27$, which is divisible by 3. So, the statement is true for $n = 1$.
>
> Induction hypothesis: Now let $n > 1$. Assume the statement is true for $n - 1$, i.e. $5^{n-1} + 2 \cdot 11^{n-1}$ is divisible by 3.
>
> Induction step: Consider the case of n:
>
> $$\begin{aligned} 5^n + 2 \cdot 11^n &= 5^{n-1} + 4 \cdot 5^{n-1} + 2 \cdot 11^{n-1} + 20 \cdot 11^{n-1} \\ &= 5^{n-1} + 2 \cdot 11^{n-1} + 4(5^{n-1} + 5 \cdot 11^{n-1}) \\ &= 5^{n-1} + 2 \cdot 11^{n-1} + 4(5^{n-1} + 2 \cdot 11^{n-1} + 3 \cdot 11^{n-1}) \\ &= 3y + 4(3y + 3 \cdot 11^{n-1}) \\ &= 3(5y + 4 \cdot 11^{n-1}) \end{aligned}$$
>
> which is divisible by 3.

Supplementary Explanation

Answer. $3 \mid (5^n + 2 \cdot 11^n)$ for all integers $n \geq 1$.

Derivation/proof. Base n=1:5+2·11=27, divisible by 3. Assume 5^k+2·11^k=3t for some integer t. Then $5^{k+1} + 2 \cdot 11^{k+1} = 5(5^k + 2 \cdot 11^k) + 12 \cdot 11^k = 15t + 12 \cdot 11^k = 3(5t + 4 \cdot 11^k)$. The final parenthesis is an integer, proving the next instance. Ordinary induction completes the required divisibility proof.

The base is $27 = 3 \cdot 9$. The supplementary step is $5(5^k + 2 \cdot 11^k) + 12 \cdot 11^k = 3(5t + 4 \cdot 11^k)$ with an integer in parentheses, explicitly using the induction hypothesis.

N015

Modular Arithmetic

2022 Midterm | Question 6(a) | Source page 3 | Difficulty 2 | 8 marks
Q-2022-MID-P06-a | Answer provenance: official

> Assume p is a prime and n is a positive integer such that $p \mid (4n^2 + 1)$.
>
> Show that $p \nmid n$.

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (a) We use proof by contradiction. Suppose $n = kp$ for some integer k. Then $4n^2 + 1 = 4(kp)^2 + 1 = 4k^2p^2 + 1 = pk'$ for some integer k', i.e., $p(k' - 4k^2p) = 1$, which is impossible for prime p.

Supplementary Explanation

Answer. p does not divide n.

Derivation/proof. Assume p|n. Then n=pt for an integer t, so p divides $4n^2=4p^2t^2$. By hypothesis p also divides $4n^2+1$. Subtracting, p divides1, which is impossible because a prime $p\geq 2$. Thus the assumption fails and p∤n. This directly proves the coprimality needed in the following Fermat argument.

Writing $n = pt$ gives $p \mid (4n^2 + 1) - 4n^2 = 1$, impossible, so $p \nmid n$.

N016

Modular Arithmetic

2019 Midterm | Question 6(a) | Source page 8 | Difficulty 2 | 3 marks
Q-2019-MID-P06-a | Answer provenance: official

> (a) For all integers a, b, c, if $a \mid c$ and $b \mid c$, then $ab \mid c^2$.

Official Answer

Source: physical PDF page 8 (2019-midterm.pdf).

True. If $c = ak$ and $c = bl$, then $c^2 = ab(kl)$, so $ab \mid c^2$.

Supplementary Explanation

Answer. True.

Derivation/proof. The premises give integers k, l with $c = ak$ and $c = bl$. Multiplying these equalities produces $c^2 = (ak)(bl) = ab(kl)$. Since kl is an integer, this is exactly the divisibility witness for $ab \mid c^2$. No coprimality assumption is required.

No division is used, so zero values are also covered under the divisibility convention permitting $0 \mid 0$.

There is no division by a possibly zero integer in the product argument: expand the integer factors directly. Thus the reasoning remains valid even when one of the integers is zero.

N017

Modular Arithmetic

2018 Final | Question 4(a) | Source page 6 | Difficulty 2
Q-2018-FINAL-P04-a | Answer provenance: official

> (a) Fill in the following recursive definition so that X is the set of all positive integers.
>
> Basis: $____ \in X$.
>
> Recursive step: If $a \in X$, then $a + 3 \in X$.
>
> For this question, no proof is required.

Official Answer

Source: physical PDF page 6 (2018-final.pdf).

$\{1, 2, 3\} \in X$.

Supplementary Explanation

Answer. Use basis elements $1, 2, 3 \in X$.

Derivation/proof. These three elements generate $1 + 3k, 2 + 3k, 3 + 3k$ for all integers $k \geq 0$, covering every positive integer by its remainder modulo three. The rule never produces a nonpositive value. At least one basis representative from each of the three residue classes is necessary; the source's braces mean three elements, not one set-valued element.

N018

Modular Arithmetic

2016 Final | Question 4(b)(ii) | Source page 8 | Difficulty 2
Q-2016-FINAL-P04-b-ii | Answer provenance: official

> (b) Is each of the following true? Please explain.
>
> (ii) Is "$(21^{70} \bmod 71) = (21^{71} \bmod 72)$"?

Official Answer

Source: PDF physical page 9 (2016-final.pdf).

(ii) No.

Consider LHS. We have

$$21^{70} \bmod 71 = 1$$

Consider RHS $= 21^{71} \bmod 72$.

By Euclid's Division Theorem, the above expression could be written as follows.

$$21^{71} = 72q + r$$

where q and r are two unique integers.

Note that $r = 21^{71} \bmod 72$. Thus,

$$21^{71} = 72q + (21^{71} \bmod 72)$$

$$\begin{aligned} 21^{71} \bmod 72 &= 21^{71} - 72q \\ &= 3(7 \cdot 21^{70} - 24q) \end{aligned}$$

which is divisible by 3. Thus, RHS is divisible by 3.

Note that LHS is equal to 1 which is not divisible by 3.

Thus, LHS is not equal to RHS.

Supplementary Explanation

Answer. False. $21^{70} \bmod 71 = 1$, while $21^{71} \bmod 72$ is divisible by 3 (in fact, it is 45).

Derivation/proof. The prime 71 does not divide 21, so Fermat's theorem gives $21^{70} \equiv 1 \pmod{71}$. Write $21^{71} = 72q + r$ with $0 \le r < 72$. Then $r = 3(7 \cdot 21^{70} - 24q)$, a multiple of 3, and hence $r \neq 1$. A numerical check uses $21^2 \equiv 9$, $21^3 \equiv 45$, and $45 \cdot 9 \equiv 45 \pmod{72}$; thus every odd power from the third onward, including the 71st, has remainder 45.

N019

Modular Arithmetic

2015 Final | Question 5(a) | Source page 10 | Difficulty 2 | 4 marks
Q-2015-FINAL-P05-a | Answer provenance: official

> Let p be a prime number. For any two integers x and y from $\mathbb{Z}_p$, if
>
> $$x = y^2 \bmod p,$$
>
> then we say that x is a quadratic residue mod p, and that y is a square root of x mod p.
>
> (a) Find all integers in $\mathbb{Z}_{11}$ that are quadratic residues mod 11.

Official Answer

Source: PDF physical page 10 (2015-final.pdf).

(a) $0\ (y=0),\ 1\ (y=1,10),\ 3\ (y=5,6),\ 4\ (y=2,9),\ 5\ (y=4,7),\ 9\ (y=3,8)$.

(Note on grading: Only the value of x is required.)

Supplementary Explanation

Answer. $\{0,1,3,4,5,9\}$.

Derivation/proof. Squaring $y = 0,1,2,3,4,5$ gives residues $0,1,4,9,5,3$. The remaining roots $6,7,8,9,10$ are congruent to $-5,-4,-3,-2,-1$ and repeat the same squares. This covers every residue class.

The source definition permits the root zero, so zero belongs in the answer.

N020

Modular Arithmetic

2015 Final | Question 5(c) | Source page 10 | Difficulty 2 | 3 marks
Q-2015-FINAL-P05-c | Answer provenance: official

> Let p be a prime number. For any two integers x and y from $\mathbb{Z}_p$, if
>
> $$x = y^2 \bmod p,$$
>
> then we say that x is a quadratic residue mod p, and that y is a square root of x mod p.
>
> (c) Show that if y is a square root of x mod p, so is $p - y$.

Official Answer

Source: PDF physical page 10 (2015-final.pdf).

(c) $(p-y)^2 \bmod p = (p^2 - 2py + y^2) \bmod p = y^2 \bmod p = x$

Supplementary Explanation

Answer. $(p-y)^2 \equiv x \pmod p$.

Derivation/proof. Given $y^2 \equiv x \pmod p$, $(p-y)^2 = p^2 - 2py + y^2 \equiv y^2 \equiv x \pmod p$. As a member of $\mathbb{Z}_p$, the new root is represented by $(p-y) \bmod p$; this includes the $y = 0$ case.

N021

Modular Arithmetic

2025 Spring Midterm | Question 6(a) | Source page 10 | Difficulty 3
Q-2025S-MID-P06-a | Answer provenance: official

> (a) Define $2 \uparrow n$ by $2 \uparrow 0 = 1$ and $2 \uparrow n = 2^{2 \uparrow (n-1)}$. In other words,
>
> $$2 \uparrow n = \underbrace{2^{2^{\cdot^{\cdot^{2}}}}}_{n}$$
>
> where there are n 2's in the tower. Compute $2 \uparrow 100 \pmod{17}$. Hint: Use Fermat's Little Theorem.

Official Answer

Source: physical PDF page 10 (COMP2711_S2025_Midterm.pdf).

(a) Note that 17 is a prime number. So $2^{16} \equiv 1 \pmod{17}$. So we need to calculate $2 \uparrow (99) \pmod{16}$. It is easy to see that $2 \uparrow x \equiv 0 \pmod{16}$ for $x \geq 3$. So $2 \uparrow (99) \equiv 0 \pmod{16}$.

So we can conclude that $2 \uparrow 100 \equiv 2^{16 \cdot k} \equiv (2^{16})^k \equiv 1 \pmod{17}$.

Supplementary Explanation

Answer. 1.

Derivation/proof. Since 17 is prime and coprime to 2, Fermat's Little Theorem gives $2^{16} \equiv 1 \pmod{17}$. We have $2 \uparrow 2 = 4$ and $2 \uparrow 3 = 16$. For every $t \geq 3$, the exponent in $2 \uparrow t = 2^{2 \uparrow (t-1)}$ is at least four, so $16 \mid 2 \uparrow t$. Write $2 \uparrow 99 = 16k$. Then $2 \uparrow 100 = 2^{16k} \equiv (2^{16})^k \equiv 1 \pmod{17}$. Only the outer exponent was reduced; the tower height is not an ordinary exponent.

N022

Modular Arithmetic

2025 Spring Midterm | Question 6(b) | Source page 10 | Difficulty 3
Q-2025S-MID-P06-b | Answer provenance: official

> (b) Calculate $47^{37^{27}} \bmod 11$?

Official Answer

Source: physical PDF page 10 (COMP2711_S2025_Midterm.pdf).

(b) $47 \equiv 3 \pmod{11}$. We also know that $3^{10} \equiv 1 \pmod{11}$ (based on little Fermat theorem).

We need to calculate $37^{27} \pmod{10}$.

$$37^{27} \equiv 7^{27} \equiv 7 \cdot 7^{26} \equiv 7 \cdot 49^{13} \equiv 7 \cdot 9^{13} \equiv 7 \cdot 9 \equiv 63 \equiv 3 \pmod{10}$$

$$3^{37^{27}} \equiv 3^{10k+3} \equiv 27 \cdot (3^{10})^k \equiv 27 \equiv 5 \pmod{11}$$

Supplementary Explanation

Answer. 5.

Derivation/proof. The prime 11 is coprime to 3, so Fermat's Little Theorem gives $3^{10} \equiv 1 \pmod{11}$. Since $47 \equiv 3 \pmod{11}$, compute $37^{27} \bmod 10$ separately. Modulo ten, $37 \equiv 7$, $7^2 \equiv 9$ and $7^4 \equiv 1$, hence $7^{27} = (7^4)^6 \cdot 7^3 \equiv 343 \equiv 3 \pmod{10}$. Write $37^{27} = 10k + 3$. Therefore $47^{37^{27}} \equiv 3^{10k+3} \equiv (3^{10})^k 3^3 \equiv 27 \equiv 5 \pmod{11}$. The intermediate residue three belongs to the exponent; the final residue five belongs to the entire power.

N023

Modular Arithmetic

2024 Fall Final | Question 5(c) | Source page 15 | Difficulty 3 | 3 marks
Q-2024F-FINAL-P05-c | Answer provenance: official

> (c) Assume a, b and c to be three integer numbers. Prove that the following equation has no integer solution.
>
> $$a^2 + b^2 - 8c = 6$$
>
> Hint: Try to use (a) and (b).

Official Answer

Source: physical PDF page 15 (COMP2711_F2024_Final.pdf).

> (c) $a^2 + b^2 = 8c + 6$ Which means $a^2 + b^2 \equiv 6 \pmod 8$. From (a) and (b) we can see that the possible remainder of a square number mod 8 is $\{0, 1, 4\}$. So it is not possible to pick two values among this numbers such that the remainder mod 8 is equal to 6.

Supplementary Explanation

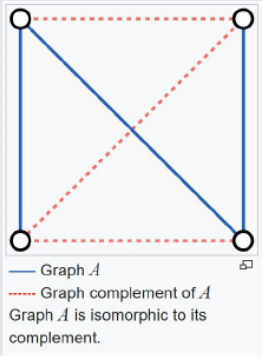
Answer. There is no integer solution.

Derivation/proof. If a solution existed, rearrangement modulo eight would give $a^2 + b^2 \equiv 6 \pmod 8$. Parts (a) and (b) restrict all square residues to $\{0, 1, 4\}$. The unordered pair sums are $0 + 0 \equiv 0$, $0 + 1 \equiv 1$, $0 + 4 \equiv 4$, $1 + 1 \equiv 2$, $1 + 4 \equiv 5$, and $4 + 4 \equiv 0$. Thus sums can only have residues $0, 1, 2, 4, 5$, never six. This contradiction rules out every integer triple a, b, c.

N024

Modular Arithmetic

2024 Fall Final | Question 11(c) | Source page 27 | Difficulty 3 | 3 marks
Q-2024F-FINAL-P11-c | Answer provenance: official

> The complement or inverse of a graph G is a graph H on the same vertices such that two distinct vertices of H are adjacent if and only if they are not adjacent in G. A self-complementary graph is a graph which is isomorphic to its complement. As an example, the following graph is self-complementary.
>
> 
>
>
> Assume a graph G is a self-complementary graph with n nodes. Consider H as its complement. Assume S to be the sum of all the nodes degrees in G.
>
> (c) Prove that $n \equiv 0 \pmod 4$ or $n \equiv 1 \pmod 4$. You can use part (b) as a fact even if you could not solve it.

Official Answer

Source: physical PDF page 27 (COMP2711_F2024_Final.pdf).

> (c) Note that $n(n-1) = 2S$ and S is even. So $4 \mid (n)(n-1)$. Also note that between n and $n-1$ one is odd and one is even. So either $4 \mid n$ or $4 \mid n-1$.

Supplementary Explanation

Answer. $n \equiv 0$ or $1 \pmod 4$.

Derivation/proof. By (a), S is even, so $S = 2e$ for an integer e. By (b), $n(n-1) = 2S = 4e$, hence $4 \mid n(n-1)$. Residues zero and one for n give product zero modulo four. If n has residue two, then $n(n-1) \equiv 2 \cdot 1 = 2$. If it has residue three, then $n(n-1) \equiv 3 \cdot 2 = 6 \equiv 2$. The last two cases contradict divisibility by four, leaving only residues zero and one. This proves the necessary condition.

N025

Modular Arithmetic

2024 Spring Midterm | Question 7 | Source page 9 | Difficulty 3 | 10 marks
Q-2024S-MID-P07 | Answer provenance: official

> Let k be an even integer. Prove that it is not possible to write 1 as the sum of the reciprocals of k odd integers. Note: As an example, if $k = 4$, then the k odd integers selected could be 3, 3, 5, and 7. Note that, in this case, the sum of the reciprocals of these k odd integers is given by
>
> $$\frac{1}{3} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7},$$
>
> which is not equal to 1. In this problem, you need to show that no matter how we pick the k odd integers, where k is an arbitrary even integer, the sum of the reciprocals of these k odd integers cannot equal 1. Hint: Try to prove this by contradiction.

Official Answer

Source: physical PDF page 9 (COMP2711_S2024_Midterm.pdf).

> Answer. Fix an even integer k. Suppose for the sake of contradiction that there exist k odd integers $a_1, a_2, \ldots, a_k$ where $\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_k} = 1$. Fractions seem to complicate the problem, so let's eliminate them by multiplying both sides by the product $a_1 \cdot a_2 \cdots a_k$:
>
> $$\frac{a_1 \cdots a_k}{a_1} + \frac{a_1 \cdots a_k}{a_2} + \cdots + \frac{a_1 \cdots a_k}{a_k} = a_1 \cdot a_2 \cdots a_k. \quad (*)$$
>
> Note that every term on the LHS is an integer because the denominator is always present in the numerator, and is therefore cancelled out. We can use the fact that k is even and that each a_i is odd to observe the following: (a) The RHS of (??) is the product of odd numbers, which must be odd. (b.1) Each term in the LHS is of the form $\frac{a_1 \cdots a_k}{a_i} = a_1 \cdots a_{i-1} \cdot a_{i+1} \cdots a_k$, which is the product of odd integers, and hence must be odd. (b.2) The LHS is the sum of k terms, each of which is odd by the previous observation. Since k is even, that sum must be even. Observation (a) tells us that the RHS of (??) is odd, but observation (b.2) tells us that the LHS is even. This contradicts the fact that LHS and RHS are equal. Therefore, we conclude that our hypothesis was wrong and that such k integers $a_1, \ldots, a_k$ do not exist.

Supplementary Explanation

Answer. The reciprocals of any even number of odd integers cannot sum to 1.

Derivation/proof. Suppose odd integers $a_1, \ldots, a_k$ satisfy $\sum_i 1/a_i = 1$. Odd integers are nonzero, so multiply by $D = \prod_i a_i$ to obtain $\sum_i (D/a_i) = D$. Every $D/a_i = \prod_{j \neq i} a_j$ is odd. There are an even number k of terms, so the left side is even, while the right side D is odd: a contradiction. Negative odd integers and repeated denominators do not change the parity argument. For $k = 0$ the empty sum is zero, not one; a negative k cannot describe a number of terms and yields no such representation either.

N026

Modular Arithmetic

2023 Fall Final | Question 6(a) | Source page 10 | Difficulty 3 | 6 marks
Q-2023F-FINAL-P06-a | Answer provenance: official

> Let a, b, c, d be integers. Show that the product
>
> $$(a-b)(a-c)(a-d)(b-c)(b-d)(c-d)$$
>
> is divisible by 12.
>
> (Hint: You might want to prove that there exist at least two numbers among a, b, c, d such that they have the same remainder mod 3 and this might help you with the main proof)

Official Answer

Source: physical PDF page 10 (COMP2711_F2023_Final.pdf).

> (a) Out of 4 different numbers, two of them have the same remainder mod 3 based on pigeonhole principle. So one of those six elements are divisible by 3. Now out of these four numbers, either we have two odd and two even or there exists at least three numbers with same oddity. In each case we can get two different pair of numbers that their difference is an even number. So it would be dividable by 4. So the total would be dividable by 12.

Supplementary Explanation

Answer. $12 \mid (a-b)(a-c)(a-d)(b-c)(b-d)(c-d)$.

Derivation/proof. Among four integers, two share a residue modulo 3, so their difference is a factor divisible by 3. For parity, either there are two even and two odd numbers, giving one even difference in each pair, or at least three integers have the same parity, giving at least two distinct even differences among those three. Thus two factors supply a factor of 4. Since $\gcd(3,4)=1$, the product is divisible by 12. Repeated integer values cause a zero factor, and 12 divides zero, so distinctness is not required.

There are only three residue classes modulo three, so two inputs share a class. The parity cases provide two even factors, whose product is divisible by four. If two inputs coincide, the product is zero and is still divisible by twelve.

N027

Modular Arithmetic

2023 Spring Midterm | Question 4 | Source page 7 | Difficulty 3 | 13 marks
Q-2023S-MID-P04 | Answer provenance: official

> Find all integer pairs (x, y) such that $2x^2 - 5y^2 = 7$, or prove that such integer pairs do not exist. [Hint: Consider mod 8.]

Official Answer

Source: physical PDF page 7 (COMP2711_S2023_Midterm.pdf).

> First of all, obviously y should be odd, otherwise left hand side will be even. Let $y = 2k + 1$. Then $y^2 = 4k(k + 1) + 1 \equiv 1 \pmod 8$, so $5y^2 \equiv 5 \pmod 8$. Doing mod 8 on both sides of the equation, we have
>
> $$2x^2 - 5 \equiv 7 \pmod 8,$$
>
> $$2x^2 \equiv 4 \pmod 8,$$
>
> Then consider the parity of x. a. If x is even. Then $2x^2 \equiv 0 \pmod 8$, not possible. b. If x is odd. Then $2x^2 = 2(2k + 1)^2 = 8k^2 + 8k + 2 \equiv 2 \pmod 8$, not possible, either. So such pair (x, y) doesn't exist.

Supplementary Explanation

Answer. There are no integer pairs (x, y).

Derivation/proof. Reducing the equation modulo 2 yields $y^2 \equiv 1 \pmod 2$, so $y = 2k + 1$ is odd. Since $k(k + 1)$ is even, $y^2 = 4k(k + 1) + 1 \equiv 1 \pmod 8$. The equation then forces $2x^2 \equiv 7 + 5 \equiv 4 \pmod 8$. If x is even, $2x^2 \equiv 0 \pmod 8$; if x is odd, $x^2 \equiv 1 \pmod 8$ and $2x^2 \equiv 2 \pmod 8$. Neither is 4, a contradiction.

The two parity cases exhaust all integers. For even $x = 2h$, the calculation is $2x^2 = 8h^2$. In particular, $y^2 \equiv 1 \pmod 8$.

N028

Modular Arithmetic

2022 Midterm | Question 7(b) | Source page 3 | Difficulty 3 | 5 marks
Q-2022-MID-P07-b | Answer provenance: official

> Calculate $3^{3^{2022}} \bmod 10$. Show your steps.

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (b) We use a similar idea but can't use Fermat's Little Theorem since 10 is not a prime. But observe that $3^4 \equiv 81 \equiv 1 \pmod{10} \implies 3^{4t} \equiv 1 \pmod{10}$ for any integer t. So it suffices to find $3^{2022} \bmod 4$. Since $3^2 \equiv 1 \pmod 4$, $3^{3022} \equiv 1 \pmod 4$. Then $3^{3^{2022}} = 3^{4t+1} \equiv 3 \pmod{10}$.

Supplementary Explanation

Answer. 3.

Derivation/proof. Modulo10, 3^4=81≡1, so the outer exponent only needs its residue modulo 4. Since 3≡−1 mod 4 and 2022 is even, 3^2022≡(−1)^2022=1 mod 4. Thus the exponent is 4t+1 and 3^(3^2022)≡(3^4)^t·3≡3 mod 10. The exponent is a power tower, not 3·2022 or 3^2022 as the entire expression.

The actual inner exponent satisfies $3^{2022} \equiv (-1)^{2022} = 1 \pmod 4$. Writing $3^{2022} = 4t + 1$ gives $3^{3^{2022}} = (3^4)^t \cdot 3$.

N029

Modular Arithmetic

2020 Final Part 1 | Question 4(a) | Source page 3 | Difficulty 3 | 3 marks
Q-2020-FINAL1-P04-a | Answer provenance: official

> How many integers $0 \leq n < 1000$ are there such that
>
> n is divisible by 2, 3 or 5?

Official Answer

Source: physical PDF page 3 (2020-final-part-1.pdf).

Let $f(k)$ be the number of integers $0 \leq n < 1000$ that are divided by k, then $f(k) = 1 + \lfloor 999/k \rfloor$. We have $f(2) = 500$, $f(3) = 334$, $f(5) = 200$, $f(6) = 167$, $f(10) = 100$, $f(15) = 67$ and $f(30) = 34$. By inclusion-exclusion principle, the answer is $f(2) + f(3) + f(5) - f(6) - f(10) - f(15) + f(30) = 734$.

Supplementary Explanation

Answer. 734.

Derivation/proof. For d>0 the number of multiples in [0,999] is $f(d) = 1 + \lfloor 999/d \rfloor$, including zero. The counts for 2,3,5 are 500,334,200; pairwise intersections use 6,10,15 and give 167,100,67; the triple uses 30 and gives 34. Inclusion-exclusion yields $500 + 334 + 200 - 167 - 100 - 67 + 34 = 734$. Zero is counted once after cancellation.

The coefficient of zero in inclusion-exclusion is $3 - 3 + 1 = 1$.

N030

Modular Arithmetic

2020 Final Part 1 | Question 4(c) | Source page 3 | Difficulty 3 | 7 marks
Q-2020-FINAL1-P04-c | Answer provenance: official

> How many integers $0 \leq n < 1000$ are there such that
>
> n is divisible by 9 but it has no digit 9?

Official Answer

Source: physical PDF page 3 (2020-final-part-1.pdf).

Let $n = 100x_1 + 10x_2 + x_3$, i.e. n is represented by 3 digits $\overline{x_1x_2x_3}$. Then $x_i \in \{0, 1, \cdots, 8\}$ for $i = 0, 1, 2$, and $9 \mid 100x_1 + 10x_2 + x_3$, or equivalently $9 \mid x_1 + x_2 + x_3$. Since $0 \leq x_1 + x_2 + x_3 \leq 24$, we have $x_1 + x_2 + x_3 = 0, 9$ or 18. (1) $x_1 + x_2 + x_3 = 0$. There is only one solution: $x_1 = x_2 = x_3 = 0$. (2) $x_1 + x_2 + x_3 = 9$. By the combinations with repetition theorem, there are $\binom{9+3-1}{3-1} = 55$ non-negative solutions to this equation. However we need to exclude three solutions $(9, 0, 0)$, $(0, 9, 0)$ and $(0, 0, 9)$. Therefore the number of solutions is 52. (3) $x_1 + x_2 + x_3 = 18$. First note that there is no solution when $x_3 = 0$ or 1. Suppose $x_3 = i$ where $i \in \{2, 3, \cdots, 8\}$, the solutions to the equation $x_1 + x_2 = 18 - i$ are $(j, 18 - i - j)$ where $j \in \{10 - i, 11 - i, \cdots, 8\}$, i.e. there are $i - 1$ solutions to this equation. Therefore the number of solutions to $x_1 + x_2 + x_3 = 18$ is $\sum_{i=2}^{8}(i - 1) = 28$. Therefore we conclude that the answer is $1 + 52 + 28 = 81$. Alternative Solution: We also find the number of solutions to $9 \mid x_1 + x_2 + x_3$ where $x_i \in \{0, 1, \cdots, 8\}$ for $i = 1, 2, 3$. For any x_1 and x_2, $9 \mid x_1 + x_2 + x_3$ is equivalent to $x_3 \equiv 9 - x_1 - x_2 \pmod 9$, which has exactly one solution in $\{0, 1, \cdots, 8\}$. Therefore the number of solutions is $9 \times 9 = 81$.

Supplementary Explanation

Answer. 81.

Derivation/proof. Represent each integer uniquely as a three-digit string abc with leading zeros allowed. This does not introduce any digit 9. Each digit lies in {0,...,8}. Since $100 \equiv 10 \equiv 1 \pmod 9$, divisibility is $a + b + c \equiv 0$. For each of the 9^2 choices of a,b there is exactly one c in {0,...,8} satisfying this congruence. Therefore 81 strings, including 000, qualify.

After choosing the first two residues, the third must satisfy $x_3 \equiv -x_1 - x_2 \pmod 9$. Each residue has one representative in the allowed range, so there are $9^2 = 81$ choices in total.

N031

Modular Arithmetic

2019 Final | Question 7 | Source page 9 | Difficulty 3 | 8 marks
Q-2019-FINAL-P07 | Answer provenance: official

> Let P be a set of 5 distinct points in the plane with integer coordinates. Draw a line segment between every two points of P (note that a total of 10 line segments will be drawn). Show that at least one of these line segments has its midpoint with integer coordinates.

Official Answer

Source: physical PDF page 9 (2019-final.pdf).

The midpoint of the segment whose endpoints are (a, b) and (c, d) is $((a + c)/2, (b + d)/2)$. We are concerned only with integer values of the original coordinates. Clearly the coordinates of these fractions will be integers as well if and only if a and c have the same parity (both odd or both even) and b and d have the same parity. Thus what matters in this problem is the parities of the coordinates. There are four possible pairs of parities: (odd, odd), (odd, even), (even, odd), and (even, even). Since we are given five points, the pigeonhole principle guarantees that at least two of them will have the same pair of parities. The midpoint of the segment joining these two points will therefore have integer coordinates.

Supplementary Explanation

Answer. At least one of the ten segments has an integer-coordinate midpoint.

Derivation/proof. Classify a point (a, b) by $(a \bmod 2, b \bmod 2)$. There are four classes: $(0, 0), (0, 1), (1, 0), (1, 1)$. Five points force two distinct points $(a, b), (c, d)$ in one class by pigeonhole principle. Then a+c and b+d are even, so the midpoint $((a + c)/2, (b + d)/2)$ has both coordinates integral. The required segment is present because every pair is joined.

Every pair of points is joined in the question, so the guaranteed same-parity pair supplies one of the ten drawn segments.

If $a \equiv b \pmod 2$ and $b \equiv c \pmod 2$, then adding the two even differences gives $a \equiv c \pmod 2$. This verifies transitivity of the parity relation.

N032

Modular Arithmetic

2019 Midterm | Question 12 | Source page 14 | Difficulty 3 | 8 marks
Q-2019-MID-P12 | Answer provenance: official

> Evaluate the following expression by repeated squaring method. Show the steps.
>
> $$8^{1027} \bmod 22$$

Official Answer

Source: physical PDF page 14 (2019-midterm.pdf).

By the repeated squaring method, we have

$$\begin{aligned}
8^{2^0} &\equiv 8 \pmod{22} \\
8^{2^1} &\equiv 8^2 \equiv 20 \pmod{22} \\
8^{2^2} &\equiv 20^2 \equiv 4 \pmod{22} \\
8^{2^3} &\equiv 4^2 \equiv 16 \pmod{22} \\
8^{2^4} &\equiv 16^2 \equiv 14 \pmod{22} \\
8^{2^5} &\equiv 14^2 \equiv 20 \pmod{22}.
\end{aligned}$$

We have $1027 = 2^0 + 2^1 + 2^{10}$. $8^{2^{10}} \equiv 4 \pmod{22}$ since $8^{2^i} \equiv 4 \pmod{22}$ if $i \equiv 2 \pmod{4}$.

$$\begin{aligned}
8^{1027} &\equiv 8^{2^0+2^1+2^{10}} \pmod{22} \\
&\equiv 8^{2^0} \cdot 8^{2^1} \cdot 8^{2^{10}} \pmod{22} \\
&\equiv 8 \cdot 20 \cdot 4 \pmod{22} \\
&\equiv 2 \pmod{22}.
\end{aligned}$$

Therefore, we have $8^{1027} \bmod 22 = 2$.

Supplementary Explanation

Answer. 2.

Derivation/proof. The residues for 8^{2^i} at $i = 0, 1, \ldots, 10$ are $8, 20, 4, 16, 14, 20, 4, 16, 14, 20, 4$, obtained each time by squaring the previous residue modulo 22. For example $8^2 \equiv 20$, $20^2 \equiv 4$, $4^2 \equiv 16$, $16^2 \equiv 14$, $14^2 \equiv 20$. Since $1027 = 2^{10} + 2^1 + 2^0$, multiply the selected residues: $4 \cdot 20 \cdot 8 = 640 \equiv 2 \pmod{22}$. Direct modular exponentiation independently gives 2.

The full table for $i = 0, 1, \ldots, 10$ is $8, 20, 4, 16, 14, 20, 4, 16, 14, 20, 4$. The final product is $640 = 29 \cdot 22 + 2$. Repeated squaring works despite $\gcd(8, 22) \neq 1$.

N033

Modular Arithmetic

2016 Final | Question 4(a) | Source page 8 | Difficulty 3 | 5 marks
Q-2016-FINAL-P04-a | Answer provenance: official

> (a) Evaluate $6^{1161} \bmod 602$.

Official Answer

Source: PDF physical page 8 (2016-final.pdf).

(a)

$$\lfloor \log_2 1161 \rfloor = 10$$

$$\begin{aligned}
R_1 &= 6^2 \bmod 602 = 6 \times 6 \bmod 602 = 36 \\
R_2 &= 6^4 \bmod 602 = R_1 \times R_1 \bmod 602 = 92 \\
R_3 &= 6^8 \bmod 602 = R_2 \times R_2 \bmod 602 = 36 \\
R_4 &= 6^{16} \bmod 602 = R_3 \times R_3 \bmod 602 = 92 \\
R_5 &= 6^{32} \bmod 602 = R_4 \times R_4 \bmod 602 = 36 \\
R_6 &= 6^{64} \bmod 602 = R_5 \times R_5 \bmod 602 = 92 \\
R_7 &= 6^{128} \bmod 602 = R_6 \times R_6 \bmod 602 = 36 \\
R_8 &= 6^{256} \bmod 602 = R_7 \times R_7 \bmod 602 = 92 \\
R_9 &= 6^{512} \bmod 602 = R_8 \times R_8 \bmod 602 = 36 \\
R_{10} &= 6^{1024} \bmod 602 = R_9 \times R_9 \bmod 602 = 92
\end{aligned}$$

$$\begin{aligned}
6^{1161} \bmod 602 &= 6^{1024+128+8+1} \bmod 602 \\
&= 6^{1024} \cdot 6^{128} \cdot 6^8 \cdot 6^1 \bmod 602 \\
&= 92 \cdot 36 \cdot 36 \cdot 6 \bmod 602 \\
&= 216
\end{aligned}$$

Supplementary Explanation

Answer. 216.

Derivation/proof. Since $6^2 \equiv 36$, $6^4 \equiv 36^2 \equiv 92$, and $6^8 \equiv 92^2 \equiv 36 \pmod{602}$, successive squarings alternate 36 and 92. As $1161 = 1024 + 128 + 8 + 1$, $6^{1161} \equiv 92 \cdot 36 \cdot 36 \cdot 6 \pmod{602}$. Reducing the products gives $92 \cdot 36 \equiv 302$, $302 \cdot 36 \equiv 36$, then $36 \cdot 6 = 216$.

N034

Modular Arithmetic

2016 Final | Question 4(b)(i) | Source page 8 | Difficulty 3
Q-2016-FINAL-P04-b-i | Answer provenance: official

> (b) Is each of the following true? Please explain.
>
> (i) Is "$(178^{88} \bmod 89) = [(401^{3,243,554} + 1,197,491) \bmod 97]$"?

Official Answer

Source: PDF physical page 8 (2016-final.pdf); PDF physical page 9 (2016-final.pdf).

(b) (i) No.

Consider LHS.

$$\begin{aligned} 178^{88} \bmod 89 &= (178 \bmod 89)^{88} \bmod 89 \\ &= 0^{88} \bmod 89 \\ &= 0 \end{aligned}$$

Consider RHS.

$$\begin{aligned} &(401^{3,243,554} + 1,197,491) \bmod 97 \\ &= [((401 \bmod 97)^{3,243,554} \bmod 97) + (1,197,491 \bmod 97)] \bmod 97 \\ &= [(13^{3,243,554} \bmod 97) + 26] \bmod 97 \\ &= [(13^{3,243,554 \bmod (97-1)} \bmod 97) + 26] \bmod 97 \\ &\qquad \text{(By Fermat's Little Theorem)} \\ &= [(13^2 \bmod 97) + 26] \bmod 97 \\ &= (72 + 26) \bmod 97 \\ &= 1 \end{aligned}$$

Thus, LHS is not equal to RHS.

Supplementary Explanation

Answer. False: the left side is 0 and the right side is 1.

Derivation/proof. Because $178 \equiv 0 \pmod{89}$, the left side is 0. For the right side, $401 \equiv 13$, $1,197,491 \equiv 26 \pmod{97}$, and $3,243,554 \equiv 2 \pmod{96}$. As 97 is prime and $13 \not\equiv 0$, Fermat gives $13^{3,243,554} \equiv 13^2 \equiv 72 \pmod{97}$. The right residue is $(72 + 26) \bmod 97 = 1$, which differs from 0.

The final right-side reduction is $72 + 26 \equiv 1 \pmod{97}$.

N035

Modular Arithmetic

2015 Final | Question 10 | Source page 17 | Difficulty 3 | 10 marks
Q-2015-FINAL-P10 | Answer provenance: derived

> Prove the following statement by induction
>
> $$\big(n(n-1)\cdots(2)(1)+1\big) \bmod m = 1$$
>
> for all integer $n > 1$ and for any $m \in \{2, 3, \cdots, n\}$.
>
> Note: No marks will be awarded if you do not use the principle of induction.

Official Answer

Source: PDF physical page 17 (2015-final.pdf).

Let $p(n)$ denotes $(n(n-1)\cdots(2)(1)+1) \bmod m = 1$.

Base case ($n = 2$): $(2+1) \bmod 2 = 1$, so $p(2)$ is true.

Inductive hypothesis: Assume $p(n-1)$ is true for $n - 1 > 1$.

$$((n-1)\cdots(2)(1)+1) \bmod m = 1$$

Inductive step (for $n > 2$):

$$\begin{aligned}
&((n-1)\cdots(2)(1)+1) \bmod m = 1 \\
\rightarrow\ &(n(n-1)\cdots(2)(1)+n) \bmod m = n \bmod m \\
\rightarrow\ &(n(n-1)\cdots(2)(1)+1+(n-1)) \bmod m = n \bmod m \\
\rightarrow\ &(n(n-1)\cdots(2)(1)+1) \bmod m = (n-(n-1)) \bmod m \\
\rightarrow\ &(n(n-1)\cdots(2)(1)+1) \bmod m = 1
\end{aligned}$$

Inductive conclusion: By the principle of mathematical induction, $p(n)$ is true for all integers $n > 1$.

Supplementary Explanation

Answer. $(n! + 1) \bmod m = 1$ for all the specified n, m.

Derivation/proof. Let $P(n)$ assert the statement for all $2 \le m \le n$. Base $n = 2$: $3 \bmod 2 = 1$. Assume $P(n-1)$ for $n > 2$. If $2 \le m \le n-1$, the hypothesis implies $(n-1)! \equiv 0 \pmod m$, hence $n! \equiv 0 \pmod m$. For the newly allowed $m = n$, $n! = n(n-1)!$ is directly divisible by n. In both cases $n! + 1 \equiv 1 \pmod m$, and $m \ge 2$ makes 1 the actual remainder. This fills the new-modulus case left implicit in the printed proof.

N036

Modular Arithmetic

2024 Spring Midterm | Question 8 | Source page 11 | Difficulty 4 | 10 marks
Q-2024S-MID-P08 | Answer provenance: official

> The professor gave a mathematics exam to five students. The marks that the five students received (listed in increasing order) were 71, 76, 80, 82, and 91. The professor entered the scores in random order into a spreadsheet, which recalculated the class average after each score was entered. The professor noticed that after each score was entered, the average was always an integer. Based on this information, you need to determine the last score entered by the professor. Justify your answer fully.

Official Answer

Source: physical PDF page 11 (COMP2711_S2024_Midterm.pdf).

> Answer. We assign variables $a = 71, b = 76, c = 80, d = 82, e = 91$. Let x_i denote the i-th score entered into the spreadsheet, for $1 \leq i \leq 5$. We know that $x_1 + x_2 \equiv 0 \pmod 2$, $x_1 + x_2 + x_3 \equiv 0 \pmod 3$, and $x_1 + x_2 + x_3 + x_4 \equiv 0 \pmod 4$. There are several approaches to find the last number. Solution 1: First, observe that there are two odd numbers and three even numbers. Since the sum of the first 4 numbers is divisible by 4, it follows that both odd numbers must exist among the first 4 numbers. The sum of these two odd numbers modulo 4 is 2. So the other two even numbers cannot be 76 and 80 (since these are both 0 modulo 4). It follows that the last number has to be either 76 or 80. Now notice that the first two numbers either have to be both odd or both even. But they cannot be both odd because in that case, the sum of first three numbers cannot be divisible by 3 (since the sum of two available odd numbers is divisible by 3 and none of the even numbers is a multiple of 3). So the first two numbers have to be even such that 82 is among them. So in the first three numbers, one is 82, another is an odd number and another is an even number such that each has a remainder 1 mod 3. So the only options will be 76 and 91. So the last number x_5 has to be 80. Solution 2: There exists an alternative argument that is simpler. The sequence of numbers $a,, b, c, d, e$ modulo 3 is $2, 1, 2, 1, 1$. Thus, there is only way to choose 3 numbers out of this set whose sum is $0 \pmod 3$, namely, by choosing the three numbers b, d, e. Next observe that this same sequence of numbers modulo 4 is $3, 0, 0, 2, 3$. Note that $b + d + e \pmod 4$ is $5 \equiv 1$. Since the sum of the first 4 numbers is $0 \pmod 4$, it follows that the 4th number selected must be a (since $a \equiv 3 \pmod 4$). Thus, the last number selected must be c, namely 80.

Supplementary Explanation

Answer. The last score is 80.

Derivation/proof. The five residues modulo three are $2, 1, 2, 1, 1$. A triple summing to zero modulo three must consist of three ones or three twos; either mixed triple has residue one or two. Only two twos exist, so the first three scores must be $\{76, 82, 91\}$, with sum $249 \equiv 1 \pmod 4$. The fourth score must have residue three modulo four. Of those remaining, $71 \equiv 3 \pmod 4$ and $80 \equiv 0 \pmod 4$, so the fourth is 71 and the last is 80. The sequence $76, 82, 91, 71, 80$ has prefix averages $76, 79, 83, 80, 80$, all integral, verifying feasibility.

N037

Modular Arithmetic

2020 Final Part 1 | Question 2 | Source page 1 | Difficulty 4 | 8 marks
Q-2020-FINAL1-P02 | Answer provenance: official

> Prove that for any $n \in \mathbb{N}$, there exists an n-digit number A_n such that all its digits are either 1 or 2, and $2^n \mid A_n$. [Hint: Use induction.]

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf); physical PDF page 2 (2020-final-part-1.pdf).

We use induction on n. Base case: $n = 1$. We can set $A_1 = 2$, $2^1 \mid A_1$. Induction hypothesis: Assume it is correct for $n = k$, which means that there exists A_k such that $2^k \mid A_k$. Now consider $n = k + 1$: As $2^k \mid A_k$, there are two possibilities for the remainder of A_k dividing 2^{k+1}. $A_k = 2^k \cdot l$, where l is even, which means $A_k \equiv 0 \pmod{2^{k+1}}$, or $A_k = 2^k \cdot (1 + l')$, where l' is even, which means $A_k \equiv 2^k \pmod{2^{k+1}}$.

1. $A_k \equiv 0 \pmod{2^{k+1}}$. In this scenario we add the digit 2 to the left side of A_k. So $A_{k+1} = \overline{2A_k}$. We show that $2^{k+1} \mid A_{k+1}$. $A_{k+1} = 2 \cdot 10^k + A_k = 2 \cdot 2^k \cdot 5^k + A_k = 2^{k+1} \cdot 5^k + A_k$. And we know that $2^{k+1} \cdot 5^k \equiv 0 \pmod{2^{k+1}}$ and $A_k \equiv 0 \pmod{2^{k+1}}$. So $A_{k+1} \equiv 0 \pmod{2^{k+1}}$.
2. $A_k \equiv 2^k \pmod{2^{k+1}}$. In this scenario we add the digit 1 to the left side of A_k. So $A_{k+1} = \overline{1A_k}$. We show that $2^{k+1} \mid A_{k+1}$. $A_{k+1} = 1 \cdot 10^k + A_k = 2^k \cdot 5^k + A_k$. And we know that $A_k \equiv 2^k \pmod{2^{k+1}}$. So $A_{k+1} \equiv 2^k + 2^k \cdot 5^k \equiv 2^k \cdot (1 + 5^k) \pmod{2^{k+1}}$. $1 + 5^k$ is an even number so $A_{k+1} \equiv 0 \pmod{2^{k+1}}$.

Supplementary Explanation

Answer. Such a number exists for every positive digit length n.

Derivation/proof. Base n=1: take A1=2. Suppose Ak has k digits in {1,2} and $A_k = 2^k t$. Choose a leading digit d∈{1,2} with d+t even: d=2 when t is even and d=1 when t is odd. Set $A_{k+1} = d10^k + A_k = 2^k(d5^k + t)$. Because 5^k is odd, the parenthesis is even by the choice of d. Hence $2^{k+1} \mid A_{k+1}$. Adding a nonzero leading digit creates exactly k+1 digits without changing the allowed suffix digits, completing induction.

The source notation $\overline{2A_k}$ and $\overline{1A_k}$ denotes digit concatenation, not multiplication.

The initial value is $A_1 = 2$. If $A_k = 2^k t$ with integer t, the recurrence supplies one more factor of two in the next term; this establishes the divisibility induction.

N038

GCDs and Congruences

2025 Spring Final | Question 4(a) | Source page 5 | Difficulty 1 | 1 mark
Q-2025S-FINAL-P04-a | Answer provenance: official

> The least common multiple (LCM) of two integers x and y is the smallest positive integer that is a multiple of both x and y. As an example, the least common multiple of 8 and 10 is 40.
>
> (a) [1 pt] Assume $x = 2^3 \cdot 5^6$ and $y = 2^4 \cdot 5^3$. What is the least common multiple of x and y?

Official Answer

Source: physical PDF page 5 (COMP2711_S2025_Final.pdf).

(a) Their LCM is $2^{\max(3,4)} \cdot 5^{\max(6,3)} = 2^4 \cdot 5^6$.

Supplementary Explanation

Answer. $2^4 \cdot 5^6$.

Derivation/proof. Divisibility by both x and y requires the exponent of two to be at least $\max(3, 4) = 4$ and that of five at least $\max(6, 3) = 6$. Using these least permitted exponents gives $2^4 \cdot 5^6$. Its quotient by x is two and by y is 5^3, verifying it is a common multiple. Every positive common multiple contains these prime powers, so none can be smaller.

N039

GCDs and Congruences

2019 Midterm | Question 10(a) | Source page 12 | Difficulty 1 | 2 marks
Q-2019-MID-P10-a | Answer provenance: official

> Solve each of these congruences with $0 \leq x \leq 103$ and $0 \leq y \leq 103$ using the result $15 \cdot 7 \equiv 1 \pmod{104}$.
>
> (a) $7x \equiv 31 \pmod{104}$

Official Answer

Source: physical PDF page 12 (2019-midterm.pdf).

$$\begin{aligned} 7x &\equiv 31 \pmod{104} \\ \rightarrow 15 \cdot 7x &\equiv 15 \cdot 31 \pmod{104} \\ \rightarrow x &\equiv 49 \pmod{104}. \end{aligned}$$

Supplementary Explanation

Answer. $x = 49$.

Derivation/proof. Multiply both sides by the inverse 15: $x \equiv 15 \cdot 31 = 465 \equiv 49 \pmod{104}$ since $465 = 4 \cdot 104 + 49$. Direct verification gives $7 \cdot 49 = 343 = 3 \cdot 104 + 31$. The representative 49 lies in the required range.

N040

GCDs and Congruences

2014 Final | Question 2(b) | Source page 4 | Difficulty 1
Q-2014-FINAL-P02-b | Answer provenance: official

> Let a, m and n be three positive integers that are larger than 1. Is each of the following statements true or false? If it is true, give a proof. If it is false, give a counterexample.
>
> (b) $(a \bmod mn) \bmod n = a \bmod m$.

Official Answer

Source: PDF physical page 4 (2014-final.pdf).

(b) This is false. Let $a = 8, m = 2$ and $n = 3$. We have

$$(a \bmod mn) \bmod n = (8 \bmod 6) \bmod 3 = 2,$$

$$a \bmod m = 8 \bmod 2 = 0.$$

Supplementary Explanation

Answer. False: $a = 8, m = 2, n = 3$ gives $2 \neq 0$.

Derivation/proof. Here $(8 \bmod 6) \bmod 3 = 2 \bmod 3 = 2$, but $8 \bmod 2 = 0$. One concrete allowed triple disproves the universal claim.

All three selected parameters are positive integers larger than one, as required.

N041

GCDs and Congruences

2014 Final | Question 2(d) | Source page 4 | Difficulty 1
Q-2014-FINAL-P02-d | Answer provenance: official

> Let a, m and n be three positive integers that are larger than 1. Is each of the following statements true or false? If it is true, give a proof. If it is false, give a counterexample.
>
> (d) If $gcd(a, n) = 1$, then $a \bmod n = 1$.

Official Answer

Source: PDF physical page 4 (2014-final.pdf).

(d) This is false. For example, $\gcd(5, 3) = 1$, but $5 \bmod 3 = 2$.

Supplementary Explanation

Answer. False: $\gcd(5, 3) = 1$ but $5 \bmod 3 = 2$.

Derivation/proof. Coprimality only guarantees that a is invertible modulo n; it does not say that its residue equals one. The displayed pair is a counterexample.

Take the unused parameter $m = 2$. All domain restrictions then hold and the implication has a true antecedent and false consequent.

For this counterexample, the actual residue is $2 \neq 1$.

N042

GCDs and Congruences

2025 Spring Final | Question 2(b) | Source page 3 | Difficulty 2 | 3 marks
Q-2025S-FINAL-P02-b | Answer provenance: official

> (b) [3 pts] Use Fermat's Little Theorem to prove that 63 is not a prime number. Note that simply factorizing 63, e.g., as 3×21, is not a valid solution here. You must apply Fermat's Little Theorem explicitly. Hint: You might find part (a) useful here.

Official Answer

Source: physical PDF page 3 (COMP2711_S2025_Final.pdf).

(b) We know that if p is prime, Then $a^{p-1} \equiv 1 \pmod{p}$. (if a is not a multiple of p)

So we can conclude that $a^{p-1} \not\equiv 1 \pmod{p} \rightarrow p$ is not prime.

As $2^{62} \not\equiv 1 \pmod{63}$ we can conclude that 63 is not prime!

Supplementary Explanation

Answer. 63 is not prime.

Derivation/proof. Part (a) gives $2^{62} \equiv 4 \pmod{63}$. Suppose 63 were prime. Since $63 \nmid 2$, Fermat's Little Theorem would force $2^{63-1} \equiv 1 \pmod{63}$. However one and four are not congruent modulo sixty-three, since $63 \nmid (4-1) = 3$. This contradicts the computed remainder. The proof explicitly uses the required theorem rather than replacing it with factorization.

N043

GCDs and Congruences

2025 Spring Midterm | Question 8(a) | Source page 12 | Difficulty 2
Q-2025S-MID-P08-a | Answer provenance: derived

> (a) Show that for any positive integers a, b, c, d,
>
> $$ad \equiv bd \pmod{cd}$$
>
> if and only if
>
> $$a \equiv b \pmod{c}.$$

Official Answer

Source: physical PDF page 12 (COMP2711_S2025_Midterm.pdf).

(a) We know that $cd \mid ad - bd$. In other words $d(a-b) = cd \cdot k$. We can divide both sides by d. So $(a-b) = c \cdot k$, which means $c \mid a - b$. This is the same as $a \equiv b \pmod{c}$.

Supplementary Explanation

Answer. The two congruences are equivalent.

Derivation/proof. Forward direction: $ad \equiv bd \pmod{cd}$ means $d(a-b) = cdk$ for an integer k. Since $d > 0$, divide the integer equality by d to obtain $a - b = ck$, hence $a \equiv b \pmod{c}$. Reverse direction: if $a \equiv b \pmod{c}$, then $a - b = ck$ for an integer k. Multiplying by d gives $ad - bd = cdk$, hence $ad \equiv bd \pmod{cd}$. Both directions hold. The reverse direction completes the printed one-direction argument.

N044

GCDs and Congruences

2024 Fall Final | Question 6(a) | Source page 17 | Difficulty 2 | 4 marks
Q-2024F-FINAL-P06-a | Answer provenance: official

> Assume that a, b are integers and p is a prime number of form $4k + 3$ (with an integer k). As an example, 19 is a prime of this form while 17 is not of this form. Assume that $p \mid a^2 + b^2$.
>
> (a) Prove that if $p \mid a$ then $p \mid b$.

Official Answer

Source: physical PDF page 17 (COMP2711_F2024_Final.pdf).

> Answer: (a) Note that if $p \mid a$ then $p \mid a^2$ which means $p \mid (a^2 + b^2) - a^2 \implies p \mid b^2 \implies p \mid b$.

Supplementary Explanation

Answer. $p \mid a \Rightarrow p \mid b$.

Derivation/proof. From $p \mid a$, write $a = pt$, giving $p \mid a^2$. The shared premise says $p \mid a^2 + b^2$. Divisibility is closed under subtraction, so $p \mid [(a^2 + b^2) - a^2] = b^2$. Since p is prime, Euclid's lemma applied to $b^2 = b \cdot b$ yields $p \mid b$. This part needs primality only; the condition $p \equiv 3 \pmod 4$ is used in the next part.

N045

GCDs and Congruences

2024 Fall Final | Question 7(1) | Source page 19 | Difficulty 2
Q-2024F-FINAL-P07-1 | Answer provenance: official

> Solve the following system of congruences and show both steps:
>
> $$3x \equiv 4 \pmod 5, \qquad x \equiv 2 \pmod 4, \qquad 3x \equiv 6 \pmod 7.$$
>
> 1. Remove the coefficients of x.

Official Answer

Source: physical PDF page 19 (COMP2711_F2024_Final.pdf).

> Answer: Note that the first step is to remove the coefficients. Note that the inverse of 3 (mod 5) is 2. So if we multiply 2 by both sides of the first equation we have: $2 \cdot 3x \equiv 2 \cdot 4 \implies x \equiv 3 \pmod 5$. Note that as $\gcd(3, 7) = 1$ we can simply divide both sides by 3 which gives us $x \equiv 2 \pmod 7$. So we need to solve the following system of congruences:
>
> $$x \equiv 3 \pmod 5, \qquad x \equiv 2 \pmod 4, \qquad x \equiv 2 \pmod 7.$$
>
> As all the modulos are coprime with each other. We can simply use chinese remainder theorem to find the solutions x.

Supplementary Explanation

Answer. $x \equiv 3 \pmod 5$, $x \equiv 2 \pmod 4$, $x \equiv 2 \pmod 7$.

Derivation/proof. Modulo five, $3 \cdot 2 = 6 \equiv 1$, so multiply the first congruence by two to obtain $x \equiv 8 \equiv 3 \pmod 5$. The second already has unit coefficient and remains $x \equiv 2 \pmod 4$. Modulo seven, $3 \cdot 5 = 15 \equiv 1$, so multiplying the third by five gives $x \equiv 30 \equiv 2 \pmod 7$. Both multiplications are invertible because three is coprime to each modulus. The normalized system is completely equivalent to the original, with no solutions gained or lost.

N046

GCDs and Congruences

2024 Spring Midterm | Question 12(a) | Source page 18 | Difficulty 2 | 5 marks
Q-2024S-MID-P12-a | Answer provenance: official

> (a) Let a and m be positive integers. Prove that no positive integer smaller than $\gcd(a, m)$ can be expressed as an integer linear combination of a and m.

Official Answer

Source: physical PDF page 18 (COMP2711_S2024_Midterm.pdf).

> Answer. (a) Let $d = \gcd(a, m)$. Let y be a positive integer that is an integer linear combination of a and m, that is, $y = ak + mk'$, for some integers k and k'. By definition of gcd, we have $d \mid a$ and $d \mid m$. It follows that $d \mid ak + mk'$. Since $y = ak + mk'$ is a positive integer, it cannot be smaller than d.

Supplementary Explanation

Answer. Every positive integer linear combination is at least $\gcd(a, m)$.

Derivation/proof. Let $d = \gcd(a, m) > 0$. There are integers u, v with $a = du, m = dv$. For any integer linear combination $y = ra + sm$, with $r, s \in \mathbb{Z}$, we have $y = d(ru + sv)$, hence $d \mid y$. If $y > 0$, the integer $ru + sv$ is at least one, giving $y \geq d$. Thus no integer with $0 < y < d$ can be represented, as claimed.

N047

GCDs and Congruences

2024 Spring Midterm | Question 12(b) | Source page 18 | Difficulty 2 | 5 marks
Q-2024S-MID-P12-b | Answer provenance: official

> (b) Let a and m be positive integers. Use the fact given in part (a) to prove that if $\gcd(a, m) > 1$, then a does not have a multiplicative inverse modulo m.

Official Answer

Source: physical PDF page 18 (COMP2711_S2024_Midterm.pdf).

> (b) Assume that a has a multiplicative inverse modulo m. Let us denote it a^{-1}. By definition, $aa^{-1} \equiv 1 \pmod{m}$. It follows that $aa^{-1} - mk = 1$ for some integer k. In other words, 1 can be obtained as an integer linear combination of a and m. But 1 is smaller than $\gcd(a, m)$, which contradicts part (a). This completes the proof.

Supplementary Explanation

Answer. If $\gcd(a, m) > 1$, then a has no multiplicative inverse modulo m.

Derivation/proof. Let $d = \gcd(a, m) > 1$ and suppose an inverse b exists, so $ab \equiv 1 \pmod{m}$. The congruence means $ab - 1 = tm$ for some integer t, hence $1 = ba - tm$. This is a linear combination of a, m with integer coefficients $b, -t$. By part (a), any positive such combination is at least d, whereas $0 < 1 < d$. This contradiction shows no such b exists.

N048

GCDs and Congruences

2023 Fall Final | Question 1(a)(i) | Source page 3 | Difficulty 2 | 1 mark
Q-2023F-FINAL-P01-a-i | Answer provenance: official

> In this question the domain is the set of all integers. Consider the following predicates:
>
> $P(x)$: x is prime.
>
> $Q(x, y)$: x, y are relatively prime (co-prime).
>
> $E(x, y, z)$: x, y are congruent mod z (i.e., $x \equiv y \pmod{z}$).
>
> (a) Determine whether the following logical expressions are True or False. No justification is needed. ($a \mid b$ means b is divisible by a)
>
> (i) $\forall x \forall a \forall b : \ x \mid ab \rightarrow (x \mid a \vee x \mid b)$

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Final.pdf).

> (a) (i) False

Supplementary Explanation

Answer. False.

Derivation/proof. Take $x = 6$, $a = 2$, and $b = 3$. Then $ab = 6$, so $6 \mid ab$ is true. But $6 \nmid 2$ and $6 \nmid 3$, making the disjunction false. Thus the implication is false for this one integer triple, which falsifies the universally quantified expression. The counterexample also shows precisely why a primality assumption would matter.

Take $x = 6, a = 2, b = 3$. The implication is $T \rightarrow F$, since $6 \mid 6$ but $6 \nmid 2$ and $6 \nmid 3$. The divisor is not restricted to primes.

N049

GCDs and Congruences

2023 Fall Final | Question 1(a)(iii) | Source page 3 | Difficulty 2 | 1 mark
Q-2023F-FINAL-P01-a-iii | Answer provenance: official

> In this question the domain is the set of all integers. Consider the following predicates:
>
> $P(x)$: x is prime.
>
> $Q(x, y)$: x, y are relatively prime (co-prime).
>
> $E(x, y, z)$: x, y are congruent mod z (i.e., $x \equiv y \pmod{z}$).
>
> (a) Determine whether the following logical expressions are True or False. No justification is needed. ($a \mid b$ means b is divisible by a)
>
> (iii) $\forall x \forall a \forall b : \ (x \mid ab \to (x \mid a \vee x \mid b)) \to P(x)$

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Final.pdf).

> (iii) False

Supplementary Explanation

Answer. False.

Derivation/proof. Let $x = 1$, and for concreteness take $a = b = 1$. Both $1 \mid ab$ and $1 \mid a \vee 1 \mid b$ are true, so their implication is true. But $P(1)$ is false because 1 is not prime. The outer implication is therefore $T \to F = F$, falsifying the universal formula. This respects the actual printed placement of the a,b quantifiers outside the whole implication.

For $x = 1, a = b = 1$, the divisibility implication is true but $P(1) = F$. The outer implication therefore has value $T \to F = F$.

N050

GCDs and Congruences

2022 Midterm | Question 7(a) | Source page 3 | Difficulty 2 | 5 marks
Q-2022-MID-P07-a | Answer provenance: official

> Calculate $2^{542} \bmod 109$. Show your steps.

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (a) 109 is a prime number. By using Fermat's Little Theorem: $2^{108} \equiv 1 \pmod{109} \implies (2^{108})^5 \equiv 1^5 \implies 2^{540} \equiv 1 \pmod{109}$. So $2^{542} = 2^2 \cdot 2^{540} \equiv 2^2 \equiv 4 \pmod{109}$.

Supplementary Explanation

Answer. 4.

Derivation/proof. 109 is prime and 2 is coprime to it, so 2^108≡1 mod 109. Divide the exponent: 542=5·108+2. Then 2^542=(2^108)^5·2²≡1·4≡4 mod 109. The representative4 already lies betweeno and 108.

Specifically $542 = 5 \cdot 108 + 2$ and $2^{542} = (2^{108})^5 \cdot 2^2$.

N051

GCDs and Congruences

2020 Midterm | Question 7 | Source page 4 | Difficulty 2 | 10 marks
Q-2020-MID-P07 | Answer provenance: official

> Solve $10x + 4 \equiv 0 \pmod{23}$.

Official Answer

Source: physical PDF page 4 (2020-midterm.pdf).

Solve $10x \equiv -4 \pmod{23}$. As $\gcd(10, 23) = 1$ we can multiply both sides by the inverse of $10 \pmod{23}$. We calculate the inverse of $10 \pmod{23}$ using extended Euclidean algorithm.

$$\begin{aligned} 23 &= 10 \cdot 2 + 3 \\ 10 &= 3 \cdot 3 + 1 \\ 3 &= 1 \cdot 3 + 0. \end{aligned}$$

Rewriting:

$$3 = 23 - 10 \cdot 2, \qquad 1 = 10 - 3 \cdot 3.$$

Substituting:

$$\begin{aligned} 1 &= 10 - 3 \cdot 3 \\ 1 &= 10 - (23 - 10 \cdot 2) \cdot 3 \\ 1 &= 23 \cdot (-1) + 10 \cdot 7. \end{aligned}$$

7 is the modular inverse of $10 \pmod{23}$. We multiply both sides by 7. $10x \equiv -4 \pmod{23} \implies 10x \cdot 10^{-1} \equiv -4 \cdot 10^{-1} \implies x \equiv -4 \cdot 7 \equiv -28 \equiv 18$. $x \equiv 18 \pmod{23}$ is the final answer.

Supplementary Explanation

Answer. $x \equiv 18 \pmod{23}$.

Derivation/proof. Move the constant to get $10x \equiv -4$. Since $23 = 2 \cdot 10 + 3$ and $10 = 3 \cdot 3 + 1$, back substitution gives $1 = 7 \cdot 10 - 3 \cdot 23$. Therefore multiply by 7: $x \equiv -28 \equiv 18 \pmod{23}$. Check $10 \cdot 18 + 4 = 184 = 8 \cdot 23$. Every solution is $18 + 23k$ for integer k.

N052

GCDs and Congruences

2020 Midterm | Question 8 | Source page 4 | Difficulty 2 | 10 marks
Q-2020-MID-P08 | Answer provenance: official

> Show that $\gcd(21n+4, 14n+3) = 1$, for any $n \in \mathbb{N}$.

Official Answer

Source: physical PDF page 4 (2020-midterm.pdf).

Using the Euclid's algorithm, we have $\gcd(21n+4, 14n+3) = \gcd(14n+3, 7n+1) = \gcd(7n+1, 1) = 1$.

Supplementary Explanation

Answer. The gcd equals 1 for every permitted n.

Derivation/proof. Subtract the second argument from the first: $(21n+4) - (14n+3) = 7n+1$. Subtract twice this remainder from the second: $(14n+3) - 2(7n+1) = 1$. Any common divisor therefore divides 1, so it is 1. Equivalently $3(14n+3) - 2(21n+4) = 1$ is a Bezout certificate valid for every integer n, including the requested naturals.

The independent Bezout certificate is $3(14n+3) - 2(21n+4) = 1$, also valid at $n = 0$.

The Euclidean reductions end in $\gcd(7n+1, 1) = 1$. This final gcd is one for every allowed n, so no exceptional natural-number case remains.

N053

GCDs and Congruences

2019 Midterm | Question 10(b) | Source page 12 | Difficulty 2 | 3 marks
Q-2019-MID-P10-b | Answer provenance: official

> Solve each of these congruences with $0 \leq x \leq 103$ and $0 \leq y \leq 103$ using the result $15 \cdot 7 \equiv 1 \pmod{104}$.
>
> (b) $15x + 41 \equiv 18 \pmod{104}$

Official Answer

Source: physical PDF page 12 (2019-midterm.pdf).

$$\begin{aligned} 15x + 41 &\equiv 18 \pmod{104} \\ \rightarrow 15x &\equiv -23 \pmod{104} \\ \rightarrow 7 \cdot 15x &\equiv 7(-23) \pmod{104} \\ \rightarrow x &\equiv 7(-23) \pmod{104} \\ \rightarrow x &\equiv 47 \pmod{104}. \end{aligned}$$

Supplementary Explanation

Answer. $x = 47$.

Derivation/proof. Subtract 41 to get $15x \equiv -23 \pmod{104}$. Since 7 is inverse to 15, $x \equiv 7(-23) = -161 \equiv 47 \pmod{104}$. Check $15 \cdot 47 + 41 = 746 = 7 \cdot 104 + 18$. Thus 47 is the allowed representative.

N054

GCDs and Congruences

2019 Midterm | Question 10(c) | Source page 12 | Difficulty 2 | 3 marks
Q-2019-MID-P10-c | Answer provenance: official

> Solve each of these congruences with $0 \leq x \leq 103$ and $0 \leq y \leq 103$ using the result $15 \cdot 7 \equiv 1 \pmod{104}$.
>
> (c) $49x \equiv 11 \pmod{104}$

Official Answer

Source: physical PDF page 12 (2019-midterm.pdf).

$$\begin{aligned} 49x &\equiv 11 \pmod{104} \\ \rightarrow 15 \cdot 7 \cdot 7x &\equiv 15 \cdot 11 \pmod{104} \\ \rightarrow 7x &\equiv 61 \pmod{104} \\ \rightarrow 15 \cdot 7x &\equiv 15 \cdot 61 \pmod{104} \\ \rightarrow x &\equiv 83 \pmod{104}. \end{aligned}$$

Supplementary Explanation

Answer. $x = 83$.

Derivation/proof. Write $49 = 7^2$. Multiplying once by 15 gives $7x \equiv 165 \equiv 61 \pmod{104}$. Multiplying again gives $x \equiv 15 \cdot 61 = 915 \equiv 83 \pmod{104}$. Check $49 \cdot 83 = 4067 = 39 \cdot 104 + 11$, confirming the result.

Each multiplication uses an invertible residue, so no solutions are lost or introduced.

N055

GCDs and Congruences

2018 Final | Question 1(b) | Source page 3 | Difficulty 2
Q-2018-FINAL-P01-b | Answer provenance: official

> For each of the following predicates, decide whether it is true or false. If it is true, prove it. If it is false, state it is false; you do not have to prove it.
>
> (b) (Prime number p; x, y, z in $\mathbb{Z}_p$; $y \neq 0$) $\forall x \forall y \exists z (x + yz \equiv 0 \pmod{p})$

Official Answer

Source: physical PDF page 3 (2018-final.pdf).

True. The inverse of y must exist in $\mathbb{Z}_p$, so we can set $z \equiv -y^{-1}x \pmod{p}$ to satisfy the equation for any x and y.

Supplementary Explanation

Answer. True; choose $z \equiv -y^{-1}x \pmod{p}$.

Derivation/proof. Because p is prime and $y \not\equiv 0 \pmod{p}$, $\gcd(y, p) = 1$, so a multiplicative inverse y^{-1} exists. Choose the representative of $-y^{-1}x$ in $\mathbb{Z}_p$. Then $x + yz \equiv x - yy^{-1}x \equiv x - x \equiv 0$. The witness may depend on both universally chosen values.

Choosing the second variable equal to the first makes the difference $x - x = 0$, independent of the value of x. This supplies the existential witness for every universal choice.

N056

GCDs and Congruences

2018 Midterm | Question 3(a) | Source page 5 | Difficulty 2 | 2 marks
Q-2018-MID-P03-a | Answer provenance: official

> Consider the set $S = \{1, 2, \ldots, 2n\}$, and subsets $P = \{p_1, p_2, \ldots, p_m\} \subseteq S$ such that they satisfy the following criterion:
>
> $\forall p_i \in P \forall p_j \in P (i \neq j \rightarrow p_i \nmid p_j)$
>
> In the above, $p_i \nmid p_j$ means p_j is not divisible by p_i.
>
> (a) Find a subset $P \subseteq S$ satisfying the above criterion such that $|P| = n$.

Official Answer

Source: physical PDF page 5 (2018-midterm.pdf).

$P = \{n + 1, n + 2, \ldots, 2n\}$. Answer for a specific n is not accepted. (2 marks) Other answers are possible.

Supplementary Explanation

Answer. $P = \{n + 1, n + 2, \ldots, 2n\}$.

Derivation/proof. This set contains exactly $2n - (n + 1) + 1 = n$ elements. If $a < b$ belong to it and $a \mid b$, their distinct positive values imply $b \geq 2a \geq 2(n + 1) > 2n$, contradicting $b \leq 2n$. Also a larger positive integer cannot divide a smaller positive integer, so both ordered directions of the criterion hold.

The set size is $2n - (n + 1) + 1 = n$. A larger positive integer cannot divide a smaller positive integer either, so the ordered-pair criterion holds in both directions.

N057

GCDs and Congruences

2018 Midterm | Question 3(b) | Source page 5 | Difficulty 2 | 3 marks
Q-2018-MID-P03-b | Answer provenance: official

> Consider the set $S = \{1, 2, \ldots, 2n\}$, and subsets $P = \{p_1, p_2, \ldots, p_m\} \subseteq S$ such that they satisfy the following criterion:
>
> $\forall p_i \in P \forall p_j \in P (i \neq j \rightarrow p_i \nmid p_j)$
>
> In the above, $p_i \nmid p_j$ means p_j is not divisible by p_i.
>
> (b) For any subset P satisfying the above criterion, we can write every $p_i \in P$ as $c_i \cdot 2^{k_i}$, where c_i is odd. Prove that the following is true: $\forall p_i \in P \forall p_j \in P (i \neq j \rightarrow c_i \neq c_j)$.

Official Answer

Source: physical PDF page 5 (2018-midterm.pdf).

WLOG consider $p_i < p_j \in P$. If $i \neq j$, then $p_i \nmid p_j$. However, if $c_i = c_j$, then $p_i = c_i \cdot 2^{k_i} \mid c_i \cdot 2^{k_i + (k_j - k_i)} = c_i \cdot 2^{k_j} = p_j$. This is a contradiction, so $c_i \neq c_j$. (3 marks)

Supplementary Explanation

Answer. Distinct elements of P have distinct odd parts.

Derivation/proof. Suppose $c_i = c_j = c$. Since the elements differ, their exponents differ; assume $k_i < k_j$. Then $p_j = c2^{k_j} = p_i 2^{k_j - k_i}$, so $p_i \mid p_j$, contradicting the defining criterion. Thus equal odd parts are impossible.

The thin vertical stroke in the printed chain is divisibility $\mid$, not equality.

N058

GCDs and Congruences

2018 Midterm | Question 5(b) | Source page 7 | Difficulty 2
Q-2018-MID-P05-b | Answer provenance: official

> (b) Let x, y be positive integers and p be a positive prime integer. Show that, if $p \mid xy$, then $p \mid x$ or $p \mid y$. (Note: You may use the result from (a) even if you did not solve that part.)

Official Answer

Source: physical PDF page 7 (2018-midterm.pdf).

Since p is prime, it follows that $\gcd(p, x)$ is either p or 1, because these are the only divisors of p. If $\gcd(p, x) = p$, then x is a multiple of p, that is, $p \mid x$. Else, $\gcd(p, x) = 1$, that is, x, p are relatively prime. In this case, using (a), we get that $p \mid y$.

Supplementary Explanation

Answer. At least one of x, y is divisible by p.

Derivation/proof. If $p \mid x$, the conclusion already holds. Otherwise, because the only positive divisors of p are $1, p$, we have $\gcd(p, x) = 1$. Applying part (a) to $p \mid xy$ with the coprime factor x gives $p \mid y$. These two cases exhaust all possibilities.

Primality is essential to the coprime case; the conclusion does not extend to arbitrary composite divisors.

Because p is prime, $p \nmid x$ forces $\gcd(p, x) = 1$. The preceding coprime divisor result then transfers the divisibility of the product to its other factor.

N059

GCDs and Congruences

2017 Midterm | Question 5(a) | Source page 7 | Difficulty 2
Q-2017-MID-P05-a | Answer provenance: official

> Use the Euclidean algorithm to find $\gcd(15, 221)$. Show the steps.

Official Answer

Source: physical PDF page 7 (2017-midterm.pdf).

$\gcd(221, 15)$:

$$\begin{aligned} 221 &= 15 \cdot 14 + 11 \\ 15 &= 11 \cdot 1 + 4 \\ 11 &= 4 \cdot 2 + 3 \\ 4 &= 3 \cdot 1 + 1 \\ 3 &= 1 \cdot 3 + 0. \end{aligned}$$

So, $\gcd(221, 15) = 1$.

Supplementary Explanation

Answer. $\gcd(15, 221) = 1$.

Derivation/proof. The divisions are $221 = 14 \cdot 15 + 11$, $15 = 11 + 4$, $11 = 2 \cdot 4 + 3$, $4 = 3 + 1$, and $3 = 3 \cdot 1 + 0$. The last nonzero remainder is 1, so the gcd is 1. Each step preserves common divisors by replacing the larger number by its remainder.

The Euclidean divisions begin with $221 = 15 \cdot 14 + 11$ and end with $4 = 3 \cdot 1 + 1$. The final nonzero remainder is therefore one, so back-substitution supplies the required inverse.

N060

GCDs and Congruences

2015 Final | Question 7(b) | Source page 13 | Difficulty 2 | 4 marks
Q-2015-FINAL-P07-b | Answer provenance: official

> (b) Use Part (a) to calculate
>
> $$4^{385} \bmod 105.$$

Official Answer

Source: PDF physical page 13 (2015-final.pdf).

(b) By the result of (a)

$$\begin{aligned} 4^{385} \bmod 105 &= 4^{(7*49)+42} \bmod 105 \\ &= ((4^{7*49} \bmod 105) \times 4^{42}) \bmod 105 \\ &= ((4^{7} \times 4^{42}) \bmod 105 \\ &= 4^{49} \bmod 105 = 4 \bmod 105 = 4. \end{aligned}$$

Supplementary Explanation

Answer. 4.

Derivation/proof. The base 4 satisfies the coprimality condition. Since $385 = 7 \cdot 49 + 42$, $4^{385} = (4^{49})^7 \cdot 4^{42} \equiv 4^7 \cdot 4^{42} = 4^{49} \equiv 4 \pmod{105}$.

The base four is divisible by none of $3, 5, 7$. The final four is already the standard remainder; this argument does not reduce arbitrary exponents modulo 49.

N061

GCDs and Congruences

2014 Final | Question 2(a) | Source page 4 | Difficulty 2
Q-2014-FINAL-P02-a | Answer provenance: official

> Let a, m and n be three positive integers that are larger than 1. Is each of the following statements true or false? If it is true, give a proof. If it is false, give a counterexample.
>
> (a) $(a \bmod mn) \bmod n = a \bmod n$.

Official Answer

Source: PDF physical page 4 (2014-final.pdf).

(a) This is true. According to Euclid's Division Theorem, there exist q_1 and r_1 such that

$$a = q_1 mn + r_1, \quad 0 \le r_1 < mn,$$

where $r_1 = (a \bmod mn)$. Similarly, there exist q_2 and r_2 such that

$$r_1 = q_2 n + r_2, \quad 0 \le r_2 < n,$$

where $r_2 = r_1 \bmod n = (a \bmod mn) \bmod n$. Combining the two equations, we get

$$a = (q_1 m + q_2)n + r_2, \quad 0 \le r_2 < n.$$

Hence, $a \bmod n = r_2$. Consequently,

$$(a \bmod mn) \bmod n = r_2 = a \bmod n.$$

Supplementary Explanation

Answer. True.

Derivation/proof. Write $a = q_1 mn + r_1$ with $r_1 = a \bmod mn$, then $r_1 = q_2 n + r_2$ with $r_2 = r_1 \bmod n$. Thus $a = (q_1 m + q_2)n + r_2$ and $0 \le r_2 < n$, so $r_2 = a \bmod n$.

N062

GCDs and Congruences

2014 Final | Question 2(c) | Source page 4 | Difficulty 2
Q-2014-FINAL-P02-c | Answer provenance: official

> Let a, m and n be three positive integers that are larger than 1. Is each of the following statements true or false? If it is true, give a proof. If it is false, give a counterexample.
>
> (c) If $a \bmod n = 1$, then $gcd(a, n) = 1$.

Official Answer

Source: PDF physical page 4 (2014-final.pdf).

(c) This is true. $a \bmod n = 1$ implies that there exist q such that $a = qn + 1$, or $a - qn = 1$. The latter equation implies that any common divisor of a and n must divide 1, hence must be 1. Therefore, $\gcd(a, n) = 1$.

Supplementary Explanation

Answer. True.

Derivation/proof. $a \bmod n = 1$ means $a = qn + 1$. Any common divisor of a and n divides $a - qn = 1$, hence it is one.

If d is a positive common divisor, then $d \mid a$ and $d \mid qn$, so $d \mid (a - qn) = 1$. The only positive divisor of one is one.

N063

GCDs and Congruences

2014 Final | Question 4(b) | Source page 7 | Difficulty 2
Q-2014-FINAL-P04-b | Answer provenance: official

> In this question we consider the following modular equations:
>
> $$x \bmod m = a, \qquad x \bmod n = b.$$
>
> For each of the following parts, does there exist an integer x, $0 \leq x < mn$ that satisfies the two equations?
>
> If yes, find the solution and show the computational steps. Otherwise, prove by contradiction that there does not exist a solution.
>
> In this question, it is up to you whether to use the extended GCD algorithm to find multiplicative inverses. If you choose to use it, there is no need to show the steps.
>
> (b) $m = 30, n = 16, a = 5, b = 2$.

Official Answer

Source: PDF physical page 7 (2014-final.pdf).

(b) Assume there is a solution x. Then there exist integers q_1 and q_2 such that

$$x = q_1 \cdot 30 + 5,$$

$$x = q_2 \cdot 16 + 2.$$

Hence we have

$$q_2 \cdot 16 - q_1 \cdot 30 = 5 - 2 = 3.$$

This equation cannot be satisfied because the left hand side is divisible by 2, while the right hand side is not. Therefore, there is no solution.

Supplementary Explanation

Answer. No solution exists.

Derivation/proof. If a solution existed, $x = 30q_1 + 5 = 16q_2 + 2$, so $16q_2 - 30q_1 = 3$. The left side is even while the right side is odd, a contradiction.

The contradiction excludes every integer solution, and therefore every solution in the specified interval.

N064

GCDs and Congruences

2013 Final | Question 1(b) | Source page 3 | Difficulty 2
Q-2013-FINAL-P01-b | Answer provenance: official

> (b) For each number in Z_8, state if it has a multiplicative inverse mod 8, and if it has, state its inverse. There is no need to explain your answer.

Official Answer

Source: PDF physical page 3 (2013-final.pdf).

(b) 0 has no inverse; 1's inverse is 1; 2 has no inverse; 3's inverse is 3; 4 has no inverse; 5's inverse is 5; 6 has no inverse; 7's inverse is 7.

Supplementary Explanation

Answer. $1^{-1} = 1, 3^{-1} = 3, 5^{-1} = 5, 7^{-1} = 7$; $0, 2, 4, 6$ have none.

Derivation/proof. A residue has an inverse modulo 8 exactly when it is coprime to 8. The odd residues are coprime; direct products $1^2, 3^2, 5^2, 7^2$ are all $1 \pmod 8$.

The even residues $0, 2, 4, 6$ have gcd greater than one with 8. The odd squares are $1, 9, 25, 49$, all congruent to one modulo eight.

N065

GCDs and Congruences

2025 Fall Final | Question 6 | Source page 10 | Difficulty 3 | 8 marks
Q-2025F-FINAL-P06 | Answer provenance: official

> Use Chinese Remainder Theorem to solve the following system of congruences:
>
> $$x \equiv 1 \pmod 2, \quad x \equiv 1 \pmod 3, \quad x \equiv 3 \pmod 4, \quad x \equiv 1 \pmod 5.$$
>
> You must show all the steps. If you just write down your answer, you will get 0 points even if the answer is correct.
> Hint: Note that one of the above congruences is redundant and can be omitted.

Official Answer

Source: physical PDF page 10 (COMP2711_F2025_Final.pdf), physical PDF page 11 (COMP2711_F2025_Final.pdf).

Note that the first equation is already included in the third one. So we can simply ignore the first equation. And just consider the following congruence system:

$$\begin{aligned} x &\equiv 1 \pmod 3 \\ x &\equiv 3 \pmod 4 \\ x &\equiv 1 \pmod 5. \end{aligned}$$

We solve the system

$$\begin{aligned} x &\equiv 1 \pmod 3 \\ x &\equiv 3 \pmod 4 \\ x &\equiv 1 \pmod 5 \end{aligned}$$

using the explicit CRT formula. The moduli are pairwise coprime and $M = 3 \cdot 4 \cdot 5 = 60$.

Define $M_1 = M/3 = 20$, $M_2 = M/4 = 15$, $M_3 = M/5 = 12$.

Find the inverses:

- y_1 such that $20y_1 \equiv 1 \pmod 3$: $20 \equiv 2 \pmod 3$, inverse of 2 modulo 3 is 2 (because $2 \cdot 2 = 4 \equiv 1$). $\Rightarrow y_1 = 2$.
- y_2 such that $15y_2 \equiv 1 \pmod 4$: $15 \equiv 3 \pmod 4$, inverse of 3 modulo 4 is 3 (because $3 \cdot 3 = 9 \equiv 1$). $\Rightarrow y_2 = 3$.
- y_3 such that $12y_3 \equiv 1 \pmod 5$: $12 \equiv 2 \pmod 5$, inverse of 2 modulo 5 is 3 (because $2 \cdot 3 = 6 \equiv 1$). $\Rightarrow y_3 = 3$.

Now apply the formula:

$$\begin{aligned} x &= a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \\ &= 1 \cdot 20 \cdot 2 + 3 \cdot 15 \cdot 3 + 1 \cdot 12 \cdot 3 \\ &= 40 + 135 + 36 \\ &= 211. \end{aligned}$$

Reduce modulo $M = 60$:

$$211 \equiv 31 \pmod{60} \qquad (211 - 3 \cdot 60 = 211 - 180 = 31).$$

Thus the unique solution is

$$x \equiv 31 \pmod{60}.$$

Supplementary Explanation

Answer. $x \equiv 31 \pmod{60}$.

Derivation/proof. Since $x \equiv 3 \pmod 4$ implies $x \equiv 1 \pmod 2$, the modulo-two condition is redundant. The remaining moduli 3, 4 and 5 are pairwise coprime, with product $M = 60$. Set $M_1 = 20, M_2 = 15, M_3 = 12$. Their corresponding inverses are 2, 3 and 3, since $20 \cdot 2 \equiv 1 \pmod 3$, $15 \cdot 3 \equiv 1 \pmod 4$ and $12 \cdot 3 \equiv 1 \pmod 5$. The CRT formula gives $x \equiv 1 \cdot 20 \cdot 2 + 3 \cdot 15 \cdot 3 + 1 \cdot 12 \cdot 3 = 211 \equiv 31 \pmod{60}$. Checking 31 modulo 2, 3, 4 and 5 gives the residues 1, 1, 3, 1, respectively, so all original conditions hold. CRT guarantees one residue class modulo 60.

N066

GCDs and Congruences

2025 Spring Final | Question 3(b) | Source page 4 | Difficulty 3 | 6 marks
Q-2025S-FINAL-P03-b | Answer provenance: official

> For a positive integer p, define the positive integer n to be p-safe if n differs in absolute value by more than 2 from all multiples of p. For example, the set of 10-safe numbers are $3, 4, 5, 6, 7, 13, 14, 15, 16, 17, \cdots$.
>
> (b) [6 pts] Find the number of positive integers less than 1001 ($1001 = 7 \cdot 11 \cdot 13$) that are simultaneously 7-safe, 11-safe, and 13-safe. Please justify your answer.

Official Answer

Source: physical PDF page 4 (COMP2711_S2025_Final.pdf).

(b) Consider the mapping from x to the 3-tuple $(a = x \bmod 7, b = x \bmod 11, c = x \bmod 13)$. For any x that is 7-safe, 11-safe, and 13-safe, we must have $a \in \{3, 4\}$, $b \in \{3, 4, 5, 6, 7, 8\}$, $c \in \{3, 4, 5, 6, 7, 8, 9, 10\}$. By the Chinese Remainder Theorem, for each such triple (a, b, c), there is also a unique positive integer $x < 1001$ such that $a = x \bmod 7, b = x \bmod 11, c = x \bmod 13$. So there is a one-to-one correspondence between such triples and the set of positive integers less 1001 that are 7-safe, 11-safe, and 13-safe. Thus, the answer is

$$2 \cdot 6 \cdot 8 = 96.$$

Supplementary Explanation

Answer. 96.

Derivation/proof. The allowed residues are $\{3, 4\}$ modulo seven, $\{3, 4, 5, 6, 7, 8\}$ modulo eleven, and $\{3, 4, 5, 6, 7, 8, 9, 10\}$ modulo thirteen. These sets have two, six and eight members. The moduli are pairwise coprime, so CRT assigns each triple a unique class modulo 1001 and distinct triples cannot overlap. Zero satisfies none of the allowed residue conditions; hence each class has exactly one representative with $1 \leq x < 1001$. The product rule yields $2 \cdot 6 \cdot 8 = 96$.

N067

GCDs and Congruences

2025 Spring Final | Question 4(b) | Source page 5 | Difficulty 3 | 5 marks
Q-2025S-FINAL-P04-b | Answer provenance: official

> The least common multiple (LCM) of two integers x and y is the smallest positive integer that is a multiple of both x and y. As an example, the least common multiple of 8 and 10 is 40.
>
> (b) [5 pts] Determine the number of ordered pairs of positive integers (a, b) such that the least common multiple of a and b is $2^3 \cdot 5^7 \cdot 11^{13}$. Briefly explain your answer.

Official Answer

Source: physical PDF page 5 (COMP2711_S2025_Final.pdf).

(b) Both a and b are divisors of $2^3 5^7 11^{13}$, and so $a = 2^x 5^y 11^z$ and $b = 2^s 5^t 11^u$ for some nonnegative integers x, y, z, s, t, u. Because $2^3 5^7 11^{13}$ is the least common multiple, $\max(x, s) = 3$, $\max(y, t) = 7$, and $\max(z, u) = 13$. Hence (x, s) can be equal to $(0, 3), (1, 3), (2, 3), (3, 3), (3, 2), (3, 1)$, or $(3, 0)$, so there are 7 choices for (x, s). Similarly, there are 15 and 27 choices for (y, t) and (z, u), respectively. By the product rule, there are $7 \times 15 \times 27 = 2835$ ordered pairs of positive integers (a, b) having $2^3 5^7 11^{13}$ as their least common multiple.

Supplementary Explanation

Answer. 2835.

Derivation/proof. Since both numbers divide the target, write $a = 2^u 5^v 11^w$ and $b = 2^s 5^t 11^z$ with nonnegative exponents. Require $\max(u, s) = 3$, $\max(v, t) = 7$ and $\max(w, z) = 13$. For target exponent e, there are $(e + 1)^2$ candidate pairs; subtract the e^2 pairs with both exponents smaller than e. This leaves $(e + 1)^2 - e^2 = 2e + 1$. The three primes therefore contribute seven, fifteen and twenty-seven ordered choices, giving $7 \cdot 15 \cdot 27 = 2835$. No other prime may appear, since that would change the LCM.

N068

GCDs and Congruences

2025 Spring Midterm | Question 8(b) | Source page 12 | Difficulty 3
Q-2025S-MID-P08-b | Answer provenance: derived

> (b) Solve the following congruence. When finding the inverse, you must use the extended Euclidean algorithm and show the steps.
>
> $$28x \equiv 32 \pmod{46}$$

Official Answer

Source: physical PDF page 12 (COMP2711_S2025_Midterm.pdf).

(b) As we can see, we cannot find the inverse of 28 modulo 46 as they are not co-prime. Note that $28x \equiv 32 \pmod{46}$ is equivalent to $14x \equiv 16 \pmod{23}$. Now we calculate the inverse of 14 modulo 23 (Euclidean algorithm), which is 5.

$$70x \equiv x \equiv 80 \equiv 11 \pmod{23}$$

Supplementary Explanation

Answer. $x \equiv 11 \pmod{23}$, or $x = 11 + 23t$ for $t \in \mathbb{Z}$.

Derivation/proof. The condition $46 \mid 28x - 32$ is equivalent to $23 \mid 14x - 16$. The extended Euclidean algorithm is

$$23 = 1 \cdot 14 + 9, \quad 14 = 1 \cdot 9 + 5, \quad 9 = 1 \cdot 5 + 4, \quad 5 = 1 \cdot 4 + 1.$$

Back-substitution gives $1 = 5 - 4 = 2 \cdot 5 - 9 = 2 \cdot 14 - 3 \cdot 9 = 5 \cdot 14 - 3 \cdot 23$, so the inverse of fourteen is five. Multiply by five to get $x \equiv 80 \equiv 11 \pmod{23}$. Substituting every $x = 11 + 23t$ into the original expression gives $28x - 32 = 276 + 644t = 46(6 + 14t)$. Thus all these integers solve it; modulo forty-six they form the two classes eleven and thirty-four.

N069

GCDs and Congruences

2024 Fall Final | Question 7(2) | Source page 19 | Difficulty 3
Q-2024F-FINAL-P07-2 | Answer provenance: derived

> Solve the following system of congruences and show both steps:
>
> $$3x \equiv 4 \pmod 5, \qquad x \equiv 2 \pmod 4, \qquad 3x \equiv 6 \pmod 7.$$
>
> 2. Use Chinese Remainder Theorem to find the solution.

Official Answer

Source: physical PDF page 19 (COMP2711_F2024_Final.pdf).

> Answer: Note that the first step is to remove the coefficients. Note that the inverse of 3 (mod 5) is 2. So if we multiply 2 by both sides of the first equation we have: $2 \cdot 3x \equiv 2 \cdot 4 \implies x \equiv 3 \pmod 5$. Note that as $\gcd(3, 7) = 1$ we can simply divide both sides by 3 which gives us $x \equiv 2 \pmod 7$. So we need to solve the following system of congruences:
>
> $$x \equiv 3 \pmod 5, \qquad x \equiv 2 \pmod 4, \qquad x \equiv 2 \pmod 7.$$
>
> As all the modulos are coprime with each other. We can simply use chinese remainder theorem to find the solutions x.

Supplementary Explanation

Answer. $x = 58 + 140t$, $t \in \mathbb{Z}$.

Derivation/proof. Use the preceding system $x \equiv 3 \pmod 5, x \equiv 2 \pmod 4, x \equiv 2 \pmod 7$. Set $M = 5 \cdot 4 \cdot 7 = 140$, with partial products $28, 35, 20$. Their inverses in the corresponding moduli are $2, 3, 6$: $28 \cdot 2 \equiv 1 \pmod 5$, $35 \cdot 3 \equiv 1 \pmod 4$, and $20 \cdot 6 \equiv 1 \pmod 7$. Thus $x \equiv 3 \cdot 28 \cdot 2 + 2 \cdot 35 \cdot 3 + 2 \cdot 20 \cdot 6 = 618 \equiv 58 \pmod{140}$. Checking the original system, $3 \cdot 58 = 174$ has residues four modulo five and six modulo seven, while 58 has residue two modulo four. CRT uniqueness gives all solutions $58 + 140t$.

N070

GCDs and Congruences

2024 Spring Final | Question 8 | Source page 9 | Difficulty 3 | 10 marks
Q-2024S-FINAL-P08 | Answer provenance: official

> Let n be a positive integer such that $n = p^2q^4r^2$, where p, q, and r are three distinct prime numbers. Prove that the number of integers from 1 to n that are relatively prime with n is given by the expression:
>
> $$n\left(1-\frac{1}{p}\right)\left(1-\frac{1}{q}\right)\left(1-\frac{1}{r}\right).$$

Official Answer

Source: physical PDF page 9 (COMP2711_S2024_Final.pdf), physical PDF page 10 (COMP2711_S2024_Final.pdf).

> Solution: To find the number of integers from 1 to n that are relatively prime to n, we need to count the number of integers from 1 to n that are not divisible by p, q, or r. Let $\phi(n)$ denote this quantity. Let $S = \{1, 2, \ldots, n\}$. Define: A_p as the set of integers in S divisible by p, A_q as the set of integers in S divisible by q, A_r as the set of integers in S divisible by r. Using the principle of inclusion-exclusion, the number of integers in S that are divisible by p, q, or r is given by:
>
> $$|A_p \cup A_q \cup A_r| = |A_p| + |A_q| + |A_r| - |A_p \cap A_q| - |A_p \cap A_r| - |A_q \cap A_r| + |A_p \cap A_q \cap A_r|.$$
>
> We need to determine the size of each of these sets:
>
> $$|A_p| = \frac{n}{p}, \quad |A_q| = \frac{n}{q}, \quad |A_r| = \frac{n}{r}, \quad |A_p \cap A_q| = \frac{n}{pq}, \quad |A_p \cap A_r| = \frac{n}{pr}, \quad |A_q \cap A_r| = \frac{n}{qr}, \quad |A_p \cap A_q \cap A_r| = \frac{n}{pqr}.$$
>
> Substituting these values, we get:
>
> $$|A_p \cup A_q \cup A_r| = \frac{n}{p} + \frac{n}{q} + \frac{n}{r} - \frac{n}{pq} - \frac{n}{pr} - \frac{n}{qr} + \frac{n}{pqr}.$$
>
> The number of integers in S that are relatively prime to n is: $\phi(n) = n - |A_p \cup A_q \cup A_r|$. Substituting the expression for $|A_p \cup A_q \cup A_r|$, and simplifying, we get:
>
> $$\begin{aligned}\phi(n) &= n - \left(\frac{n}{p} + \frac{n}{q} + \frac{n}{r} - \frac{n}{pq} - \frac{n}{pr} - \frac{n}{qr} + \frac{n}{pqr}\right) \\ &= n\left(1 - \left(\frac{1}{p} + \frac{1}{q} + \frac{1}{r} - \frac{1}{pq} - \frac{1}{pr} - \frac{1}{qr} + \frac{1}{pqr}\right)\right) \\ &= n\left(1 - \frac{1}{p} - \frac{1}{q} - \frac{1}{r} + \frac{1}{pq} + \frac{1}{pr} + \frac{1}{qr} - \frac{1}{pqr}\right) \\ &= n\left(1-\frac{1}{p}\right)\left(1-\frac{1}{q}\right)\left(1-\frac{1}{r}\right).\end{aligned}$$

Supplementary Explanation

Answer. $n(1-1/p)(1-1/q)(1-1/r)$.

Derivation/proof. Within $S = \{1, \ldots, n\}$, let A_p, A_q, A_r be the multiples of the three respective primes. Only these primes can divide both an integer and n, so the target is $n - |A_p \cup A_q \cup A_r|$. The individual sizes are $n/p, n/q, n/r$; pairwise intersections have sizes $n/(pq), n/(pr), n/(qr)$ and the triple intersection has size $n/(pqr)$, since each product divides n. Inclusion-exclusion gives $n - n/p - n/q - n/r + n/(pq) + n/(pr) + n/(qr) - n/(pqr)$. Factor out n to obtain exactly the expansion of $(1-1/p)(1-1/q)(1-1/r)$.

N071

GCDs and Congruences

2024 Spring Final | Question 9(a) | Source page 11 | Difficulty 3 | 5 marks
Q-2024S-FINAL-P09-a | Answer provenance: official

> (a) Let p be a prime number and k be an integer such that $1 \leq k \leq p-1$. Prove that the binomial coefficient $\binom{p}{k}$ is divisible by p.

Official Answer

Source: physical PDF page 11 (COMP2711_S2024_Final.pdf).

> Solution: (a) Consider the binomial coefficient: $\binom{p}{k} = \frac{p!}{k!(p-k)!}$. The numerator $p!$ can be written as $p \cdot (p-1) \cdots 2 \cdot 1$, which is clearly divisible by p. The denominator is the product of two factorials, $k!$ and $(p-k)!$. Since $1 \leq k \leq p-1$, both k and $p-k$ are integers strictly less than p. Thus, the denominator $k!(p-k)!$ is the product of integers all of which are non-zero and strictly less than p. This means that neither $k!$ nor $(p-k)!$ is divisible by p. Since $\binom{p}{k}$ is an integer, and it is expressed as a fraction where the numerator is divisible by p but the denominator is not, we can conclude that $\binom{p}{k}$ itself must be divisible by p.

Supplementary Explanation

Answer. $p \mid \binom{p}{k}$ for $1 \leq k \leq p-1$.

Derivation/proof. Let $C = \binom{p}{k} \in \mathbb{Z}$. Then $C\,k!(p-k)! = p!$, whose right side is divisible by p. Since $k, p-k < p$, every positive denominator factor lies between 1 and $p-1$, so none is divisible by p. The prime product property gives $p \nmid k!(p-k)!$. Applying the same property to $C\,k!(p-k)!$ gives $p \mid C$. At $k = 0, p$ the coefficient is one, showing why these endpoints must be excluded.

N072

GCDs and Congruences

2024 Spring Final | Question 9(b) | Source page 11 | Difficulty 3 | 10 marks
Q-2024S-FINAL-P09-b | Answer provenance: official

> (b) Let p be a prime number. Use mathematical induction to prove that for any positive integer a, the number $a^p - a$ is divisible by p. In other words, you need to prove Fermat's Little Theorem using induction on the integer a. Note: If you give the number-theoretic proof of Fermat's Little Theorem presented in class, you will receive 0 marks. Hint: In your proof of part (b), you may find the claim given in part (a) useful. Note that in part (b), you are permitted to use this claim from part (a), even if you were unable to prove it.

Official Answer

Source: physical PDF page 11 (COMP2711_S2024_Final.pdf), physical PDF page 12 (COMP2711_S2024_Final.pdf).

> (b) The base case is when $a = 1$, in which case we have $a^p - a = 1^p - 1 = 0$, which is divisible by p, and hence the base case holds. Now supposing that the claim holds for an arbitrary a, i.e., $p \mid a^p - a$, we wish to show that it holds for $a + 1$, i.e., $p \mid (a+1)^p - (a+1)$. Expanding $(a+1)^p - (a+1)$ using the binomial formula, we get
>
> $$\begin{aligned}
> (a+1)^p - (a+1) &= \left(\sum_{i=0}^{p} \binom{p}{i} \cdot a^i \cdot 1^{p-i}\right) - (a+1) \\
> &= \left(\sum_{i=0}^{p} \binom{p}{i} \cdot a^i\right) - (a+1) \\
> &= \left(\sum_{i=1}^{p-1} \binom{p}{i} \cdot a^i\right) + \binom{p}{0} \cdot a^0 + \binom{p}{p} \cdot a^p - (a+1) \\
> &= \left(\sum_{i=1}^{p-1} \binom{p}{i} \cdot a^i\right) + 1 + a^p - (a+1) \\
> &= \left(\sum_{i=1}^{p-1} \underbrace{\binom{p}{i}}_{\text{divisible by } p \text{ from part (a)}} \cdot a^i\right) + \underbrace{a^p - a}_{\text{divisible by } p \text{ by IH}}.
> \end{aligned}$$
>
> Thus, $(a+1)^p - (a+1)$ is equal to the sum of terms each of which is divisible by p, and therefore we have $p \mid (a+1)^p - (a+1)$, which completes our inductive proof.

Supplementary Explanation

Answer. $p \mid a^p - a$ for every positive integer a.

Derivation/proof. Fix the prime p. Base case: $a = 1$ gives $1^p - 1 = 0$, divisible by p. The inductive hypothesis is $p \mid a^p - a$. By the binomial theorem, $(a+1)^p - (a+1) = (a^p - a) + \sum_{i=1}^{p-1} \binom{p}{i} a^i$: the constant one cancels the subtracted one, and $a^p - a$ forms the first term. Part (a) makes every internal coefficient, $1 \le i \le p-1$, divisible by p, so the entire sum is divisible by p. The first term is divisible by p by the inductive hypothesis. Thus $p \mid (a+1)^p - (a+1)$, completing induction without assuming Fermat's theorem.

N073

GCDs and Congruences

2024 Spring Midterm | Question 10 | Source page 14 | Difficulty 3 | 10 marks
Q-2024S-MID-P10 | Answer provenance: official

> Find all the solutions in the set of integers to the following system of equations:
>
> $$x \equiv 2 \pmod 3, \qquad x \equiv 4 \pmod 5, \qquad x \equiv 3 \pmod 7.$$
>
> Show all your calculations. You may use any facts and theorems taught in class.

Official Answer

Source: physical PDF page 14 (COMP2711_S2024_Midterm.pdf).

> Answer. We can use the Chinese remainder theorem (CRT) to solve this problem.
>
> $$m = \prod_{i=1}^{3} m_i = 3 \times 5 \times 7 = 105.$$
>
> $$M_1 = \frac{m}{m_1} = \frac{105}{3} = 35, \qquad M_2 = \frac{m}{m_2} = \frac{105}{5} = 21, \qquad M_3 = \frac{m}{m_3} = \frac{105}{7} = 15.$$
>
> $$\begin{aligned} y_1 &\equiv (M_1)^{-1} \equiv (35)^{-1} \equiv 2^{-1} \equiv 2 \pmod 3, \\ y_2 &\equiv (M_2)^{-1} \equiv (21)^{-1} \equiv 1^{-1} \equiv 1 \pmod 5, \\ y_3 &\equiv (M_3)^{-1} \equiv (15)^{-1} \equiv 1^{-1} \equiv 1 \pmod 7. \end{aligned}$$
>
> $$\begin{aligned} x &\equiv \sum_{i=1}^{3} a_i M_i y_i \pmod m \\ &\equiv 2 \times 35 \times 2 + 4 \times 21 \times 1 + 3 \times 15 \times 1 \pmod{105} \\ &\equiv 140 + 84 + 45 \pmod{105} \\ &\equiv 269 \pmod{105} \\ &\equiv 59 \pmod{105}. \end{aligned}$$
>
> Therefore, the solutions are $x = 105n + 59$ $(n \in \mathbb{Z})$.

Supplementary Explanation

Answer. $x = 59 + 105k$, $k \in \mathbb{Z}$.

Derivation/proof. The full modulus is $M = 3 \cdot 5 \cdot 7 = 105$, and $M_1 = 35$, $M_2 = 21$, $M_3 = 15$. Their inverses modulo $3, 5, 7$ are respectively $2, 1, 1$, since $35 \cdot 2 \equiv 1 \pmod 3$, $21 \equiv 1 \pmod 5$, and $15 \equiv 1 \pmod 7$. Therefore $x \equiv 2 \cdot 35 \cdot 2 + 4 \cdot 21 + 3 \cdot 15 = 269 \equiv 59 \pmod{105}$. Checking 59 gives residues two, four and three as requested. CRT uniqueness shows that all and only the solutions are $59 + 105k$, allowing every positive, negative or zero integer k.

N074

GCDs and Congruences

2024 Spring Midterm | Question 11 | Source page 16 | Difficulty 3 | 15 marks
Q-2024S-MID-P11 | Answer provenance: official

> Let $p \geq 7$ be a prime. Prove that the number with decimal representation
>
> $$\underbrace{1111\cdots1}_{\text{p-1 times}}$$
>
> is divisible by p. Note: For example, in the special case $p = 7$, note that 111111 is divisible by 7, since $111111 = 7 \times 15873$. Hint 1: Express the number as the sum of powers of 10. Hint 2: You may find Fermat's Little Theorem helpful.

Official Answer

Source: physical PDF page 16 (COMP2711_S2024_Midterm.pdf), physical PDF page 17 (COMP2711_S2024_Midterm.pdf).

> Answer. Call the given number S. We want to prove that $p \mid S$. Following Hint 1, note that S can be written as: $S = 10^0 + 10^1 \cdots + 10^{p-2}$. This is the sum of the first $p-1$ terms of a geometric series, for which we can apply the formula: $r^0 + r^1 \cdots + r^{n-1} = \frac{r^n-1}{r-1}$, where $r \neq 1$. Substituting $r := 10, n := p-1$, we obtain:
>
> $$S = 10^0 + 10^1 \ldots 10^{p-2} = \frac{10^{p-1}-1}{10-1} = \frac{10^{p-1}-1}{9}.$$
>
> Rearranging, we get that: $9S = 10^{p-1} - 1$. Taking this equation mod p, we get: $9S \equiv 10^{p-1} - 1 \pmod{p}$. (*) Since p is a prime greater than or equal to 7, it follows that p and 10 have no common factors. Following Hint 2, we note that Fermat's Little Theorem is applicable here, and we can use it to obtain $10^{p-1} \equiv 1 \pmod{p}$. (**) Substituting from (??) into (??), we get: $9S \equiv 10^{p-1} - 1 \equiv 1 - 1 \equiv 0 \pmod{p}$, which implies that $p \mid 9S$. Since p is a prime and $p \mid 9S$, it follows that either $p \mid 9$ or $p \mid S$, but since $9 = 3 \cdot 3$ and $p \geq 7$, it cannot be the case that $p \mid 9$. Therefore, we conclude that $p \mid S$.

Supplementary Explanation

Answer. p divides the repunit with $p-1$ digits.

Derivation/proof. Let $S = 1 + 10 + \cdots + 10^{p-2}$, with $p-1$ terms. Geometric summation gives $9S = 10^{p-1} - 1$. Since $p \geq 7$ is prime, $p \nmid 10$, and Fermat's Little Theorem gives $10^{p-1} \equiv 1 \pmod{p}$. Thus $9S \equiv 0 \pmod{p}$. Also $p \nmid 9$, so the prime product property implies $p \mid S$ from $p \mid 9S$. The two exclusions respectively justify applying Fermat and cancelling nine, completing the proof.

N075

GCDs and Congruences

2023 Fall Final | Question 1(a)(ii) | Source page 3 | Difficulty 3 | 1 mark
Q-2023F-FINAL-P01-a-ii | Answer provenance: official

> In this question the domain is the set of all integers. Consider the following predicates:
>
> $P(x)$: x is prime.
>
> $Q(x, y)$: x, y are relatively prime (co-prime).
>
> $E(x, y, z)$: x, y are congruent mod z (i.e., $x \equiv y \pmod{z}$).
>
> (a) Determine whether the following logical expressions are True or False. No justification is needed. ($a \mid b$ means b is divisible by a)
>
> (ii) $\forall x \forall a \forall b : \ P(x) \rightarrow (x \mid ab \rightarrow (x \mid a \vee x \mid b))$

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Final.pdf).

> (ii) True

Supplementary Explanation

Answer. True.

Derivation/proof. If $P(x)$ is false, the outer implication is true. Suppose x is prime and $x \mid ab$. If $x \mid a$, the desired disjunction holds. Otherwise $\gcd(x, a) = 1$, so integers u,v satisfy $ux + va = 1$. Multiply by b to get $uxb + vab = b$. Both terms on the left are divisible by x, since $x \mid ab$; therefore $x \mid b$. This proves the assertion for all integer a and b.

When x is prime and $x \nmid a$, Bézout gives $ux + va = 1$. Multiplication gives $uxb + vab = b$, so $x \mid b$ if $x \mid ab$. If x is not prime, or if $x \nmid ab$, the corresponding implication is vacuously true.

N076

GCDs and Congruences

2023 Fall Final | Question 1(b) | Source page 3 | Difficulty 3 | 2 marks
Q-2023F-FINAL-P01-b | Answer provenance: official

> In this question the domain is the set of all integers. Consider the following predicates:
>
> $P(x)$: x is prime.
>
> $Q(x, y)$: x, y are relatively prime (co-prime).
>
> $E(x, y, z)$: x, y are congruent mod z (i.e., $x \equiv y \pmod{z}$).
>
> (a) Determine whether the following logical expressions are True or False. No justification is needed. ($a \mid b$ means b is divisible by a)
>
> Write down Fermat's little theorem using quantifiers and the predicates given above. You are also allowed to use power. As an example $E(x^2, y, z)$ is acceptable.
>
> (Fermat's little theorem: If p is prime and a is any integer not divisible by p, then $a^{p-1} \equiv 1 \pmod{p}$.)

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Final.pdf).

> (b) $\forall x\forall y:(P(x)\land Q(y,x))\implies E(y^{x-1},1,x).

Supplementary Explanation

Answer. $\forall p \forall a \big((P(p) \land Q(a, p)) \to E(a^{p-1}, 1, p) \big)$.

Derivation/proof. The quantified variables p and a range over integers. $P(p)$ expresses that p is prime. Under this hypothesis, $p \nmid a$ is equivalent to $\gcd(a, p) = 1$, namely $Q(a, p)$, since a prime has only the positive divisors 1 and p. The predicate $E(a^{p-1}, 1, p)$ encodes exactly the congruence in Fermat's theorem. Connecting the two hypotheses to that conclusion with implication yields the stated quantified translation.

Using p, a instead of x, y, the two necessary hypotheses are $P(p)$ and $Q(a, p)$; the conclusion is $E(a^{p-1}, 1, p)$. For prime p, coprimality is equivalent to $p \nmid a$.

N077

GCDs and Congruences

2023 Fall Final | Question 7 | Source page 11 | Difficulty 3 | 11 marks
Q-2023F-FINAL-P07 | Answer provenance: official

> Solve this system of linear congruences:
>
> $$3x \equiv 1 \pmod{10}, \qquad 5x \equiv 3 \pmod{6}.$$
>
> Show all the steps.
>
> (Hint: You might want to get rid of the coefficients somehow first)

Official Answer

Source: physical PDF page 11 (COMP2711_F2023_Final.pdf).

> One approach is to simplify each equation first by multiplying each side by the inverse of the used coefficient. In first equation the inverse of 3 mod 10 exists as they are relatively prime and the inverse is 7. In second equation the inverse of 5 mod 6 exists as they are relatively prime and the inverse is 5. So the equation converts to:
>
> $$x \equiv 7 \pmod{10}, \qquad x \equiv 3 \pmod{6}.$$
>
> Now we can not use CRT directly as 6, 10 are not relatively prime. So, we use the same idea as in homework to break them down them into separate cases that are pairwise relatively prime. Which means we need to solve:
>
> $$x \equiv 2 \pmod{5}, \qquad x \equiv 1 \pmod{2}, \qquad x \equiv 0 \pmod{3}.$$
>
> $$m = 30 \quad M_1 = 6 \quad M_2 = 15 \quad M_3 = 10.$$
>
> $$y_1 = 1, \quad y_2 = 1, \quad y_3 = 1.$$
>
> $$x = 2 \cdot 6 \cdot 1 + 1 \cdot 15 \cdot 1 + 0 \cdot 10 \cdot 1 = 27.$$
>
> $$x \equiv 27 \pmod{30}.$$

Supplementary Explanation

Answer. $x \equiv 27 \pmod{30}$.

Derivation/proof. Because $3 \cdot 7 \equiv 1 \pmod{10}$ and $5 \cdot 5 \equiv 1 \pmod{6}$, the equations become $x \equiv 7 \pmod{10}$ and $x \equiv 15 \equiv 3 \pmod{6}$. Set $x = 7 + 10k$. Then $7 + 10k \equiv 3 \pmod{6}$ gives $4k \equiv 2 \pmod{6}$, equivalently $2k \equiv 1 \pmod{3}$, so $k \equiv 2 \pmod{3}$. Thus $x = 7 + 10(2 + 3t) = 27 + 30t$. Check $3 \cdot 27 \equiv 1 \pmod{10}$ and $5 \cdot 27 \equiv 3 \pmod{6}$. The parameterization captures all integer solutions.

Alternatively, substitute $x = 7 + 10k$ to get $4k \equiv 2 \pmod{6}$. The equivalence $6 \mid (4k - 2) \iff 3 \mid (2k - 1)$ gives $k = 2 + 3t$ and therefore $x = 27 + 30t$.

N078

GCDs and Congruences

2023 Spring Midterm | Question 5 | Source page 8 | Difficulty 3 | 13 marks
Q-2023S-MID-P05 | Answer provenance: official

> Solve this system of linear congruences:
>
> $$x \equiv 1 \pmod 3, \qquad 4x \equiv 1 \pmod 7, \qquad 3x \equiv 1 \pmod{11}.$$

Official Answer

Source: physical PDF page 8 (COMP2711_S2023_Midterm.pdf).

> $$1^{-1} \equiv 1 \pmod 3$$
> $$4^{-1} \equiv 2 \pmod 7$$
> $$3^{-1} \equiv 4 \pmod{11}$$
>
> Multiply the above inverse on the equation's both sizes respectively:
>
> $$1x \equiv 1 \pmod 3 \to x \equiv 1 \pmod 3$$
> $$4x \equiv 1 \pmod 7 \to x \equiv 2 \pmod 7$$
> $$3x \equiv 1 \pmod{11} \to x \equiv 4 \pmod{11}$$
>
> Use Chinese Remainder Theorem:
>
> $$M_1 = 77,\ M_2 = 33,\ M_3 = 21$$
> $$y_1 \equiv 77^{-1} \equiv 2 \pmod 3$$
> $$y_2 \equiv 33^{-1} \equiv 3 \pmod 7$$
> $$y_3 \equiv 21^{-1} \equiv 10 \pmod{11}$$
>
> We have:
>
> $$x = 1 \cdot 77 \cdot 2 + 2 \cdot 33 \cdot 3 + 4 \cdot 21 \cdot 10 = 1192 \equiv 37 \pmod{231}.$$

Supplementary Explanation

Answer. $x \equiv 37 \pmod{231}$, or $x = 37 + 231k$ for $k \in \mathbb{Z}$.

Derivation/proof. Because $4 \cdot 2 \equiv 1 \pmod 7$ and $3 \cdot 4 \equiv 1 \pmod{11}$, the system is $x \equiv 1, 2, 4$ modulo $3, 7, 11$, respectively. Let $M = 231$ and use $M_i = 77, 33, 21$ with inverses $2, 3, 10$ in the corresponding moduli. CRT gives $x \equiv 77 \cdot 2 + 2 \cdot 33 \cdot 3 + 4 \cdot 21 \cdot 10 = 1192 \equiv 37 \pmod{231}$. Check: $37 \equiv 1 \pmod 3$, $4 \cdot 37 \equiv 1 \pmod 7$, and $3 \cdot 37 \equiv 1 \pmod{11}$.

The original-system checks include $4 \cdot 37 = 148 \equiv 1 \pmod 7$ and $3 \cdot 37 = 111 \equiv 1 \pmod{11}$. CRT ensures uniqueness modulo the product 231.

N079

GCDs and Congruences

2023 Spring Midterm | Question 6 | Source page 9 | Difficulty 3 | 12 marks
Q-2023S-MID-P06 | Answer provenance: official

> Calculate $\gcd(2^{100}-1, 2^{65}-1)$.

Official Answer

Source: physical PDF page 9 (COMP2711_S2023_Midterm.pdf).

> In fact, for any a, m, n, we can apply the Euclidean algorithm:
>
> $$\gcd(a^m-1, a^n-1) = \gcd(a^m-a^n, a^n-1) = \gcd(a^n(a^{m-n}-1), a^n-1)$$
>
> Writing a^m-a^n as $a^n(a^{m-n}$, we have
>
> $$\gcd(a^m-1, a^n-1) = \gcd(a^n(a^{m-n}-1), a^n-1)$$
>
> Since $\gcd(a^n, a^n-1) = 1$, we get:
>
> $$\gcd(a^m-1, a^n-1) = \gcd(a^{m-n}-1, a^n-1).$$
>
> We can now use this rule iteratively, in a way similar to the original Euclidean algorithm:
>
> $$\begin{aligned}\gcd(2^{100}-1, 2^{65}-1) &= \gcd(2^{65}-1, 2^{35}-1)\\ &= \gcd(2^{35}-1, 2^{30}-1)\\ &= \gcd(2^{30}-1, 2^5-1)\\ &= \gcd(2^{25}-1, 2^5-1)\\ &= \cdots = \gcd(2^5-1, 2^5-1) = 2^5-1 = 31.\end{aligned}$$

Supplementary Explanation

Answer. 31.

Derivation/proof. For $m > n$, subtracting 2^n-1 from 2^m-1 gives $2^n(2^{m-n}-1)$. Since $\gcd(2^n, 2^n-1) = 1$, this proves

$$\gcd(2^m-1, 2^n-1) = \gcd(2^{m-n}-1, 2^n-1).$$

The Euclidean reductions of exponents are $(100, 65) \to (65, 35) \to (35, 30) \to (30, 5) \to (5, 0)$. Hence the gcd is $2^5-1 = 31$. Both original exponents are multiples of 5, consistent with divisibility by 31.

The final form is $\gcd(2^5-1, 0) = 31$.

N080

GCDs and Congruences

2022 Final Part 2 | Question 3 | Source page 2 | Difficulty 3 | 6 marks
Q-2022-FINAL2-P03 | Answer provenance: official

Solve the following system of congruences:

$$5x + 9y \equiv 5 \pmod{11}, \qquad 3x - 4y \equiv 2 \pmod{11}.$$

Official Answer

Source: physical PDF page 2 (2022-final-part-2.pdf).

$$x \equiv 9 \pmod{11}, \qquad y \equiv 9 \pmod{11}.$$

Supplementary Explanation

Answer. $x \equiv 9 \pmod{11}$, $y \equiv 9 \pmod{11}$.

Derivation/proof. Multiply the first equation by 3 and the second by 5: 15x+27y≡15 and 15x−20y≡10. Subtract to obtain47y≡5, or3y≡5 mod 11. Since3·4≡1, y≡20≡9. Substitute in 3x−4y≡2 to get 3x≡38≡5, hence x≡9. Verify5·9+9·9=126≡5 and 3·9−4·9=−9≡2. The invertible coefficients show this residue pair is unique.

The elimination gives $47y \equiv 5$, and $3 \cdot 4 = 12 \equiv 1 \pmod{11}$. Both resulting coefficients are invertible, so the residue pair is unique; all integer solutions are $x = 9 + 11s, y = 9 + 11t$ with $s, t \in \mathbb{Z}$.

N081

GCDs and Congruences

2022 Midterm | Question 6(b) | Source page 3 | Difficulty 3 | 10 marks
Q-2022-MID-P06-b | Answer provenance: official

> Assume p is a prime and n is a positive integer such that $p \mid (4n^2 + 1)$.
>
> Show that $p \equiv 1 \pmod 4$. (Hint: Analyze $(2n)^{p-1}$ using Fermat's Little Theorem and part (a))

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (b) $4n^2 + 1$ is an odd number so we know that p has to be odd. So either $p \equiv 1 \pmod 4$ or $p \equiv 3 \pmod 4$. Assume $p \equiv 3 \pmod 4$ which means $p = 4k + 3$. Since $p \nmid n$ and p is an odd prime, so $p \nmid 2n$. By Fermat's Little Theorem, we have $(2n)^{p-1} \equiv (2n)^{4k+2} \equiv (4n^2)^{2k+1} \equiv 1 \pmod p$. Recall that $p \mid 4n^2 + 1 \implies 4n^2 \equiv -1 \pmod p$. So $(4n^2)^{2k+1} \equiv (-1)^{2k+1} \equiv -1 \pmod p$. Now we have $1 \equiv -1 \pmod p$, which is impossible since $p \neq 2$. This proves that $p \equiv 1 \pmod 4$.

Supplementary Explanation

Answer. $p \equiv 1 \pmod 4$.

Derivation/proof. Because 4n²+1 is odd, p is odd, and p∤n implies gcd(2n,p)=1. If p≡3 mod 4, write p=4k+3. Fermat gives (2n)^(p−1)≡1 modp, but (2n)^(4k+2)=(4n²)^(2k+1)≡(−1)^(2k+1)=−1 modp by the divisibility hypothesis. Hence p divides2, contradicting oddness. The only other odd residue modulo 4 is 1, proving the result.

The required hypothesis is $\gcd(2n, p) = 1$. The power $(4n^2)^{2k+1}$ is -1 modulo p, so comparison with Fermat gives $p \mid 2$, contradicting oddness.

N082

GCDs and Congruences

2021 Midterm | Question 3 | Source page 2 | Difficulty 3 | 10 marks
Q-2021-MID-P03 | Answer provenance: official

> Show that there is no largest prime number.

Official Answer

Source: physical PDF page 2 (2021-midterm.pdf).

> We prove this by contradiction.
>
> Suppose there is a largest prime number n. Note that $n! + 1$ has no factors between 1 and n. Thus, if all primes were n or less, $n! + 1$ would have no prime factors, and it would be prime itself. Therefore, $n! + 1$ is a prime number greater than n, which is a contradiction.

Supplementary Explanation

Answer. There is no largest prime.

Derivation/proof. Assume a largest prime p exists and form N=p!+1>1. Every integer greater than one has a prime divisor q (take its smallest divisor greater than one, which must be prime). If q≤p, then q divides p! as well as N and thus divides their difference 1, impossible. Therefore q>p, contradicting maximality. We use a prime divisor of N; we do not assume N itself is prime.

The least divisor $q > 1$ of N is prime: if it were composite, a proper divisor strictly between 1 and q would also divide N, contradicting minimality. The factorial construction need not itself be prime; for example $5! + 1 = 121 = 11^2$.

If $q \le p$, then $q \mid (N - p!) = 1$, impossible; hence $q > p$.

N083

GCDs and Congruences

2021 Midterm | Question 7 | Source page 3 | Difficulty 3 | 10 marks
Q-2021-MID-P07 | Answer provenance: official

> Show that $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$ for any two distinct prime numbers p and q.

Official Answer

Source: physical PDF page 3 (2021-midterm.pdf).

> By Fermat's little theorem, $p^{q-1} \equiv 1 \pmod{q}$. Clearly, $q^{p-1} \equiv 0 \pmod{q}$. Therefore, $p^{q-1} + q^{p-1} \equiv 1 + 0 \equiv 1 \pmod{q}$. Similarly, $p^{q-1} + q^{p-1} \equiv 0 + 1 \equiv 1 \pmod{p}$. By the Chinese remainder theorem, we have $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$.

Supplementary Explanation

Answer. The congruence holds.

Derivation/proof. Modulo q, p is nonzero because the primes are distinct, so Fermat gives $p^{q-1} \equiv 1$. Also $q^{p-1} \equiv 0$ since p−1≥1. Their sum is 1 modulo q. Similarly it is 1 modulo p. Thus p and q both divide the sum minus one; because they are coprime, pq divides it. This proves the desired congruence, including the case where one prime is 2.

With $A = p^{q-1} + q^{p-1}$, the separate congruences give $q \mid A - 1$ and $p \mid A - 1$; coprimality gives $pq \mid A - 1$.

N084

GCDs and Congruences

2020 Final Part 1 | Question 4(b) | Source page 3 | Difficulty 3 | 4 marks
Q-2020-FINAL1-P04-b | Answer provenance: official

> How many integers $0 \leq n < 1000$ are there such that
>
> $n \equiv 1 \pmod{2}$, $n \equiv 2 \pmod{3}$, and $n \equiv 3 \pmod{5}$?

Official Answer

Source: physical PDF page 3 (2020-final-part-1.pdf).

By Chinese remainder theorem, we know that $n \equiv 1 \pmod{2}$, $n \equiv 2 \pmod{3}$, and $n \equiv 3 \pmod{5}$ is equivalent to $n = 23 + 30k$ for an integer k. Since $0 \leq n < 1000$, we have $0 \leq k \leq 32$. Therefore the answer is 33.

Supplementary Explanation

Answer. 33.

Derivation/proof. The last condition gives residues 3,8,13,18,23,28 modulo 30. Requiring oddness leaves 3,13,23; requiring remainder 2 modulo 3 leaves only 23. As 2,3,5 are pairwise coprime this is the unique CRT class: $n = 23 + 30k$. The range imposes $0 \leq k \leq 32$, giving 33 integers (last 983).

The allowed integer parameter runs from zero through thirty-two inclusive. Therefore its count is $32 - 0 + 1 = 33$, retaining both endpoints.

N085

GCDs and Congruences

2020 Midterm | Question Bonus Problem | Source page 5 | Difficulty 3 | 10 marks
Q-2020-MID-BONUS | Answer provenance: official

> A quasi-square number $n \in \mathbb{N}$ is one that is divisible by a square number. For example, 24 is a quasi-square number because it is divisible by $4 = 2^2$, while 15 is not a quasi-square number.
>
> Prove that for any $k \in \mathbb{N}$, there exist k consecutive numbers all of which are quasi-square numbers. [Hint: Use the Chinese Remainder Theorem.]

Official Answer

Source: physical PDF page 5 (2020-midterm.pdf).

We give a constructive proof. Let p_i be the ith prime number. We know that $\gcd(p_i^2, p_j^2) = 1$ for $i \neq j$. Consider the following system of linear congruences:

$$\begin{aligned} x &\equiv -1 \pmod{p_1^2} \\ x &\equiv -2 \pmod{p_2^2} \\ &\cdots \\ x &\equiv -k \pmod{p_k^2}. \end{aligned}$$

Based on the Chinese Remainder Theorem, there is a solution $x_0 \in \mathbb{Z}_n$ to this system for $n = p_1^2 \cdots p_k^2$. We know that

$$p_1^2 \mid x_0 + 1, \quad p_2^2 \mid x_0 + 2, \quad \ldots, \quad p_k^2 \mid x_0 + k.$$

So $x_0 + 1, \ldots, x_0 + k$ are all quasi-square numbers.

Supplementary Explanation

Answer. For every positive k, there is a block $x_0 + 1, \ldots, x_0 + k$ such that $p_i^2 \mid x_0 + i$ for distinct primes $p_1, \ldots, p_k$.

Derivation/proof. Fix k≥1 and choose k distinct primes $p_1, \ldots, p_k$. Their squares are pairwise coprime: a common prime divisor of p_i^2 and p_j^2 would force $p_i = p_j$. By the Chinese Remainder Theorem, the simultaneous congruences $x \equiv -i \pmod{p_i^2}$ for i=1,...,k have a solution modulo $M = \prod_{i=1}^{k} p_i^2$. Choose a nonnegative representative x_0, or add a multiple of M if a larger positive starting value is desired. For each i, the congruence is exactly $p_i^2 \mid x_0 + i$. Thus all k numbers are positive, consecutive, and divisible by square numbers strictly greater than 1. This proves the stronger nontrivial-square claim, so it also proves the literal existence statement despite the defective example. If a convention includes k=0 in the naturals, the empty block case is vacuous. Independently, for k=3 choose primes 2,3,5: $x_0 = 547$ satisfies residues 3 mod 4, 7 mod 9, and 22 mod 25; the block 548,549,550 equals $4 \cdot 137, 9 \cdot 61, 25 \cdot 22$.

Choose a nonnegative representative x_0 so all $x_0 + i$ are positive. If $k = 0$ is permitted, the empty block is automatic. For $k = 3$, primes $2, 3, 5$ and $x_0 = 547$ give $548 = 4 \cdot 137$, $549 = 9 \cdot 61$, and $550 = 25 \cdot 22$.

N086

GCDs and Congruences

2020 Midterm | Question 6 | Source page 3 | Difficulty 3 | 10 marks
Q-2020-MID-P06 | Answer provenance: official

> Use the extended Euclid algorithm to find the inverse of $53 \pmod{180}$. Show the steps of the algorithm.

Official Answer

Source: physical PDF page 3 (2020-midterm.pdf); physical PDF page 4 (2020-midterm.pdf).

Calculate $\gcd(180, 53)$ using euclidean algorithm.

$$\begin{aligned} 180 &= 53 \cdot 3 + 21 \\ 53 &= 21 \cdot 2 + 11 \\ 21 &= 11 \cdot 1 + 10 \\ 11 &= 10 \cdot 1 + 1 \\ 10 &= 1 \cdot 10 + 0. \end{aligned}$$

So $\gcd(180, 53) = 1$. Rewriting:

$$\begin{aligned} 21 &= 180 - 53 \cdot 3 \\ 11 &= 53 - 21 \cdot 2 \\ 10 &= 21 - 11 \cdot 1 \\ 1 &= 11 - 10 \cdot 1. \end{aligned}$$

Substituting:

$$\begin{aligned} 1 &= 11 - 10 \cdot 1 \\ 1 &= 11 - (21 - 11 \cdot 1) \cdot 1 \\ 1 &= 21 \cdot (-1) + 11 \cdot 2 \\ 1 &= 21 \cdot (-1) + (53 - 21 \cdot 2) \cdot 2 \\ 1 &= 53 \cdot 2 + 21 \cdot (-5) \\ 1 &= 53 \cdot 2 + (180 - 53 \cdot 3) \cdot (-5) \\ 1 &= 180 \cdot (-5) + 53 \cdot 17. \end{aligned}$$

So 17 is the modular inverse of $53 \pmod{180}$.

Supplementary Explanation

Answer. $53^{-1} \equiv 17 \pmod{180}$.

Derivation/proof. Euclidean divisions: $180 = 3 \cdot 53 + 21$, $53 = 2 \cdot 21 + 11$, $21 = 11 + 10$, $11 = 10 + 1$, $10 = 10 \cdot 1$. Back substitution gives $1 = 11 - 10 = 2 \cdot 11 - 21 = 2 \cdot 53 - 5 \cdot 21 = 17 \cdot 53 - 5 \cdot 180$. Thus the coefficient of 53 is the inverse. Check $53 \cdot 17 = 901 = 5 \cdot 180 + 1$.

N087

GCDs and Congruences

2020 Midterm | Question 9 | Source page 4 | Difficulty 3 | 10 marks
Q-2020-MID-P09 | Answer provenance: official

> Solve $3^{5x-2} \equiv 9 \pmod{23}$. The solution is not unique, any solution is acceptable. [Hint: Use Fermat's Little Theorem.]

Official Answer

Source: physical PDF page 5 (2020-midterm.pdf).

Multiplying 3^{-2} (inverse taken with modulo 23) on both sides, we obtain:

$$3^{5x-4} \equiv 1 \pmod{23}.$$

By Fermat's Little Theorem, if $5x-4$ is a multiple of 22, then this congruence will hold, namely $5x-4 \equiv 0 \pmod{22}$, i.e.,

$$5x \equiv 4 \pmod{22}.$$

The inverse of 5 is $9 \pmod{22}$, so $x \equiv 4 \cdot 9 \equiv 36 \equiv 14 \pmod{22}$. So $x = 22k + 14$ for any natural number k.

Supplementary Explanation

Answer. One solution is $x = 14$; indeed every $x = 14 + 22k$ with $k \geq 0$ works.

Derivation/proof. Because 23 is prime and coprime to 3, $3^{22} \equiv 1 \pmod{23}$. It suffices to make $5x - 2 \equiv 2 \pmod{22}$, or $5x \equiv 4 \pmod{22}$. The inverse of 5 modulo 22 is 9, since $5 \cdot 9 = 45 \equiv 1$. Hence $x \equiv 36 \equiv 14$. At x=14 the exponent is 68=3·22+2, so $3^{68} \equiv 3^2 = 9$. This constructs acceptable solutions without claiming that Fermat's period is minimal.

For $x = 14$, $5x - 2 = 68 = 3 \cdot 22 + 2$ and hence $3^{68} \equiv (3^{22})^3 \cdot 3^2 \equiv 9 \pmod{23}$.

Using Fermat's exponent twenty-two is not a claim of the least period. It is a sufficient period, and constructing one compatible exponent class suffices for the requested solution.

N088

GCDs and Congruences

2019 Final | Question 2 | Source page 4 | Difficulty 3 | 4 marks
Q-2019-FINAL-P02 | Answer provenance: official

> Solve the following system of linear congruences by the Chinese remainder theorem. Show all steps except the extended Euclidean algorithm.
>
> $$\begin{cases} x \equiv 4 \pmod 9 \\ x \equiv 8 \pmod{14} \\ x \equiv 3 \pmod 5. \end{cases}$$

Official Answer

Source: physical PDF page 4 (2019-final.pdf).

Let $m = 9 \cdot 14 \cdot 5 = 630$. $M_1 = 14 \cdot 5 = 70$, $M_2 = 9 \cdot 5 = 45$, $M_3 = 9 \cdot 14 = 126$. By the extended Eucildean algorithm, we find that 4 is an inverse of $M_1 \pmod 9$, 5 is an inverse of $M_2 \pmod{14}$ and 1 is an inverse of $M_3 \pmod 5$. Hence, $x = 4 \cdot 70 \cdot 4 + 8 \cdot 45 \cdot 5 + 3 \cdot 126 \cdot 1 = 3298 \equiv 148 \pmod{630}$.

Supplementary Explanation

Answer. $x \equiv 148 \pmod{630}$.

Derivation/proof. The moduli 9,14,5 are pairwise coprime and have product $M = 630$. Partial products are $M_1 = 70$, $M_2 = 45$, $M_3 = 126$. Their respective inverses are 4,5,1 because $70 \cdot 4 \equiv 1 \pmod 9$, $45 \cdot 5 \equiv 1 \pmod{14}$, and $126 \equiv 1 \pmod 5$. CRT gives $x \equiv 4 \cdot 70 \cdot 4 + 8 \cdot 45 \cdot 5 + 3 \cdot 126 = 3298 \equiv 148 \pmod{630}$. Check $148 = 16 \cdot 9 + 4 = 10 \cdot 14 + 8 = 29 \cdot 5 + 3$. CRT ensures uniqueness modulo 630.

The full integer solution set is $148 + 630k$, $k \in \mathbb{Z}$.

N089

GCDs and Congruences

2019 Midterm | Question 10(d) | Source page 12 | Difficulty 3 | 4 marks
Q-2019-MID-P10-d | Answer provenance: official

> Solve each of these congruences with $0 \leq x \leq 103$ and $0 \leq y \leq 103$ using the result $15 \cdot 7 \equiv 1 \pmod{104}$.
>
> (d) $\begin{cases} 9x + 2y \equiv 10 \pmod{104} \\ x + y \equiv 1 \pmod{104}. \end{cases}$

Official Answer

Source: physical PDF page 12 (2019-midterm.pdf).

We have $9x + 2y + (-2)(x + y) \equiv 10 - 2 \pmod{104}$, which is $7x \equiv 8 \pmod{104}$. So,

$$\begin{aligned} 7x &\equiv 8 \pmod{104} \\ \rightarrow 15 \cdot 7x &\equiv 15 \cdot 8 \pmod{104} \\ \rightarrow x &\equiv 16 \pmod{104}. \end{aligned}$$

When $x = 16$, we have $16 + y \equiv 1 \pmod{104}$, and thus $y \equiv -15 \equiv 89 \pmod{104}$.

Supplementary Explanation

Answer. $(x, y) = (16, 89)$.

Derivation/proof. Subtract twice the second congruence from the first to obtain $7x \equiv 8 \pmod{104}$. Multiply by 15: $x \equiv 120 \equiv 16$. Then $y \equiv 1 - 16 = -15 \equiv 89$. Substitution checks $9 \cdot 16 + 2 \cdot 89 = 322 \equiv 10$ and $16 + 89 = 105 \equiv 1 \pmod{104}$. The invertible coefficient makes x unique and then y unique in the required ranges.

Elimination and multiplication by an inverse are reversible, so the solution in the specified range is unique.

For the second congruence the proposed values give $x + y = 16 + 89 = 105$. Since $105 = 104 + 1$, their sum has the required remainder one modulo 104.

N090

GCDs and Congruences

2018 Final | Question 9(a) | Source page 13 | Difficulty 3
Q-2018-FINAL-P09-a | Answer provenance: official

> (a) How many elements in $\mathbb{Z}_{70}$ have multiplicative inverses? Explain your answer.

Official Answer

Source: physical PDF page 13 (2018-final.pdf).

We can write $70 = 2 \cdot 5 \cdot 7$.

$$M_7 = \{0, 7, 14, 21, 28, 35, 42, 49, 56, 63\}$$

$$M_5 = \{0, 5, 10, 15, 20, \ldots, 65\}$$

$$M_2 = \{0, 2, 4, 6, \ldots, 68\}.$$

The integers that do not have an inverse are those that are not relatively prime with 70, i.e., all the multiples of $7, 5, 2$, i.e., $M_7 \cup M_5 \cup M_2$. Note that $|M_7| = 10$, $|M_5| = 14$, $|M_2| = 35$. Moreover, $|M_7 \cap M_5| = 2$, $|M_7 \cap M_2| = 5$, and $|M_5 \cap M_2| = 7$. Finally, $|M_7 \cap M_5 \cap M_2| = 1$. By the inclusion exclusion principle, $|M_7 \cup M_5 \cup M_2| = 10 + 14 + 35 - 2 - 5 - 7 + 1 = 36$. Therefore, the number of elements that have an inverse is $70 - 36 = 24$.

Supplementary Explanation

Answer. 24.

Derivation/proof. An element is invertible exactly when coprime to $70 = 2 \cdot 5 \cdot 7$. Among residues 0 through 69, counts of multiples of 2,5,7 are 35,14,10; pairwise intersections have sizes 7,5,2 and the triple intersection has size 1. Thus nonunits number $35 + 14 + 10 - 7 - 5 - 2 + 1 = 46$, and units number $70 - 46 = 24$. The independent totient product $70(1 - 1/2)(1 - 1/5)(1 - 1/7) = 24$ agrees.

As an independent arithmetic check, $70(1 - 1/2)(1 - 1/5)(1 - 1/7) = 24$.

N091

GCDs and Congruences

2018 Midterm | Question 3(c) | Source page 5 | Difficulty 3 | 2 marks
Q-2018-MID-P03-c | Answer provenance: official

> Consider the set $S = \{1, 2, \ldots, 2n\}$, and subsets $P = \{p_1, p_2, \ldots, p_m\} \subseteq S$ such that they satisfy the following criterion:
>
> $\forall p_i \in P \forall p_j \in P (i \neq j \rightarrow p_i \nmid p_j)$
>
> In the above, $p_i \nmid p_j$ means p_j is not divisible by p_i.
>
> (c) Prove that there is no $P' \subseteq S$ satisfying the above criterion such that $|P'| > n$. (Hint: By contradiction, suppose $|P'| = n + 1$ satisfies the above criterion. Following (b), put each p_i in the pigeonhole c_i.)

Official Answer

Source: physical PDF page 5 (2018-midterm.pdf).

Let each hole be an odd number in S. The largest odd number in S is $2n - 1$, so the holes are $\{1, 3, 5, \ldots, 2n - 1\}$, so there are n holes, and the odd factor of each number in P' fits into one of these holes. Therefore, there are $n + 1$ numbers for n holes, and by the pigeonhole principle, there exists two different $p_i, p_j \in P'$ in the same hole, i.e. $c_i = c_j$. But we know that this is impossible from (b), giving a contradiction. (2 marks)

Supplementary Explanation

Answer. Every such subset has size at most n.

Derivation/proof. There are exactly n possible odd parts in S, namely $1, 3, \ldots, 2n - 1$. If P' had more than n elements, pigeonhole principle would place two distinct elements in the same odd-part class. Writing them as $c2^r, c2^s$ makes the smaller divide the larger, contradicting the criterion. Thus $|P'| \leq n$.

If there are more than n elements, select any $n + 1$ of them before applying pigeonhole principle.

N092

GCDs and Congruences

2018 Midterm | Question 5(a) | Source page 7 | Difficulty 3
Q-2018-MID-P05-a | Answer provenance: official

> (a) Let x, y, z be positive integers. Show that, if $z \mid xy$ and y, z are relatively prime, then $z \mid x$.

Official Answer

Source: physical PDF page 7 (2018-midterm.pdf).

Since y, z are relatively prime, $\gcd(y, z) = 1$. Using the extended Euclidean algorithm, we can calculate integers s, t such that

$$sy + tz = 1 \tag{1}$$

Multiplying both sides of (1) with x, we get

$$syx + tzx = x \tag{2}$$

From the problem statement, $z \mid xy$, thus $z \mid syx$. Moreover, it is obvious that $z \mid tzx$. We know that from this it follows that z divides their sum, that is, $z \mid (syx + tzx)$. But, from (2), $(syx + tzx) = x$, therefore $z \mid x$. Students may choose to explicitly prove the fact that if $a \mid b$ and $a \mid c$ then $a \mid (b + c)$. This is fine but not necessary.

Supplementary Explanation

Answer. $z \mid x$.

Derivation/proof. Since $\gcd(y, z) = 1$, Bezout's identity supplies integers s, t with $sy + tz = 1$. Multiply by x to obtain $sxy + tzx = x$. The hypothesis $z \mid xy$ gives $z \mid sxy$, and $z \mid tzx$ directly. Hence z divides their sum x.

Multiplying the Bezout equation by x gives $sxy + tzx = x$. Both summands on the left are divisible by z, so $z \mid (sxy + tzx) = x$, proving the required divisibility.

N093

GCDs and Congruences

2017 Final | Question 4(b) | Source page 6 | Difficulty 3
Q-2017-FINAL-P04-b-time | Answer provenance: official

> Recall that in RSA, we need to choose two primes p and q, and the public key is (n, e) where $n = pq$. Let $T = (p-1)(q-1)$. The private key is d, where $ed \equiv 1 \pmod{T}$. The encryption of x is $x^e \bmod n$. This means that d must be an integer in $\mathbb{Z}_T$ that is relatively prime with T. We can find d using the following algorithm:
>
> 1. Randomly pick $d \in \mathbb{Z}_T$.
> 2. If d and T are relatively prime, then return d. If they are not, then pick a new d and loop from step 1.
>
> Assume that we will never pick the same d twice.
>
> What is the worst case running time of your algorithm in terms of T for a single d? Express your time in big-O notation.

Official Answer

Source: physical PDF page 6 (2017-final.pdf); physical PDF page 7 (2017-final.pdf).

It is the Euclidean algorithm to find gcd: start with $\gcd(T, d)$, and the recursive step is to change $\gcd(a, b)$ into $\gcd(b, a \bmod b)$. It was shown in class that this takes $O(\log T)$ steps for the worst case of the Fibonacci numbers.

Supplementary Explanation

Answer. $O(\log T)$ arithmetic operations under the exam's unit-cost model.

Derivation/proof. For successive positive remainders $a > b > r$, write $a = qb + r$ with $q \geq 1$ and $r < b$. Hence $a \geq b + r > 2r$: after two remainder steps the value more than halves. Starting below or at T gives at most $2\log_2 T + O(1)$ steps. Consecutive Fibonacci numbers make the quotient repeatedly one and require logarithmically many steps, so this is the worst-case order.

N094

GCDs and Congruences

2017 Final | Question 4(c) | Source page 6 | Difficulty 3 | 3 marks
Q-2017-FINAL-P04-c | Answer provenance: derived

> Recall that in RSA, we need to choose two primes p and q, and the public key is (n, e) where $n = pq$. Let $T = (p-1)(q-1)$. The private key is d, where $ed \equiv 1 \pmod{T}$. The encryption of x is $x^e \bmod n$. This means that d must be an integer in $\mathbb{Z}_T$ that is relatively prime with T. We can find d using the following algorithm:
>
> 1. Randomly pick $d \in \mathbb{Z}_T$.
> 2. If d and T are relatively prime, then return d. If they are not, then pick a new d and loop from step 1.
>
> Assume that we will never pick the same d twice.
>
> (c) Bob says: "Instead of picking a random d, I will iterate over the primes from smallest to largest: I will test $d = 2, d = 3, d = 5, d = 7, \ldots$ until I get a valid d." Prove that in the worst case, this algorithm will terminate in $O(\log n)$ number of loops.

Official Answer

Source: physical PDF page 7 (2017-final.pdf).

Suppose $P_{\log_2 n}$ consists of the first $\log_2 n$ primes. Then $\prod_{x \in P_{\log_2 n}} x > \prod_{x \in P_{\log_2 n}} 2 > n$. So using the fundamental theorem of arithmetic, there cannot be more than $\log_2 n$ primes which are factors of n. This means that Bob will have to try at most $\log_2 n + 1 = O(\log n)$ primes.

Supplementary Explanation

Answer. At most $\lfloor \log_2 T \rfloor + 1 = O(\log n)$ prime candidates are tested.

Derivation/proof. A prime candidate fails precisely when it divides T. If r distinct primes fail, their product divides T and is at least 2^r. Therefore $2^r \le T$, giving $r \le \log_2 T$. The next tested prime succeeds, so there are at most $r + 1$ tests. Since $T = (p-1)(q-1) < pq = n$, this is $O(\log n)$. This counts candidate tests, not prime generation time.

N095

GCDs and Congruences

2017 Midterm | Question 5(b) | Source page 7 | Difficulty 3
Q-2017-MID-P05-b | Answer provenance: official

> Find $x \in \mathbb{Z}_{221}$ such that $15x \equiv 2 \pmod{221}$. Show your workings.

Official Answer

Source: physical PDF page 7 (2017-midterm.pdf).

Rewriting:

$$\begin{aligned} 11 &= 221 - 15 \cdot 14 \\ 4 &= 15 - 11 \cdot 1 \\ 3 &= 11 - 4 \cdot 2 \\ 1 &= 4 - 3 \cdot 1. \end{aligned}$$

Substituting:

$$\begin{aligned} 1 &= 4 - (11 - 4 \cdot 2) \cdot 1 \\ &= 4 \cdot 3 - 11 \cdot 1 \\ &= (15 - 11 \cdot 1) \cdot 3 - 11 \cdot 1 \\ &= 15 \cdot 3 - 11 \cdot 4 \\ &= 15 \cdot 3 - (221 - 15 \cdot 14) \cdot 4 \\ &= 15 \cdot 59 - 221 \cdot 4. \end{aligned}$$

Therefore, the inverse of 15 modulo 221 is 59.

$$15^{-1} \cdot 15x \equiv 15^{-1} \cdot 2 \pmod{221}.$$

$$x \equiv 59 \cdot 2 \equiv 118 \pmod{221}.$$

Supplementary Explanation

Answer. $x = 118$.

Derivation/proof. Using $221 = 14 \cdot 15 + 11$, $15 = 11 + 4$, $11 = 2 \cdot 4 + 3$, $4 = 3 + 1$, back-substitute: $1 = 4 - 3 = 3 \cdot 4 - 11 = 3 \cdot 15 - 4 \cdot 11 = 59 \cdot 15 - 4 \cdot 221$. Thus $15^{-1} \equiv 59 \pmod{221}$ and $x \equiv 2 \cdot 59 = 118$. Direct check: $15 \cdot 118 = 1770 = 8 \cdot 221 + 2$.

The inverse also guarantees uniqueness of the residue class, and $118 \in \mathbb{Z}_{221}$.

N096

GCDs and Congruences

2016 Final | Question 6(a) | Source page 11 | Difficulty 3 | 8 marks
Q-2016-FINAL-P06-a | Answer provenance: official

> Does there exist a solution for $x \in Z_{3000}$ such that x satisfies the above two modular equations? If yes, please give all possible solutions in Z_{3000} and show your steps. If no, please explain why there is no solution in Z_{3000}.
>
> (a) Consider the following two modular equations.
>
> $$x \bmod 990 = 733$$
>
> $$x \bmod 693 = 40$$
>
> Does there exist a solution for $x \in Z_{3000}$ such that x satisfies the above two modular equations? If yes, please give all possible solutions in Z_{3000} and show your steps. If no, please explain why there is no solution in Z_{3000}.

Official Answer

Source: PDF physical page 11 (2016-final.pdf).

(a) Yes.

Based on the two modular equations, we express them as follows.

$$x = 990q_1 + 733 \ldots\ldots\ldots\ldots (*)$$

$$x = 693q_2 + 40 \ldots\ldots\ldots\ldots (**)$$

where q_1 and q_2 are two integers.

We derive as follows.

$$\begin{aligned} 990q_1 + 733 &= 693q_2 + 40 \\ 693 &= 693q_2 - 990q_1 \\ 99 \cdot 7 &= 99 \cdot 7q_2 - 99 \cdot 10q_1 \\ 7 &= 7q_2 - 10q_1 \\ 7q_2 &= 7 + 10q_1 \\ q_2 &= 1 + \frac{10}{7}q_1 \end{aligned}$$

Since both q_1 and q_2 are integers, we derive that q_1 is a multiple of 7.

Consider (*).

When $q_1 = 0$, $x = 990 \cdot 0 + 733 = 733$.

Thus, all solution in Z_{3000} is 733.

Supplementary Explanation

Answer. The only solution in $\mathbb{Z}_{3000}$ is $x = 733$.

Derivation/proof. Write $x = 733 + 990t$. Since $733 \equiv 40 \pmod{693}$, the second condition is $990t \equiv 0 \pmod{693}$. Dividing by $\gcd(990, 693) = 99$ gives $10t \equiv 0 \pmod{7}$, hence $t \equiv 0 \pmod{7}$. All integer solutions are $733 + 6930k$. Only $k = 0$ lies in $[0, 2999]$. Check: $733 \bmod 990 = 733$ and $733 \bmod 693 = 40$.

After enforcing the interval, the only solution is 733.

N097

GCDs and Congruences

2016 Final | Question 6(b) | Source page 11 | Difficulty 3 | 7 marks
Q-2016-FINAL-P06-b | Answer provenance: official

> Does there exist a solution for $x \in Z_{3000}$ such that x satisfies the above two modular equations? If yes, please give all possible solutions in Z_{3000} and show your steps. If no, please explain why there is no solution in Z_{3000}.
>
> (b) Consider the following two modular equations.
>
> $$x \bmod 991 = 752$$
>
> $$x \bmod 997 = 40$$
>
> Does there exist a solution for $x \in Z_{3000}$ such that x satisfies the above two modular equations? If yes, please give all possible solutions in Z_{3000} and show your steps. If no, please explain why there is no solution in Z_{3000}.

Official Answer

Source: PDF physical page 11 (2016-final.pdf); PDF physical page 12 (2016-final.pdf).

(b) No.

Since gcd(991,997) = 1, by Chinese Reminder Theorem, we know that there exists a unique solution in Z_{988027} where $988027 = 991 \times 997$. However, this unique solution in Z_{988027} is equal to 447,693 which is greater than or equal to 3000.

By Euclid's Extended GCD algorithm, we derive the following.

$$991(166) + 997(-165) = 1$$

The multiplicative inverse of 991 in Z_{997} is $166 \bmod 997 = 166$.

The multiplicative inverse of 997 in Z_{991} is $-165 \bmod 991 = 826$.

Construct $y = 752 \cdot 997 \cdot 826 + 40 \cdot 991 \cdot 166 = 625868784$.

The unique solution x in Z_{988027} is $625868784 \bmod 988027 = 447,693$, which is greater than or equal to 3000. Thus, there is no solution in Z_{3000}.

Supplementary Explanation

Answer. There is no solution in $\mathbb{Z}_{3000}$.

Derivation/proof. Put $x = 752 + 991t$. Modulo 997 this requires $-6t \equiv -712$, or $6t \equiv 712$. Since $6 \cdot 166 \equiv -1 \pmod{997}$, an inverse of 6 is 831, giving $t \equiv 712 \cdot 831 \equiv 451 \pmod{997}$. Thus $x = 447693 + 988027k$. Its least nonnegative representative exceeds 2999. Independently, the only first-congruence candidates in the requested range are 752, 1743, 2734; their remainders modulo 997 are 752, 746, 740, none equal to 40.

N098

GCDs and Congruences

2015 Final | Question 7(a) | Source page 13 | Difficulty 3 | 6 marks
Q-2015-FINAL-P07-a | Answer provenance: official

> (a) Use Fermat's Little Theorem and the Chinese Remainder Theorem to show that, if an integer a is not divisible by any of 3, 5 and 7, then
>
> $$a^{49} \bmod 105 = a \bmod 105.$$

Official Answer

Source: PDF physical page 13 (2015-final.pdf).

(a) Note that $105 = 3 \times 5 \times 7$. We also have:

$$\begin{aligned} a^{49} \bmod 3 &= (a^{24*(3-1)+1}) \bmod 3 = a \bmod 3 \quad \text{by FLT} \\ a^{49} \bmod 5 &= (a^{12*(5-1)+1}) \bmod 5 = a \bmod 5 \quad \text{by FLT} \\ a^{49} \bmod 7 &= (a^{8*(7-1)+1}) \bmod 7 = a \bmod 7 \quad \text{by FLT} \end{aligned}$$

By CRT, we have

$$a^{49} \bmod 3 \times 5 = a \bmod 15.$$

Applying CRT again we get

$$a^{49} \bmod 15 \times 7 = a \bmod 105.$$

Supplementary Explanation

Answer. $a^{49} \equiv a \pmod{105}$.

Derivation/proof. For $r = 3, 5, 7$, the hypothesis gives $\gcd(a, r) = 1$, so $a^{r-1} \equiv 1 \pmod{r}$. Since $48 = 24 \cdot 2 = 12 \cdot 4 = 8 \cdot 6$, $a^{49} = a(a^{r-1})^{48/(r-1)} \equiv a \pmod{r}$. The pairwise coprime moduli multiply to 105; CRT therefore makes the two residues equal modulo 105.

CRT uniqueness makes the two standard remainders equal modulo the product 105.

N099

GCDs and Congruences

2014 Final | Question 4(a) | Source page 7 | Difficulty 3
Q-2014-FINAL-P04-a | Answer provenance: official

> In this question we consider the following modular equations:
>
> $$x \bmod m = a, \qquad x \bmod n = b.$$
>
> For each of the following parts, does there exist an integer x, $0 \leq x < mn$ that satisfies the two equations?
>
> If yes, find the solution and show the computational steps. Otherwise, prove by contradiction that there does not exist a solution.
>
> In this question, it is up to you whether to use the extended GCD algorithm to find multiplicative inverses. If you choose to use it, there is no need to show the steps.
>
> (a) $m = 15, n = 16, a = 5, b = 2$.

Official Answer

Source: PDF physical page 7 (2014-final.pdf).

(a) In this case, m and n are relatively prime. So,there is a solution. To find the solution, first compute multiplicative inverses:

$$\bar{m} \text{ s.t. } m \cdot \bar{m} = 1 \bmod n,$$

$$\bar{n} \text{ s.t. } n \cdot \bar{n} = 1 \bmod m.$$

It turns out that $\bar{m} = 15$ and $\bar{n} = 1$. Now we can find the solution:

$$\begin{aligned} x &= (an\bar{n} + bm\bar{m}) \bmod mn \\ &= (5 \cdot 16 \cdot 1 + 2 \cdot 15 \cdot 15) \bmod 15 \cdot 16 \\ &= 50 \end{aligned}$$

Supplementary Explanation

Answer. Yes, $x = 50$.

Derivation/proof. Since $15, 16$ are coprime, CRT applies. Inverses are $15^{-1} \equiv 15 \pmod{16}$ and $16^{-1} \equiv 1 \pmod{15}$. Thus $x \equiv 5 \cdot 16 \cdot 1 + 2 \cdot 15 \cdot 15 = 530 \equiv 50 \pmod{240}$.

The representative 50 lies in $0 \leq x < 240$, with $50 \bmod 15 = 5$ and $50 \bmod 16 = 2$.

N100

GCDs and Congruences

2013 Final | Question 1(a) | Source page 3 | Difficulty 3
Q-2013-FINAL-P01-a | Answer provenance: official

> (a) Let m and n be two positive integers such that m divides n, i.e., $n = sm$ for some other integer s. Show that, for any integer x, $(x \bmod n) \bmod m = x \bmod m$.

Official Answer

Source: PDF physical page 3 (2013-final.pdf).

(a) Let $x \bmod n = r_1$. Then $x = q_1 n + r_1$ for some integer q_1. Further let $r_1 \bmod m = r_2$. Then $r_1 = q_2 m + r_2$ for some integer q_2. Hence,

$$x = q_1 n + q_2 m + r_2 = q_1 s m + q_2 m + r_2 = (q_1 s + q_2) m + r_2,$$

where $0 \leq r_2 < m$. Consequently,

$$x \bmod m = r_2 = (x \bmod n) \bmod m.$$

Supplementary Explanation

Answer. The two remainders are equal.

Derivation/proof. Let $r_1 = x \bmod n$, so $x = q_1 n + r_1$. Let $r_2 = r_1 \bmod m$, so $r_1 = q_2 m + r_2$ with $0 \leq r_2 < m$. Since $n = sm$, $x = (q_1 s + q_2) m + r_2$; uniqueness of the remainder modulo m gives $x \bmod m = r_2$.

N101

GCDs and Congruences

2013 Final | Question 3 | Source page 6 | Difficulty 3 | 10 marks
Q-2013-FINAL-P03 | Answer provenance: official

> Let p and q be two different prime numbers. Let $a \in Z_p$ and $b \in Z_q$. Consider the following two modular equations:
>
> $$x \bmod p = a, \qquad x \bmod q = b.$$
>
> We have learned in class that there exists $x \in Z_{pq}$ that satisfies the two equations. Give a contrapositive proof for the following statement:
>
> There is only one $x \in Z_{pq}$ that satisfies the two equations.
>
> Note that this is the uniqueness part of the Chinese Reminder Theorem.

Official Answer

Source: PDF physical page 6 (2013-final.pdf).

Let s denote the statement. The premise is always true. So, the statement is $T \to s$. By contrapositive rule, we prove $\neg s \to F$. This means that we assume $\neg s$ and then derive some contradiction.

Assume there are two different numbers x_1 and x_2 from Z_{pq} that satisfy the equations.

Without losing generality, assume $x_1 > x_2$. Then $0 < (x_1 - x_2) < pq - 1$. Hence $(x_1 - x_2)$ is not divisible by pq.

Because both x_1 and x_2 satisfy the equations, we have

$$\begin{aligned} x_1 \bmod p = a, \quad x_1 \bmod q = b. \\ x_2 \bmod p = a, \quad x_2 \bmod q = b. \end{aligned}$$

By subtracting the second line from the first, we

$$x_1 \bmod p - x_2 \bmod p = 0, \qquad x_1 \bmod q - x_2 \bmod q = 0,$$

which imply:

$$(x_1 - x_2) \bmod p = 0, \qquad (x_1 - x_2) \bmod q = 0.$$

These two equations mean that $(x_1 - x_2)$ is divisible by p and q. Because p and q are different prime numbers, this implies that $(x_1 - x_2)$ is divisible by pq, which contradicts our earlier conclusion that $(x_1 - x_2)$ is not divisible by pq.

The statement is therefore proved. Q.E.D

Supplementary Explanation

Answer. Two distinct solutions cannot exist.

Derivation/proof. Assume distinct $x_1, x_2 \in \{0, \ldots, pq-1\}$ both satisfy both congruences. Then $p \mid (x_1 - x_2)$ and $q \mid (x_1 - x_2)$. Coprimality implies $pq \mid (x_1 - x_2)$, but $0 < |x_1 - x_2| < pq$, a contradiction.

Existence is given in the question. Excluding two distinct solutions therefore proves that there is exactly one.

N102

GCDs and Congruences

2013 Final | Question 4 | Source page 7 | Difficulty 3 | 10 marks
Q-2013-FINAL-P04 | Answer provenance: official

> Prove the following equality without using repeated squaring:
>
> $$46^{120} \bmod 77 = 1.$$
>
> [Hint: Make use of the fact that $77 = 11 \cdot 7$ and try to use Fermat's Little Theorem and the Chinese Reminder Theorem.]

Official Answer

Source: PDF physical page 7 (2013-final.pdf).

Let $r = 46^{120} \bmod 77$. Using Problem 1 (a), we get

$$r \bmod 7 = 46^{120} \bmod 7, r \bmod 11 = 46^{120} \bmod 11.$$

By Fermat's Little Theorem, we have

$$46^{120} \bmod 7 = (4^6)^{20} \bmod 7 = (4^6 \bmod 7)^{20} \bmod 7 = 1,$$
$$46^{120} \bmod 11 = (4^{10})^{12} \bmod 11 = (4^{10} \bmod 11)^{12} \bmod 11 = 1.$$

So, we have

$$r \bmod 7 = 1, r \bmod 11 = 1.$$

Obviously, we also have

$$1 \bmod 7 = 1, 1 \bmod 11 = 1.$$

It follows from the Chinese Reminder Theorem that $r = 1$. The proof is completed.

Supplementary Explanation

Answer. $46^{120} \equiv 1 \pmod{77}$.

Derivation/proof. Let $r = 46^{120} \bmod 77$. Since $77 = 7 \cdot 11$, Fermat gives $46^{120} = (46^6)^{20} \equiv 1 \pmod 7$ and $46^{120} = (46^{10})^{12} \equiv 1 \pmod{11}$. The residue 1 has the same two remainders, so CRT uniqueness yields $r = 1$ modulo 77.

Here $46 \equiv 4 \pmod 7$ and $46 \equiv 2 \pmod{11}$, so both relevant bases are nonzero.

N103

GCDs and Congruences

2024 Fall Final | Question 6(b) | Source page 17 | Difficulty 4 | 8 marks
Q-2024F-FINAL-P06-b | Answer provenance: official

> Assume that a, b are integers and p is a prime number of form $4k+3$ (with an integer k). As an example, 19 is a prime of this form while 17 is not of this form. Assume that $p \mid a^2 + b^2$.
>
> (b) Prove that if $p \nmid a$ then there will be contradictions. (In other words, if $p \mid a^2 + b^2$ we can conclude that $p \mid a$ and $p \mid b$). Hint: You might find little Fermat theorem useful here. You can also use part (a) even if you could not prove it.

Official Answer

Source: physical PDF page 17 (COMP2711_F2024_Final.pdf).

> (b) Assume that $p \nmid a$. This also implies that $p \nmid b$ (think why?). This means $\gcd(p, a) = \gcd(p, b) = 1$. $p \mid a^2 + b^2 \implies a^2 \equiv -b^2 \pmod{p}$. Note that $p - 1$ is an even number and $\frac{p-1}{2}$ is odd.
>
> $$a^2 \equiv -b^2 \pmod{p} \implies (a^2)^{\frac{p-1}{2}} \equiv (-b^2)^{\frac{p-1}{2}} \implies a^{p-1} \equiv -b^{p-1}.$$
>
> By little Fermat theorem we know that $a^{p-1} \equiv b^{p-1} \equiv 1 \pmod{p}$. So this means $1 \equiv -1 \pmod{p} \implies 2 \equiv 0 \pmod{p}$. This is impossible as p is odd and this is the contradiction.

Supplementary Explanation

Answer. $p \mid a$ and $p \mid b$.

Derivation/proof. Suppose $p \nmid a$. If $p \mid b$, subtraction from the square sum would give $p \mid a^2$ and then $p \mid a$, a contradiction. Thus $p \nmid b$. Now $a^2 \equiv -b^2 \pmod{p}$. Since $p = 4k + 3$, the exponent $(p-1)/2 = 2k + 1$ is odd, so raising both sides gives $a^{p-1} \equiv -b^{p-1} \pmod{p}$. Fermat applies to both nondivisible bases and gives $1 \equiv -1 \pmod{p}$, hence $p \mid 2$. But p is an odd prime at least three, so this is a contradiction. Therefore $p \mid a$, and part (a) gives $p \mid b$.

N104

GCDs and Congruences

2022 Midterm | Question 6(c) | Source page 3 | Difficulty 4 | 10 marks
Q-2022-MID-P06-c | Answer provenance: official

> Assume p is a prime and n is a positive integer such that $p \mid (4n^2 + 1)$.
>
> Show that there are infinitely many primes of the form $4k + 1$.

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (c) Assume we have finitely many primes of form $4k + 1$, say $p_1, p_2, \cdots, p_t$. Consider $N = 4(p_1 p_2 \dots p_t)^2 + 1$. Let p be any prime factor of N (p may be N itself if N is prime). Based on part (b), $p = 4k + 1$ for some k. Also note that p must be different from $p_1, \dots, p_t$, since none of them divides N. Then we have found a new prime p in the form of $4k + 1$, which is a contradiction.

Supplementary Explanation

Answer. Infinitely many primes are $1 \pmod 4$.

Derivation/proof. Suppose the complete finite list is p1,...,pt and put N=4(p1...pt)²+1. (The class is nonempty because5 belongs to it.) Choose a prime divisor q of N. Applying the preceding divisor result with n=p1...pt gives q≡1 mod 4. For each listed pi, N≡1 modpi, so none divides N. Thus q is in the required class but absent from the allegedly complete list, a contradiction. N itself need not be prime.

With $M = p_1 \cdots p_t$, we have $N = 4M^2 + 1 > 1$. A prime divisor satisfies $q \equiv 1 \pmod 4$, but $N \equiv 1 \pmod{p_i}$ gives $p_i \nmid N$ for every listed prime.

N105

GCDs and Congruences

2021 Final Part 1 | Question 1 | Source page 1 | Difficulty 4 | 10 marks
Q-2021-FINAL1-P01 | Answer provenance: official

> You are constructing a string of n digits by the following procedure.
>
> 1. Set all the n digits to 0.
> 2. For every x_1-th digit, replace it by 1.
> 3. For every x_2-th digit, replace it by 2.
> 4. For every x_3-th digit, replace it by 3.
>
> Below is an example with $n = 15, x_1 = 3, x_2 = 5$ and $x_3 = 7$.
>
> "001021301201032"
>
> Suppose $x_1 = 31, x_2 = 43$ and $x_3 = 53$. What is the minimum value n so that the string ends with "31020"?

Official Answer

Source: physical PDF page 1 (2021-final-part-1.pdf).

> This is equivalent to finding the smallest n that satisfies the following system of congruences.
>
> $$\begin{cases} n \equiv 3 \pmod{31} \\ n \equiv 1 \pmod{43} \\ n \equiv 4 \pmod{53} \end{cases}$$
>
> Let $m = 31 \cdot 43 \cdot 53 = 70649$, $M_1 = 43 \cdot 53 = 2279$, $M_2 = 31 \cdot 53 = 1643$ and $M_3 = 31 \cdot 43 = 1333$. By the extended Euclidean algorithm, we have 2 is an inverse of $M_1 \pmod{31}$, 24 is an inverse of $M_2 \pmod{43}$ and 20 is an inverse of M_3. By the Chinese Remainder Theorem, $n = 3 \cdot 2 \cdot M_1 + 1 \cdot 24 \cdot M_2 + 4 \cdot 20 \cdot M_3 = 159746 \equiv 18448 \pmod{70649}$. So, the minimum length is 18448.

Supplementary Explanation

Answer. $n = 18448$.

Derivation/proof. The 3 at n−4 requires n≡4 mod 53; the 1 at n−3 requires n≡3 mod 31; the 2 at n−1 requires n≡1 mod 43. The pairwise-coprime modulus product is 70649. For partial products 2279,1643,1333 the inverses modulo 31,43,53 are respectively 2,24,20: their products have remainder one. CRT gives n≡3·2·2279+1·24·1643+4·20·1333=159746≡18448. Check positions 18444,…,18448: their residues modulo (31,43,53) are (30,40,0),(0,41,1),(1,42,2),(2,0,3),(3,1,4). Thus overwrite precedence produces exactly 31020, including both zeros. Every valid n must be in this CRT class, whose least positive representative is 18448.

The inverse products can be checked explicitly: $4558 = 147 \cdot 31 + 1$, $39432 = 917 \cdot 43 + 1$, and $26660 = 503 \cdot 53 + 1$.

The CRT calculation is $159746 \equiv 18448 \pmod{70649}$; the valid value is the least positive representative.

N106

GCDs and Congruences

2018 Final | Question 3(b) | Source page 5 | Difficulty 4
Q-2018-FINAL-P03-b | Answer provenance: official

> S, a set of integers, is recursively defined as follows:
>
> Basis: 4914 and 3588 are in S.
>
> Recursive step: If $x, y \in S$, then $x - y \in S$.
>
> (Note that x and y may be the same.)
>
> (b) Let A_m be the set of all multiples of some integer m. Find the integer m such that $S = A_m$, and provide a complete proof of this equality, showing all steps.

Official Answer

Source: physical PDF page 5 (2018-final.pdf).

Part (a) shows that $4914s \in S$ and $3588t \in S$ for all s, t, so $\gcd(4914, 3588) \in S$. Therefore

$$\begin{aligned}\gcd(4914, 3588) &= \gcd(3588, 1326) = \gcd(1326, 936) = \gcd(936, 390) \\ &= \gcd(390, 156) = \gcd(156, 78) = 78 \in S.\end{aligned}$$

$A_{78} \subseteq S$: Using part (a) directly, since $78 \in S$, then $78n \in S$ for any integer n. $S \subseteq A_{78}$: 4914 and 3588 are multiples of 78. Suppose x and y are multiples of 78. Then clearly $x - y$ is a multiple of 78. By structural induction, $S \subseteq A_{78}$.

Supplementary Explanation

Answer. $m = 78$ is the positive generator, and $S = 78\mathbb{Z}$ (also $A_{-78} = A_{78}$).

Derivation/proof. Euclid gives $4914 = 3588 + 1326$, $3588 = 2 \cdot 1326 + 936$, $1326 = 936 + 390$, $936 = 2 \cdot 390 + 156$, $390 = 2 \cdot 156 + 78$, $156 = 2 \cdot 78$. Every remainder belongs to S: previous entries and their multiples belong by part (a), and differences are allowed. Thus $78 \in S$, and part (a), together with $0 \in S$, gives $78\mathbb{Z} \subseteq S$. Conversely $4914 = 63 \cdot 78$ and $3588 = 46 \cdot 78$. Differences of multiples of 78 remain multiples, so structural induction gives $S \subseteq 78\mathbb{Z}$. Both inclusions prove equality.

The basis numbers satisfy $4914 = 63 \cdot 78$ and $3588 = 46 \cdot 78$. A generator $m = -78$ gives the same set as $m = 78$.

N107

Cryptography

2019 Midterm | Question 11(c) | Source page 13 | Difficulty 1 | 3 marks
Q-2019-MID-P11-c | Answer provenance: official

> Consider Bob is using RSA algorithm for public key cryptography. Bob starts by picking the two prime numbers $p = 43$ and $q = 23$. So, $n = pq = 989$.
>
> (c) Suppose Bob's public key is $(989, 53)$ and Alice wants to send a message $m = 720$ to Bob. Write down the modular expression that computes the encrypted message c. It is not necessary to evaluate the expression.

Official Answer

Source: physical PDF page 13 (2019-midterm.pdf).

The encrypted message $c = 720^{53} \bmod 989$.

Supplementary Explanation

Answer. $c = 720^{53} \bmod 989$.

Derivation/proof. RSA encryption maps a message representative m to $m^e \bmod n$, using the recipient's public key. Substitute $m = 720, e = 53, n = 989$. The message is already in $\mathbb{Z}_{989}$, so the expression is valid without first reducing it. The private exponent is not used for this encryption request.

The private decryption exponent 401 must not replace the public exponent 53.

N108

Cryptography

2013 Final | Question 2(a) | Source page 4 | Difficulty 1
Q-2013-FINAL-P02-a | Answer provenance: official

> Suppose for applying RSA, $p = 11$, $q = 19$, and $e = 7$.
>
> (a) What is the public key (e, n)?

Official Answer

Source: PDF physical page 4 (2013-final.pdf).

(a) The public key is the pair: (e, n) where $e = 7$ and $n = p * q = 209$.

Supplementary Explanation

Answer. $(7, 209)$.

Derivation/proof. RSA modulus $n = pq = 11 \cdot 19 = 209$; the specified public exponent is $e = 7$.

The second component is the product of the two primes, not the totient 180. The question specifies the order (e, n), so the exponent comes first.

N109

Cryptography

2025 Fall Final | Question 7(i) | Source page 12 | Difficulty 2 | 3 marks
Q-2025F-FINAL-P07-i | Answer provenance: official

> This problem is on the RSA algorithm for public key cryptogtaphy. To generate his keys, Bob uses two prime numbers $p = 7, q = 11$. So, $n = pq = 77$ and $T = (p-1)(q-1) = 60$.
>
> (i) [3 pts] Write down the possible values for e if it has to be between 10 and 20. No argument is needed.

Official Answer

Source: physical PDF page 12 (COMP2711_F2025_Final.pdf).

(i) Possible e with $10 \leq e \leq 20$ and $\gcd(e, 60) = 1$:

$$e = 11, 13, 17, 19.$$

Supplementary Explanation

Answer. $e = 11, 13, 17, 19$.

Derivation/proof. Factor $60 = 2^2 \cdot 3 \cdot 5$. The condition $\gcd(e, 60) = 1$ means that e is divisible by none of 2, 3 or 5. Eliminate every even number from 10 through 20. The remaining candidates are 11, 13, 15, 17 and 19; remove 15 because it has factors 3 and 5. The four values $11, 13, 17, 19$ have no prime factor in common with 60. Each therefore has a multiplicative inverse modulo 60 and is a valid public exponent.

N110

Cryptography

2025 Fall Final | Question 7(iii) | Source page 12 | Difficulty 2 | 3 marks
Q-2025F-FINAL-P07-iii | Answer provenance: official

> This problem is on the RSA algorithm for public key cryptogtaphy. To generate his keys, Bob uses two prime numbers $p = 7, q = 11$. So, $n = pq = 77$ and $T = (p-1)(q-1) = 60$.
>
> (iii) [3 pts] Suppose Alice wants to encrypt message $m = 3$ using Bob's public key (n, e). Note that $n = 77$, $e = 7$. Use the repeated squaring technique to calculate the ciphertext $c = m^e$.

Official Answer

Source: physical PDF page 12 (COMP2711_F2025_Final.pdf).

(iii) Encrypt message $m = 3$ using Bob's public key $(n = 77, e = 7)$ by repeated squaring to compute $c = 3^7 \bmod 77$.

$$3^1 \equiv 3 \pmod{77}$$
$$3^2 \equiv 9 \pmod{77}$$
$$3^4 = (3^2)^2 \equiv 9^2 = 81 \equiv 4 \pmod{77}.$$

Thus, the ciphertext is $3^7 = 3 \times 3^2 \times 3^4 \pmod{77} = 3 \times 9 \times 4 \pmod{77} = 31$.

$$c = 31.$$

Supplementary Explanation

Answer. The RSA ciphertext is $c = 31$.

Derivation/proof. RSA encryption uses the standard residue $c \equiv m^e \pmod{n}$. Modulo 77, $3^1 \equiv 3$, $3^2 \equiv 9$ and $3^4 \equiv 9^2 = 81 \equiv 4$. Since $7 = 4 + 2 + 1$,

$$3^7 \equiv 3^4 \cdot 3^2 \cdot 3 \equiv 4 \cdot 9 \cdot 3 = 108 \equiv 31 \pmod{77}.$$

The value 31 lies between 0 and 76, so it is the ciphertext representative. Two squarings followed by multiplication implement the requested repeated-squaring method.

N111

Cryptography

2024 Fall Final | Question 8(a) | Source page 21 | Difficulty 2 | 2 marks
Q-2024F-FINAL-P08-a | Answer provenance: official

> Consider an RSA system in which $p = 5$ and $q = 19$. Answer the following questions:
>
> (a) Can we choose $e = 9$? Why?

Official Answer

Source: physical PDF page 21 (COMP2711_F2024_Final.pdf).

> Answer: (a) Note that $n = 95$ $T = 4 \cdot 18 = 72$. e has to be co-prime with T but it is not. So it cannot be chosen.

Supplementary Explanation

Answer. We cannot choose $e = 9$.

Derivation/proof. From $p = 5, q = 19$, obtain $n = 95$ and $\varphi(n) = (5-1)(19-1) = 4 \cdot 18 = 72$. Choosing $e = 9$ would require $9d \equiv 1 \pmod{72}$. But $\gcd(9, 72) = 9 > 1$ and every $9d - 72t$ is divisible by nine, so it cannot equal one. No required inverse exists, and this exponent is invalid.

N112

Cryptography

2023 Fall Final | Question 5(b) | Source page 8 | Difficulty 2 | 6 marks
Q-2023F-FINAL-P05-b | Answer provenance: official

> This problem is on the RSA algorithm for public key cryptogtaphy. To generate his keys, Bob samples two prime numbers $p = 17, q = 19$. So, $n = pq = 323$ and $T = 288$.
>
> (a) Assume that Bob chooses $e = 23$.
>
> Suppose Alice wants to encrypt message $m = 18$ using Bob's public key (n, e). Use the repeated squaring technique to calculate the ciphertext $c = m^e$.

Official Answer

Source: physical PDF page 8 (COMP2711_F2023_Final.pdf), physical PDF page 9 (COMP2711_F2023_Final.pdf).

> (b)
>
> $$\begin{aligned} 18^{2^0} \bmod 323 &= 18 \\ 18^{2^1} \bmod 323 &= 18^2 \bmod 323 = 1 \\ 18^{2^2} \bmod 323 &= 1^2 \bmod 323 = 1 \\ 18^{2^3} \bmod 323 &= 1^2 \bmod 323 = 1 \\ 18^{2^4} \bmod 323 &= 1^2 \bmod 323 = 1. \end{aligned}$$
>
> $$23 = 2^4 + 2^2 + 2^1 + 2^0.$$
>
> $$\begin{aligned} 18^{23} &\equiv 18^{2^4} \cdot 18^{2^2} \cdot 18^{2^1} \cdot 18^{2^0} \\ &\equiv 1 \cdot 1 \cdot 1 \cdot 18 \\ &\equiv 18 \pmod{323}. \end{aligned}$$

Supplementary Explanation

Answer. $c = 18 \pmod{323}$.

Derivation/proof. With the source public parameters $n = 323, e = 23$, repeated squaring gives $18^{2^0} \equiv 18$ and $18^{2^1} = 324 \equiv 1 \pmod{323}$. Squaring 1 repeatedly yields $18^{2^2} \equiv 18^{2^3} \equiv 18^{2^4} \equiv 1$. Since $23 = 16 + 4 + 2 + 1$,

$$18^{23} \equiv 18^{16}18^4 18^2 18 \equiv 1 \cdot 1 \cdot 1 \cdot 18 \equiv 18 \pmod{323}.$$

All factors were computed by the requested repeated-squaring method, not by expanding the large integer power.

Since $18^2 = 324 \equiv 1 \pmod{323}$ and $23 = 16 + 4 + 2 + 1$, the selected residues multiply to $1 \cdot 1 \cdot 1 \cdot 18$.

N113

Cryptography

2019 Midterm | Question 11(a) | Source page 13 | Difficulty 2 | 3 marks
Q-2019-MID-P11-a | Answer provenance: official

> Consider Bob is using RSA algorithm for public key cryptography. Bob starts by picking the two prime numbers $p = 43$ and $q = 23$. So, $n = pq = 989$.
>
> (a) Bob's public key is a pair $(989, e)$. Which of the following integer(s) can Bob use for e? Why?
>
> (i) 21; (ii) 7; (iii) 25; (iv) 11

Official Answer

Source: physical PDF page 13 (2019-midterm.pdf).

(iii). We have $T = (p-1)(q-1) = 42 \cdot 22 = 924$. Bob can use 25 as this is the only choice such that $\gcd(25, 924) = 1$.

Supplementary Explanation

Answer. Only (iii), $e = 25$.

Derivation/proof. The totient is $T = 42 \cdot 22 = 924 = 2^2 \cdot 3 \cdot 7 \cdot 11$. A valid e must be coprime to 924. Candidate 21 shares factors 3 and 7, 7 shares 7, and 11 shares 11. Candidate 25 has only prime factor 5, which does not divide 924, so its gcd is one and it is valid.

Since the only prime divisor of twenty-five is five and five does not divide 924, $\gcd(25, 924) = 1$. This is the RSA admissibility condition, even though twenty-five is composite.

N114

Cryptography

2018 Midterm | Question 6(a) | Source page 8 | Difficulty 2
Q-2018-MID-P06-a | Answer provenance: official

> Consider the RSA encryption with parameters $p = 19, q = 37$.
>
> (a) Which one of the integers 2,3,11 can be used as the encryption key e? Why? (Note: only one of them is an acceptable solution.)

Official Answer

Source: physical PDF page 8 (2018-midterm.pdf).

$(p-1)(q-1) = 648$. For e it must be true that $\gcd(e, 648) = 1$. Therefore, e cannot be 2 (since $648 = 2 \cdot 324$) or 3 (since $648 = 3 \cdot 216$). On the other hand $\gcd(11, 648) = 1$, therefore $e = 11$.

Supplementary Explanation

Answer. $e = 11$.

Derivation/proof. Compute $T = (19-1)(37-1) = 18 \cdot 36 = 648 = 2^3 3^4$. A valid exponent must satisfy $\gcd(e, T) = 1$. The candidates 2 and 3 divide T, so each fails. Eleven is divisible by neither prime factor of T, hence $\gcd(11, 648) = 1$ and it is valid.

The factorization $648 = 2^3 \cdot 3^4$ shows that an acceptable exponent must be coprime to both two and three. Testing the supplied candidates against those prime factors identifies precisely the valid choices.

N115

Cryptography

2018 Midterm | Question 6(b) | Source page 8 | Difficulty 2
Q-2018-MID-P06-b | Answer provenance: official

> Consider the RSA encryption with parameters $p = 19, q = 37$.
>
> (b) Compute the value of the decryption key d for your choice of e from step (a). Show all intermediate steps.

Official Answer

Source: physical PDF page 8 (2018-midterm.pdf).

The private key d is the multiplicative inverse of e modulo $(p-1)(q-1)$. Using the extended Euclidean algorithm:

$$\begin{aligned} 648 &= 58 \cdot 11 + 10 \\ 11 &= 1 \cdot 10 + 1 \\ 10 &= 10 \cdot 1. \end{aligned}$$

Substituting backwards

$$1 = 11 - 10 \cdot 1 = 11 - (648 - 58 * 11) = -1 \cdot 648 + 59 \cdot 11.$$

Since 59 is already in $\mathbb{Z}_{648}$ the answer is $d = 59$. Note that students can use a table form to obtain the answer, which should be accepted.

Supplementary Explanation

Answer. $d = 59$.

Derivation/proof. With $e = 11$, solve $11d \equiv 1 \pmod{648}$. Divide: $648 = 58 \cdot 11 + 10$, $11 = 10 + 1$, $10 = 10 \cdot 1$. Back substitution gives $1 = 11 - 10 = 11 - (648 - 58 \cdot 11) = 59 \cdot 11 - 648$. Thus $d = 59$, already in $\mathbb{Z}_{648}$. Check $11 \cdot 59 = 649 \equiv 1 \pmod{648}$.

Numerically, $11 \cdot 59 = 649 = 648 + 1$.

N116

Cryptography

2017 Final | Question 4(b) | Source page 6 | Difficulty 2
Q-2017-FINAL-P04-b-algorithm | Answer provenance: official

> Recall that in RSA, we need to choose two primes p and q, and the public key is (n, e) where $n = pq$. Let $T = (p-1)(q-1)$. The private key is d, where $ed \equiv 1 \pmod{T}$. The encryption of x is $x^e \bmod n$. This means that d must be an integer in $\mathbb{Z}_T$ that is relatively prime with T. We can find d using the following algorithm:
>
> 1. Randomly pick $d \in \mathbb{Z}_T$.
> 2. If d and T are relatively prime, then return d. If they are not, then pick a new d and loop from step 1.
>
> Assume that we will never pick the same d twice.
>
> (b) Give an efficient algorithm to test if a single d and T are relatively prime.

Official Answer

Source: physical PDF page 6 (2017-final.pdf); physical PDF page 7 (2017-final.pdf).

It is the Euclidean algorithm to find gcd: start with $\gcd(T, d)$, and the recursive step is to change $\gcd(a, b)$ into $\gcd(b, a \bmod b)$. It was shown in class that this takes $O(\log T)$ steps for the worst case of the Fibonacci numbers.

Supplementary Explanation

Answer. Run Euclid on (T, d) and accept exactly when its returned gcd is 1.

Derivation/proof. Set $a = T, b = d$. While $b \neq 0$, replace (a, b) by $(b, a \bmod b)$. Return whether $a = 1$. A common divisor of a, b also divides $a - qb$, and conversely a common divisor of $b, a - qb$ divides a. Thus every iteration preserves the gcd, and at termination $\gcd(a, 0) = a$ gives the correct test.

Nonzero remainders strictly decrease, guaranteeing termination at $(g, 0)$ with $g = \gcd(T, d)$. The procedure also correctly handles $d = 0$.

Each Euclidean iteration replaces the current pair by $(a, b) \leftarrow (b, a \bmod b)$. The gcd is invariant and the second nonnegative entry strictly decreases until it is zero; the remaining first entry is the gcd.

N117

Cryptography

2016 Final | Question 5(a) | Source page 10 | Difficulty 2
Q-2016-FINAL-P05-a-kenneth | Answer provenance: official

> Consider the RSA cryptosystem where the public key is represented in the form of (e, n). We know that the public key of Raymond is $(7, 247)$ and the public key of Kenneth is $(13, 187)$.
>
> (a) Is it possible to find the secret key of Raymond and the secret key of Kenneth? If yes, please find the two secret keys. If no, please explain the reason.
>
> Atomic focus [editorial]: Kenneth's secret key; Raymond's is a separate atom.

Official Answer

Source: PDF physical page 10 (2016-final.pdf).

(a) Yes.

Consider Kenneth's case.

Note that $11 \times 17 = 187$.

Thus, $p = 11$ and $q = 17$.

So, $T = (11 - 1)(17 - 1) = 10 \times 16 = 160$

Secret key of Kenneth = the multiplicative inverse of 13 in $Z_{160} = 37$

Supplementary Explanation

Answer. Yes. Kenneth's secret exponent is $d_K = 37$ (private key $(37, 187)$).

Derivation/proof. Since $187 = 11 \cdot 17$, $\varphi(187) = 10 \cdot 16 = 160$. Euclid gives $160 = 12 \cdot 13 + 4$, $13 = 3 \cdot 4 + 1$. Back substitution yields $1 = 37 \cdot 13 - 3 \cdot 160$, so $13^{-1} \equiv 37 \pmod{160}$. Directly, $13 \cdot 37 = 481 = 3 \cdot 160 + 1$.

N118

Cryptography

2016 Final | Question 5(a) | Source page 10 | Difficulty 2
Q-2016-FINAL-P05-a-raymond | Answer provenance: official

> Consider the RSA cryptosystem where the public key is represented in the form of (e, n). We know that the public key of Raymond is $(7, 247)$ and the public key of Kenneth is $(13, 187)$.
>
> (a) Is it possible to find the secret key of Raymond and the secret key of Kenneth? If yes, please find the two secret keys. If no, please explain the reason.
>
> Atomic focus [editorial]: Raymond's secret key; Kenneth's is a separate atom.

Official Answer

Source: PDF physical page 10 (2016-final.pdf).

(a) Yes.

Consider Raymond's case.

Note that $13 \times 19 = 247$.

Thus, $p = 13$ and $q = 19$.

So, $T = (13 - 1)(19 - 1) = 12 \times 18 = 216$

Secret key of Raymond = the multiplicative inverse of 7 in $Z_{216} = 31$

Supplementary Explanation

Answer. Yes. Raymond's secret exponent is $d_R = 31$ (private key $(31, 247)$).

Derivation/proof. Factor $247 = 13 \cdot 19$, so $\varphi(247) = 12 \cdot 18 = 216$. The Euclidean identity $216 = 30 \cdot 7 + 6$ and $7 = 6 + 1$ gives $1 = 31 \cdot 7 - 216$. Therefore $7 \cdot 31 \equiv 1 \pmod{216}$ and the RSA secret exponent is 31.

Check: $7 \cdot 31 = 217 = 216 + 1$.

N119

Cryptography

2016 Final | Question 5(b) | Source page 10 | Difficulty 2 | 3 marks
Q-2016-FINAL-P05-b | Answer provenance: official

> Consider the RSA cryptosystem where the public key is represented in the form of (e, n). We know that the public key of Raymond is $(7, 247)$ and the public key of Kenneth is $(13, 187)$.
>
> (b) Suppose that Kenneth wants to send an encrypted message to Raymond. The message is represented by a number 20. What is the value of the encrypted message? Please show your steps.

Official Answer

Source: PDF physical page 10 (2016-final.pdf).

(b) Since the encrypted message is sent to Raymond, we should use the public key of Raymond for encryption.

$$\begin{aligned}\text{The encrypted message} &= 20^7 \bmod 247 \\ &= 1280000000 \bmod 247 \\ &= 58\end{aligned}$$

Supplementary Explanation

Answer. $c = 20^7 \bmod 247 = 58$.

Derivation/proof. Use Raymond's public key $(7, 247)$. Compute $20^2 \equiv 153$, $20^4 \equiv 153^2 \equiv 191 \pmod{247}$. Then $20^7 \equiv 191 \cdot 153 \cdot 20 \equiv 77 \cdot 20 \equiv 58 \pmod{247}$. This agrees with $1,280,000,000 \bmod 247 = 58$.

N120

Cryptography

2015 Final | Question 6(a) | Source page 11 | Difficulty 2 | 3 marks
Q-2015-FINAL-P06-a | Answer provenance: official

> Assume for the RSA scheme, you are given $p = 17$ and $q = 31$.
>
> (a) Which one of the following integers can be used as the public key e: 2, 5 or 7? Explain. (Note: There is only one possible choice.)

Official Answer

Source: PDF physical page 11 (2015-final.pdf).

(a) p=17, q=31, so T=(17-1)*(31-1)=*16*30, only 7 is relatively prime to T, we will be able to find the inverse only for 7, therefore e=7

Supplementary Explanation

Answer. $e = 7$.

Derivation/proof. Compute $n = 17 \cdot 31 = 527$ and $\varphi(n) = 16 \cdot 30 = 480$. Then $\gcd(2, 480) = 2$, $\gcd(5, 480) = 5$, and $\gcd(7, 480) = 1$. Only 7 has an inverse modulo 480 and can serve as the exponent.

The inverse modulus is $\varphi(n) = (17 - 1)(31 - 1) = 480$.

N121

Cryptography

2014 Final | Question 3(c) | Source page 5 | Difficulty 2
Q-2014-FINAL-P03-c | Answer provenance: official

> Consider the following simplified version of the RSA algorithm for public cryptography:
>
> - (i) Bob's public key is a pair (n, e), where n is a prime number and e is a positive integer that is smaller than n and is relatively prime with $n - 1$.
> - (ii) Bob's private key is $d = e^{-1} \bmod (n - 1)$.
> - (iii) Alice encrypts a message m $(0 < m < n - 1)$ by calculating $c = m^e \bmod n$, and sends the ciphertext c to Bob.
> - (iv) Bob decrypts the ciphertext c by calculating $c^d \bmod n$.
>
> Suppose $n = 251$ and $e = 137$.
>
> (c) Is the system secure? Explain why or why not.

Official Answer

Source: PDF physical page 6 (2014-final.pdf).

(c) The system is not secure because the adversary can easily compute the private key d from the public key (n, e).

Supplementary Explanation

Answer. No; it is insecure because $n = 251$ is public and prime, so $n - 1 = 250$ is public and $d = e^{-1} \pmod{250}$ is easy to compute.

Derivation/proof. The public key exposes enough information to reproduce the private-key calculation directly. Real RSA relies on factoring a composite modulus being hard; that protection is absent here.

The attacker knows $\varphi(n) = n - 1 = 250$ without factorization, and can run extended Euclid on the public $e = 137$ to recover $d = 73$.

N122

Cryptography

2013 Final | Question 2(b) | Source page 4 | Difficulty 2
Q-2013-FINAL-P02-b | Answer provenance: official

> Suppose for applying RSA, $p = 11$, $q = 19$, and $e = 7$.
>
> (b) Calculate the private (secret) key d. Show the computational steps.

Official Answer

Source: PDF physical page 4 (2013-final.pdf).

(b) The private key is given by

$$d = e^{-1} \bmod (p-1)(q-1) = 7^{-1} \bmod 180.$$

To find the multiplicative inverse of 7 in Z_{180}, we run the extended GCD algorithm:

$$\begin{aligned} 180 &= 7 \times 25 + 5, \text{ thus } \gcd(7, 180) = \gcd(5, 7), \\ 7 &= 5 \times 1 + 2, \text{ thus } \gcd(5, 7) = \gcd(2, 5), \\ 5 &= 2 \times 2 + 1, \text{ thus } \gcd(2, 5) = \gcd(1, 2) = 1. \\ 2 &= 1 \times 2 + 0, \text{ thus } \gcd(1, 2) = 1. \end{aligned}$$

Now working backward, we get:

$$\begin{aligned} 1 &= 1 \times 1 + 2 \times 0 \\ &= 5 - 2 \times 2 \\ &= 5 - 2 \times (7 - 5 \times 1) = 3 \times 5 - 2 \times 7 \\ &= 3 \times (180 - 25 \times 7) - 2 \times 7 = 3 \times 180 - 77 \times 7. \end{aligned}$$

Thus $d = -77 \bmod 180 = 103$.

Supplementary Explanation

Answer. $d = 103$.

Derivation/proof. Here $\varphi(n) = (11-1)(19-1) = 180$. Euclid gives $180 = 25 \cdot 7 + 5$, $7 = 1 \cdot 5 + 2$, $5 = 2 \cdot 2 + 1$; back substitution gives $1 = 3 \cdot 180 - 77 \cdot 7$. Thus $7^{-1} \equiv -77 \equiv 103 \pmod{180}$.

Check: $7 \cdot 103 = 721 = 4 \cdot 180 + 1$.

N123

Cryptography

2025 Fall Final | Question 7(ii) | Source page 12 | Difficulty 3 | 3 marks
Q-2025F-FINAL-P07-ii | Answer provenance: official

> This problem is on the RSA algorithm for public key cryptogtaphy. To generate his keys, Bob uses two prime numbers $p = 7, q = 11$. So, $n = pq = 77$ and $T = (p-1)(q-1) = 60$.
>
> (ii) [3 pts] Assume that Bob chooses $e = 7$. Compute his private key d by running the extended Euclidean algorithm.

Official Answer

Source: physical PDF page 12 (COMP2711_F2025_Final.pdf).

(ii) Bob chooses $e = 7$. Compute d such that $7d \equiv 1 \pmod{60}$ using Extended Euclidean Algorithm:

$$\begin{aligned} 60 &= 8 \cdot 7 + 4 \\ 7 &= 1 \cdot 4 + 3 \\ 4 &= 1 \cdot 3 + 1 \\ 3 &= 3 \cdot 1 + 0. \end{aligned}$$

Back-substitute:

$$1 = 4 - 1 \cdot 3 \qquad 3 = 7 - 1 \cdot 4 \implies 1 = 4 - 1 \cdot (7 - 1 \cdot 4) = 2 \cdot 4 - 1 \cdot 7$$

$$4 = 60 - 8 \cdot 7 \implies 1 = 2 \cdot (60 - 8 \cdot 7) - 1 \cdot 7 = 2 \cdot 60 - 17 \cdot 7$$

$$\implies -17 \cdot 7 \equiv 1 \pmod{60} \implies d \equiv 43 \pmod{60} \qquad d = 43.$$

Supplementary Explanation

Answer. $d = 43$, or $d \equiv 43 \pmod{60}$.

Derivation/proof. The public exponent is coprime to the totient, and the private exponent is its multiplicative inverse modulo that totient. Extended Euclidean back-substitution constructs the inverse coefficient. The divisions are $60 = 8 \cdot 7 + 4$, $7 = 1 \cdot 4 + 3$, $4 = 1 \cdot 3 + 1$ and $3 = 3 \cdot 1 + 0$. Starting from the last nonzero remainder,

$$1 = 4 - 3 = 4 - (7 - 4) = 2 \cdot 4 - 7 = 2(60 - 8 \cdot 7) - 7 = 2 \cdot 60 - 17 \cdot 7.$$

Thus $7(-17) \equiv 1 \pmod{60}$. Add 60 to obtain the positive representative $d = -17 + 60 = 43$. Verification gives $7 \cdot 43 = 301 = 5 \cdot 60 + 1$, as required.

N124

Cryptography

2024 Fall Final | Question 8(b) | Source page 21 | Difficulty 3 | 4 marks
Q-2024F-FINAL-P08-b | Answer provenance: official

> Consider an RSA system in which $p = 5$ and $q = 19$. Answer the following questions:
>
> (b) Use Extended Euclidean algorithm and calculate d when $e = 23$. Show all the steps.

Official Answer

Source: physical PDF page 21 (COMP2711_F2024_Final.pdf).

> (b)
>
> $$72 = 3 \cdot 23 + 3, \qquad 23 = 7 \cdot 3 + 2, \qquad 3 = 1 \cdot 2 + 1, \qquad 2 = 2 \cdot 1 + 0.$$
>
> $$\begin{aligned} 1 &= 3 - 1 \cdot 2 \\ &= 3 - 1 \cdot (23 - 7 \cdot 3) \\ &= 8 \cdot 3 - 1 \cdot 23 \\ &= 8 \cdot (72 - 3 \cdot 23) - 1 \cdot 23 \\ &= 8 \cdot 72 - 25 \cdot 23. \end{aligned}$$
>
> Thus, $d \equiv -25 \pmod{72}$. Since d must be positive, we add 72 to -25: $d = 72 - 25 = 47$.

Supplementary Explanation

Answer. $d = 47$.

Derivation/proof. $\varphi(95) = 72$, so solve $23d \equiv 1 \pmod{72}$. Euclidean division gives $72 = 3 \cdot 23 + 3$, $23 = 7 \cdot 3 + 2$, $3 = 1 \cdot 2 + 1$, and $2 = 2 \cdot 1$. Back-substitution yields $1 = 3 - 2 = 3 - (23 - 7 \cdot 3) = 8 \cdot 3 - 23 = 8(72 - 3 \cdot 23) - 23 = 8 \cdot 72 - 25 \cdot 23$. Hence $d \equiv -25 \equiv 47 \pmod{72}$. Direct verification gives $23 \cdot 47 = 1081 = 15 \cdot 72 + 1$, so the positive decryption exponent is 47.

N125

Cryptography

2024 Fall Final | Question 8(c) | Source page 21 | Difficulty 3 | 4 marks
Q-2024F-FINAL-P08-c | Answer provenance: official

> Consider an RSA system in which $p = 5$ and $q = 19$. Answer the following questions:
>
> (c) Use repeated square method to calculate the encryption of $M = 3$. Show all the steps.

Official Answer

Source: physical PDF page 21 (COMP2711_F2024_Final.pdf).

> (c)
>
> $$3^2 \equiv 9 \pmod{95} \equiv 9$$
>
> $$3^4 \equiv 9^2 \pmod{95} \equiv 81 \pmod{95} \equiv 81 \equiv -14 \pmod{95}$$
>
> $$3^8 \equiv 81^2 \pmod{95} \equiv 14^2 \pmod{95} \equiv 196 \pmod{95} \equiv 6 \pmod{95}$$
>
> $$3^{16} \equiv 6^2 \pmod{95} \equiv 36 \pmod{95} \equiv 36$$
>
> Since $23 = 2^4 + 2^2 + 2^1 + 2^0$: We need to calcualte $3^{2^4} \cdot 3^{2^2} \cdot 3^2 \cdot 3 \equiv 72$.

Supplementary Explanation

Answer. The ciphertext is $C = 72$.

Derivation/proof. Use $n = 95, e = 23$, so $C \equiv 3^{23} \pmod{95}$. Repeated squaring gives $3^2 \equiv 9$, $3^4 \equiv 81 \equiv -14$, $3^8 \equiv (-14)^2 = 196 \equiv 6$, and $3^{16} \equiv 6^2 = 36$. Since $23 = 16 + 4 + 2 + 1$, $3^{23} \equiv 36 \cdot 81 \cdot 9 \cdot 3$. Reduce $36 \cdot 81 = 2916 \equiv 66$, then $66 \cdot 9 = 594 \equiv 24$, then $24 \cdot 3 = 72$. Thus the standard ciphertext representative is 72, with every reduction modulo 95.

N126

Cryptography

2024 Spring Midterm | Question 9 | Source page 13 | Difficulty 3 | 10 marks
Q-2024S-MID-P09 | Answer provenance: official

> We implement a "toy example" of the RSA cryptosystem by choosing two prime numbers $p = 23$ and $q = 37$. Assume that we use $(851, 17)$ as the public key in this example. Compute the private decryption key. Show all your calculations.

Official Answer

Source: physical PDF page 13 (COMP2711_S2024_Midterm.pdf).

> Answer. $n = p \cdot q = 23 \times 37 = 851$, $e = 17$. $t = \phi(n) = (p-1)(q-1) = (23-1) \times (37-1) = 792$. The private key d satisfies $e \cdot d \equiv 1 \pmod{t}$, which means $d = (17)^{-1} \pmod{792}$. Using the extended Euclidean algorithm.
>
> $$\begin{array}{lll} \gcd(792, 17) & 792 = 46 \times 17 + 10 & 10 = 792 - 46 \times 17 \quad (1) \\ \gcd(17, 10) & 17 = 1 \times 10 + 7 & 7 = 17 - 1 \times 10 \quad (2) \\ \gcd(10, 7) & 10 = 1 \times 7 + 3 & 3 = 10 - 1 \times 7 \quad (3) \\ \gcd(7, 3) & 7 = 2 \times 3 + 1 & 1 = 7 - 2 \times 3 \quad (4) \end{array}$$
>
> Substituting:
>
> $$\begin{aligned} 1 &= 7 - 2 \times 3 && \text{(By (4))} \\ &= 7 - 2 \times (10 - 1 \times 7) && \text{(By (3))} \\ &= (-2) \times 10 + 3 \times 7 \\ &= (-2) \times 10 + 3 \times (17 - 1 \times 10) && \text{(By (2))} \\ &= 3 \times 17 + (-5) \times 10 \\ &= 3 \times 17 + (-5) \times (792 - 46 \times 17) && \text{(By (1))} \\ &= (-5) \times 792 + 233 \times 17. \end{aligned}$$
>
> Therefore, the private key $d = 233$. Remark: d must be a positive number. Considering that we are using $C^d \bmod n$ to compute the original message M, it is necessary to compute a positive multiplicative inverse.

Supplementary Explanation

Answer. $d = 233$; the private key may be written $(851, 233)$.

Derivation/proof. $n = 23 \cdot 37 = 851$ and $\varphi(n) = 22 \cdot 36 = 792$. Solve $17d \equiv 1 \pmod{792}$. The Euclidean divisions are $792 = 46 \cdot 17 + 10$, $17 = 10 + 7$, $10 = 7 + 3$, and $7 = 2 \cdot 3 + 1$. Back-substitution gives $1 = 7 - 2(10 - 7) = 3 \cdot 7 - 2 \cdot 10 = 3 \cdot 17 - 5 \cdot 10 = 233 \cdot 17 - 5 \cdot 792$. Hence $d \equiv 233 \pmod{792}$ and we choose the positive representative 233. Directly, $17 \cdot 233 = 3961 = 5 \cdot 792 + 1$, confirming the inverse.

N127

Cryptography

2023 Fall Final | Question 5(a) | Source page 8 | Difficulty 3 | 5 marks
Q-2023F-FINAL-P05-a | Answer provenance: official

> This problem is on the RSA algorithm for public key cryptogtaphy. To generate his keys, Bob samples two prime numbers $p = 17, q = 19$. So, $n = pq = 323$ and $T = 288$.
>
> (a) Assume that Bob chooses $e = 23$.
>
> Assume that Bob chooses $e = 23$. Compute his private key d by running the extended GCD algorithm.

Official Answer

Source: physical PDF page 8 (COMP2711_F2023_Final.pdf).

> (a) Find the inverse of $e = 23$ in $\mathbb{Z}_{288}$:
>
> $$288 = 23 \cdot 12 + 12, \quad 23 = 12 \cdot 1 + 11, \quad 12 = 11 \cdot 1 + 1, \quad 11 = 1 \cdot 11 + 0.$$
>
> Hence we get:
>
> $$\begin{aligned} 1 &= 12 - 11 \cdot 1 \\ &= 12 - (23 - 1 \cdot 12) \cdot 1 \\ &= 12 \cdot 2 - 23 \cdot 1 \\ &= (288 - 23 \cdot 12) \cdot 2 - 23 \cdot 1 \\ &= 288 \cdot 2 + 23 \cdot (-25). \end{aligned}$$
>
> The private key $d = -25 \equiv 263 \pmod{288}$.

Supplementary Explanation

Answer. $d = 263 \pmod{288}$.

Derivation/proof. The divisions are $288 = 12 \cdot 23 + 12$, $23 = 12 + 11$, and $12 = 11 + 1$. Back substitution gives

$$1 = 12 - 11 = 12 - (23 - 12) = 2 \cdot 12 - 23 = 2(288 - 12 \cdot 23) - 23 = 2 \cdot 288 - 25 \cdot 23.$$

Thus $23^{-1} \equiv -25 \equiv 263 \pmod{288}$. Verify $23 \cdot 263 = 6049 = 21 \cdot 288 + 1$. This is the RSA private exponent because $T = (17 - 1)(19 - 1) = 288$.

The first division is $288 = 12 \cdot 23 + 12$, and the final identity is $1 = 2 \cdot 288 - 25 \cdot 23$. Verification gives $23 \cdot 263 = 6049 = 21 \cdot 288 + 1$.

N128

Cryptography

2023 Spring Midterm | Question 7(a) | Source page 10 | Difficulty 3 | 7 marks
Q-2023S-MID-P07-a | Answer provenance: official

> Suppose we implement the RSA cryptosystem by choosing two prime numbers $p = 31$ and $q = 43$. We further choose a number $e = 23$, which is relatively prime to $(p-1)(q-1) = 1260$.
>
> (a) Suppose we implement the RSA cryptosystem by choosing two prime numbers $p = 31$ and $q = 43$. We further choose a number $e = 23$, which is relatively prime to $(p-1)(q-1) = 1260$. What is the value of the secret key d? Show your calculations.

Official Answer

Source: physical PDF page 10 (COMP2711_S2023_Midterm.pdf).

> (a) Find the inverse of $e = 23$ in $\mathbb{Z}_{1260}$:
>
> $$\begin{aligned} 1260 &= 23 \cdot 54 + 18 \\ 23 &= 18 \cdot 1 + 5 \\ 18 &= 5 \cdot 3 + 3 \\ 5 &= 3 \cdot 1 + 2 \\ 3 &= 2 \cdot 1 + 1 \end{aligned}$$
>
> Hence we get:
>
> $$\begin{aligned} 1 &= 3 - 2 \cdot 1 \\ &= 3 - (5 - 3 \cdot 1) \cdot 1 \\ &= 3 \cdot 2 - 5 \cdot 1 \\ &= (18 - 5 \cdot 3) \cdot 2 - 5 \cdot 1 \\ &= 18 \cdot 2 - 5 \cdot 7 \\ &= 18 \cdot 2 - (23 - 18 \cdot 1) \cdot 7 \\ &= 18 \cdot 9 - 23 \cdot 7 \\ &= (1260 - 23 \cdot 54) \cdot 9 - 23 \cdot 7 \\ &= 1260 \cdot 9 - 23 \cdot 493 \end{aligned}$$
>
> The secret $d = -493 \equiv 767 \pmod{1260}$

Supplementary Explanation

Answer. $d = 767 \pmod{1260}$.

Derivation/proof. Euclid gives $1260 = 54 \cdot 23 + 18$, $23 = 18 + 5$, $18 = 3 \cdot 5 + 3$, $5 = 3 + 2$, and $3 = 2 + 1$. Back substitution yields

$$1 = 3 - 2 = 2 \cdot 3 - 5 = 2 \cdot 18 - 7 \cdot 5 = 9 \cdot 18 - 7 \cdot 23 = 9 \cdot 1260 - 493 \cdot 23.$$

Thus $23^{-1} \equiv -493 \equiv 767 \pmod{1260}$. The check $23 \cdot 767 = 17641 = 14 \cdot 1260 + 1$ confirms the private exponent.

The inverse equation is $23d \equiv 1 \pmod{\varphi(n)}$ with $\varphi(n) = (31-1)(43-1) = 1260$.

N129

Cryptography

2023 Spring Midterm | Question 7(b) | Source page 10 | Difficulty 3 | 8 marks
Q-2023S-MID-P07-b | Answer provenance: official

> Suppose we implement the RSA cryptosystem by choosing two prime numbers $p = 31$ and $q = 43$. We further choose a number $e = 23$, which is relatively prime to $(p-1)(q-1) = 1260$.
>
> (b) Suppose $m = 2$. Alice encodes her message m using her private key d to compute $x = m^d \pmod{n}$, where d is derived from (a). Show the computational steps.

Official Answer

Source: physical PDF page 11 (COMP2711_S2023_Midterm.pdf).

> (b) $p \cdot q = 1333$
>
> $$\begin{aligned}
> 2^{2^0} \bmod 1333 &= 2 \\
> 2^{2^1} \bmod 1333 &= 2^2 \bmod 1333 = 4 \\
> 2^{2^2} \bmod 1333 &= 4^2 \bmod 1333 = 16 \\
> 2^{2^3} \bmod 1333 &= 16^2 \bmod 1333 = 256 \\
> 2^{2^4} \bmod 1333 &= 256^2 \bmod 1333 = 219 \\
> 2^{2^5} \bmod 1333 &= 219^2 \bmod 1333 = -27 \\
> 2^{2^6} \bmod 1333 &= (-27)^2 \bmod 1333 = 729 \\
> 2^{2^7} \bmod 1333 &= 729^2 \bmod 1333 = 907 \\
> 2^{2^8} \bmod 1333 &= 907^2 \bmod 1333 = 188 \\
> 2^{2^9} \bmod 1333 &= 188^2 \bmod 1333 = 686
> \end{aligned}$$
>
> $$767 = 2^9 + 2^7 + 2^6 + 2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0$$
>
> $$\begin{aligned}
> 2^{767} &\equiv 2^{2^9} \cdot 2^{2^7} \cdot 2^{2^6} \cdot 2^{2^5} \cdot 2^{2^4} \cdot 2^{2^3} \cdot 2^{2^2} \cdot 2^{2^1} \cdot 2^{2^0} \\
> &\equiv 686 \cdot 907 \cdot 729 \cdot (-27) \cdot 219 \cdot 256 \cdot 16 \cdot 4 \cdot 2 \\
> &\equiv (686 \cdot 907 \cdot 729 \bmod 1333) \cdot (-27 \cdot 219 \cdot 256 \bmod 1333) \\
> &\qquad \cdot (16 \cdot 4 \cdot 2 \bmod 1333) \\
> &\equiv 16 \cdot 560 \cdot 128 \\
> &\equiv 500 \pmod{1333}.
> \end{aligned}$$

Supplementary Explanation

Answer. $x = 500 \pmod{1333}$.

Derivation/proof. Here $n = 31 \cdot 43 = 1333$ and $d = 767 = 512 + 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1$. Repeated squaring gives residues of $2^{1,2,4,8,16,32,64,128,256,512}$ as

$$2, 4, 16, 256, 219, -27, 729, 907, 188, 686 \pmod{1333}.$$

Multiply the entries for the binary expansion of 767:

$$2^{767} \equiv 686 \cdot 907 \cdot 729 \cdot (-27) \cdot 219 \cdot 256 \cdot 16 \cdot 4 \cdot 2 \equiv 16 \cdot 560 \cdot 128 \equiv 500 \pmod{1333}.$$

An independent CRT check gives $500 \equiv 4 \pmod{31}$ and $500 \equiv 27 \pmod{43}$, agreeing with 2^{767} in each prime modulus.

The negative value -27 is a convenient congruent representative; its least nonnegative residue is 1306.

N130

Cryptography

2022 Midterm | Question 8(a) | Source page 3 | Difficulty 3 | 5 marks
Q-2022-MID-P08-a | Answer provenance: official

> Consider RSA encryption with $p = 7, q = 19$ and $e = 29$. Your solution should be numbers.
>
> What is the encrypted message for plaintext $m = 12$?

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (a) $c = 12^e \bmod pq$, which is $12^{29} \bmod 133$. This can be solved by repeated squaring. Set $I_i \equiv 12^{2^i} \pmod{133}$.
>
> $$\begin{aligned} I_0 &= 12 \\ I_1 &= I_0 \cdot I_0 \bmod 133 = 11 \\ I_2 &= I_1 \cdot I_1 \bmod 133 = 121 \\ I_3 &= I_2 \cdot I_2 \bmod 133 = 11 \\ I_4 &= I_3 \cdot I_3 \bmod 133 = 121 \end{aligned}$$
>
> Note that $12^{29} = 12^{16} \cdot 12^8 \cdot 12^4 \cdot 12 = 12^{2^4} \cdot 12^{2^3} \cdot 12^{2^2} \cdot 12^{2^0}$.
>
> Thus, $12^{29} \equiv (I_4 I_3 I_2 I_0) \pmod{133} \equiv (12 \cdot 11 \cdot 121^2) \pmod{133} \equiv 122 \pmod{133}$.
>
> So the encrypted message is 122.

Supplementary Explanation

Answer. 122.

Derivation/proof. The modulus is n=7·19=133 and ciphertext is 12^29 mod 133. Repeated-square residues at exponents1,2,4,8,16 are 12,11,121,11,121: $12^2 \equiv 11, 11^2 = 121, 121^2 \equiv 11$. Since29=16+8+4+1, multiply121·11·121·12. Reduce121·11=1331≡1, leaving 121·12=1452≡122 mod 133. This is in the required message residue range.

The products are $1331 = 10 \cdot 133 + 1$ and $1452 = 10 \cdot 133 + 122$. The repeated square $121^2 = 14641 \equiv 11 \pmod{133}$ explains the alternating residues.

N131

Cryptography

2022 Midterm | Question 8(b) | Source page 3 | Difficulty 3 | 8 marks
Q-2022-MID-P08-b | Answer provenance: official

> Consider RSA encryption with $p = 7, q = 19$ and $e = 29$. Your solution should be numbers.
>
> Compute the private key d. Show your steps.

Official Answer

Source: physical PDF page 4 (2022-midterm.pdf).

> (b) The private key d is the multiplicative inverse of e modulo $(p-1)(q-1) = 108$. First find $\gcd(108, 29)$:
>
> $$\begin{aligned} 108 &= 29 \cdot 3 + 21 \\ 29 &= 21 \cdot 1 + 8 \\ 21 &= 8 \cdot 2 + 5 \\ 8 &= 5 \cdot 1 + 3 \\ 5 &= 3 \cdot 1 + 2 \\ 3 &= 2 \cdot 1 + 1 \end{aligned}$$
>
> So, $\gcd(108, 29) = 1$.
>
> Rewriting and substituting:
>
> $$\begin{aligned} 1 &= 3 - 2 \\ &= 3 - (5 - 3) \\ &= -5 + 2 \cdot 3 \\ &= -5 + 2 \cdot (8 - 5) \\ &= 2 \cdot 8 - 3 \cdot 5 \\ &= 2 \cdot 8 - 3 \cdot (21 - 2 \cdot 8) \\ &= -3 \cdot 21 + 8 \cdot 8 \\ &= -3 \cdot 21 + 8 \cdot (29 - 21) \\ &= 8 \cdot 29 - 11 \cdot 21 \\ &= 8 \cdot 29 - 11 \cdot (108 - 3 \cdot 29) \\ &= -11 \cdot 108 + 41 \cdot 29 \end{aligned}$$
>
> Multiplicative inverse of 29 modulo 108 is 41, so $d = 41$.

Supplementary Explanation

Answer. 41.

Derivation/proof. φ(n)=6·18=108. Euclid:108=3·29+21;29=21+8;21=2·8+5;8=5+3;5=3+2;3=2+1. Back-substitute $1 = 3 - 2 = 2 \cdot 3 - 5 = 2 \cdot 8 - 3 \cdot 5 = 8 \cdot 8 - 3 \cdot 21 = 8 \cdot 29 - 11 \cdot 21 = 41 \cdot 29 - 11 \cdot 108$. Hence29d≡1 mod 108 gives d=41. Check29·41=1189=11·108+1.

The complete intermediate substitutions are

$$1 = 3 - 2 = 3 - (5 - 3) = 2 \cdot 3 - 5 = 2(8 - 5) - 5 = 2 \cdot 8 - 3 \cdot 5$$
$$= 2 \cdot 8 - 3(21 - 2 \cdot 8) = 8 \cdot 8 - 3 \cdot 21$$
$$= 8(29 - 21) - 3 \cdot 21 = 8 \cdot 29 - 11 \cdot 21$$
$$= 8 \cdot 29 - 11(108 - 3 \cdot 29) = 41 \cdot 29 - 11 \cdot 108.$$

The check is $1189 = 11 \cdot 108 + 1$.

N132

Cryptography

2021 Midterm | Question 5 | Source page 2 | Difficulty 3 | 10 marks
Q-2021-MID-P05 | Answer provenance: official

> Alice and Bob share a key using the Diffie-Hellman key exchange algorithm with $a = 17, p = 31, k_1 = 34$ and $k_2 = 41$. What is the shared key? Note that the shared key must be in $\mathbb{Z}_p$. Use Fermat's Little Theorem and the repeated squaring method to simplify the calculation. Show all the computational steps.

Official Answer

Source: physical PDF page 2 (2021-midterm.pdf).

> The shared key is $a^{k_1 k_2} \bmod p = 17^{34 \cdot 41} \bmod 31$.
>
> By Fermat's Little Theorem, $17^{34 \cdot 41} \equiv 17^{34 \cdot 41 \bmod 30} \equiv 17^{4 \cdot 11 \bmod 30} \equiv 17^{14} \pmod{31}$.
>
> We compute the shared key by repeated squaring method.
>
> $$\begin{aligned} 17^{2^1} \bmod 31 &= 10 \\ 17^{2^2} \bmod 31 &= 10^2 \bmod 31 = 7 \\ 17^{2^3} \bmod 31 &= 7^2 \bmod 31 = 18 \end{aligned}$$
>
> Note that $17^{14} = 17^{2^1 + 2^2 + 2^3}$. So,
>
> $$\begin{aligned} 17^{14} &\equiv 17^{2^1} 17^{2^2} 17^{2^3} \\ &\equiv 10 \cdot 7 \cdot 18 \\ &\equiv 20 \pmod{31}. \end{aligned}$$

Supplementary Explanation

Answer. 20.

Derivation/proof. Both parties obtain $a^{k_1 k_2} \bmod p$, here $17^{1394} \bmod 31$. Since 31 is prime and 17 is nonzero modulo 31, Fermat gives period 30; 1394=14 mod 30. Repeated squaring gives $17^2 \equiv 289 \equiv 10$, $17^4 \equiv 10^2 \equiv 7$, $17^8 \equiv 7^2 \equiv 18$ modulo 31. As 14=8+4+2, multiply $18 \cdot 7 \cdot 10 = 1260 \equiv 20$. This lies in the required residue range 0,...,30.

Explicitly, $1394 = 46 \cdot 30 + 14$ and $17^{30} \equiv 1 \pmod{31}$. The base is coprime to the prime modulus, which is essential for Fermat reduction.

The final square and reduction are $17^8 \equiv 49 \equiv 18$ and $1260 \equiv 20 \pmod{31}$.

N133

Cryptography

2021 Midterm | Question 6(a) | Source page 3 | Difficulty 3 | 4 marks
Q-2021-MID-P06-a | Answer provenance: official

> Consider the RSA encryption with parameters $p = 443, q = 211$.
>
> List out all possible public keys (n, e) so that e in $[120, 130]$. Justification is not necessary.

Official Answer

Source: physical PDF page 3 (2021-midterm.pdf).

> (a) $T = (p-1)(q-1) = 442 \cdot 210 = 2 \cdot 2 \cdot 3 \cdot 5 \cdot 7 \cdot 13 \cdot 17 = 92820$. We need the values e in $[120, 130]$ that are relatively prime to T. The values cannot contain any factor in $\{2, 3, 5, 7, 13, 17\}$. Therefore, 121 and 127 are the only possible values of e.

Supplementary Explanation

Answer. $(93473, 121)$ and $(93473, 127)$.

Derivation/proof. Compute $n = 443 \cdot 211 = 93473$ and $\varphi(n) = 442 \cdot 210 = 92820 = 2^2 \cdot 3 \cdot 5 \cdot 7 \cdot 13 \cdot 17$. An exponent must be coprime to the totient. Even candidates fail; 123,129 share3; 125 shares5. Of the remaining odd values 121=11² and 127, neither contains a forbidden factor, so exactly these two qualify. Include n in each requested key pair.

N134

Cryptography

2021 Midterm | Question 6(b) | Source page 3 | Difficulty 3 | 8 marks
Q-2021-MID-P06-b | Answer provenance: official

> Consider the RSA encryption with parameters $p = 443, q = 211$.
>
> Suppose you select the smallest value e in your answer of (a) to be the public key (n, e). Compute the corresponding private key d. Show all your steps.

Official Answer

Source: physical PDF page 3 (2021-midterm.pdf).

> (b) The private key should satisfy $(ed) \bmod T = 1$. i.e. d is the multiplicative inverse of e in $\mathbb{Z}_T$. Run the extended GCD algorithm to find d:
>
> $$\begin{aligned} 92820 &= 121 \cdot 767 + 13 \\ 121 &= 13 \cdot 9 + 4 \\ 13 &= 4 \cdot 3 + 1 \\ 4 &= 1 \cdot 4 + 0 \end{aligned}$$
>
> Then,
>
> $$\begin{aligned} 1 &= 13 - 4 \cdot 3 \\ &= 13 - (121 - 13 \cdot 9) \cdot 3 \\ &= 13 \cdot 28 - 121 \cdot 3 \\ &= (92820 - 121 \cdot 767) \cdot 28 - 121 \cdot 3 \\ &= 92820 \cdot 28 - 121 \cdot 21479. \end{aligned}$$
>
> Thus, $d = -21479 \bmod 92820 = 71341$.

Supplementary Explanation

Answer. $d = 71341$.

Derivation/proof. For e= 121 and totient 92820, Euclid gives $92820 = 767 \cdot 121 + 13$, $121 = 9 \cdot 13 + 4$, $13 = 3 \cdot 4 + 1$. Thus $1 = 13 - 3(121 - 9 \cdot 13) = 28 \cdot 13 - 3 \cdot 121 = 28 \cdot 92820 - 21479 \cdot 121$. The inverse coefficient -21479 reduces to 71341 modulo 92820. Verify $121 \cdot 71341 = 8632261 = 93 \cdot 92820 + 1$.

The final Euclidean division is $4 = 4 \cdot 1 + 0$. The inverse is taken modulo 92820, not the public modulus 93473.

The inverse coefficient satisfies $-21479 \equiv 71341 \pmod{92820}$.

N135

Cryptography

2020 Midterm | Question 10 | Source page 5 | Difficulty 3 | 10 marks
Q-2020-MID-P10 | Answer provenance: official

> Recall the digital signature scheme based on RSA. Suppose your public key is (n, e) and private key is d, and you want to sign a message $x \in \mathbb{Z}_n$. You release both x and $C = x^d \bmod n$. People can then verify your signature by checking $C^e \bmod n = x$.
>
> However, you should be careful not to just sign any message people give you. Suppose an attacker asks you to sign another message $y = r^e x \bmod n$ where $r \neq 1$ is a number chosen by the attacker, and you sign it (i.e., release $y^d \bmod n$). Then the attacker can forge your signature on x, i.e., compute C without knowing d. Show how the attacker can do this. (This is known as a chosen-message-attack.)

Official Answer

Source: physical PDF page 5 (2020-midterm.pdf).

$y^d \equiv (r^e x)^d \equiv r^{ed} \cdot x^d \equiv r \cdot x^d \pmod{n}$. So the attacker can just find r^{-1} (in $\mathbb{Z}_n$), and then compute $y^d \cdot r^{-1} \equiv x^d \equiv C \pmod{n}$.

Supplementary Explanation

Answer. Choose $r \neq 1$ with $\gcd(r, n) = 1$, obtain $s = y^d \bmod n$, and output $C = sr^{-1} \bmod n$.

Derivation/proof. Because the source lets the attacker choose r, choose a unit modulo n rather than an arbitrary nonunit: compute gcd(r,n) with the public n and retain a value coprime to n. For a standard RSA modulus n>2, n−1 is already such a unit distinct from 1. Extended Euclid computes t with $rt \equiv 1 \pmod{n}$ using no private information. The attacker computes $y = r^e x \bmod n$ with the public e and requests its signature s. Since $ed = 1 + k\varphi(n)$ and r is a unit, Euler's theorem gives $r^{ed} \equiv r \pmod{n}$. Hence $s \equiv (r^e x)^d \equiv r^{ed} x^d \equiv r x^d \pmod{n}$ and $st \equiv x^d = C \pmod{n}$. No inverse of x is needed, so x itself need not be a unit. As an independent numerical check, take n=55, e= 3, d=27, x=7 and r=2. Then y=1, s=1, and $2^{-1} \equiv 28 \pmod{55}$, so the forged C is 28; indeed $7^{27} \equiv 28$ and $28^3 \equiv 7 \pmod{55}$.

A numerical check is $n = 55, e = 3, d = 27, x = 7, r = 2$. Then $y = 1, s = 1, t = 28$, yielding $C = 28$, with $7^{27} \bmod 55 = 28$ and $28^3 \bmod 55 = 7$.

Multiplying the returned signature by the publicly computed inverse gives $st \equiv r x^d t \equiv x^d = C \pmod{n}$. The cancellation uses $rt \equiv 1$, so it requires a unit r but never requires an inverse for x.

N136

Cryptography

2019 Midterm | Question 11(b) | Source page 13 | Difficulty 3 | 6 marks
Q-2019-MID-P11-b | Answer provenance: official

> Consider Bob is using RSA algorithm for public key cryptography. Bob starts by picking the two prime numbers $p = 43$ and $q = 23$. So, $n = pq = 989$.
>
> (b) Suppose Bob chooses $e = 53$. What is the private key d? Show the steps of the extended Euclidean algorithm.

Official Answer

Source: physical PDF page 13 (2019-midterm.pdf).

The private key should satisfy $ed \equiv 1 \pmod{T}$, i.e., d is the multiplicative inverse of e in $\mathbb{Z}_T$. Run the extended Euclidean algorithm to find d:

$$\begin{aligned} 924 &= 53 \cdot 17 + 23 \\ 53 &= 23 \cdot 2 + 7 \\ 23 &= 7 \cdot 3 + 2 \\ 7 &= 2 \cdot 3 + 1 \\ 2 &= 1 \cdot 2 + 0. \end{aligned}$$

Then,

$$\begin{aligned} 1 &= 7 - 2 \cdot 3 \\ &= 7 - (23 - 7 \cdot 3) \cdot 3 \\ &= 7 \cdot 10 + 23 \cdot (-3) \\ &= (53 - 23 \cdot 2) \cdot 10 + 23 \cdot (-3) \\ &= 23 \cdot (-23) + 53 \cdot 10 \\ &= (924 - 53 \cdot 17) \cdot (-23) + 53 \cdot 10 \\ &= 53 \cdot 401 + 924 \cdot (-23). \end{aligned}$$

Thus, $d = 401$.

Supplementary Explanation

Answer. $d = 401$.

Derivation/proof. Use $924 = 17 \cdot 53 + 23$, $53 = 2 \cdot 23 + 7$, $23 = 3 \cdot 7 + 2$, $7 = 3 \cdot 2 + 1$, $2 = 2 \cdot 1$. Back substitute: $1 = 7 - 3 \cdot 2 = 10 \cdot 7 - 3 \cdot 23 = 10 \cdot 53 - 23 \cdot 23 = 401 \cdot 53 - 23 \cdot 924$. Hence $53^{-1} \equiv 401 \pmod{924}$. Check $53 \cdot 401 = 21253 = 23 \cdot 924 + 1$.

N137

Cryptography

2018 Midterm | Question 6(c) | Source page 8 | Difficulty 3
Q-2018-MID-P06-c | Answer provenance: official

> Consider the RSA encryption with parameters $p = 19, q = 37$.
>
> (c) Assume you want to encrypt the message 3. What is the resulting encrypted ciphertext? Use repeated squaring and show all intermediate steps.

Official Answer

Source: physical PDF page 8 (2018-midterm.pdf).

The encrypted ciphertext is computed as

$$3^e \bmod pq = 3^{11} \bmod 703.$$

To compute this, we first compute

$$\begin{aligned} 3^2 \bmod 703 &= 3 \cdot 3 \bmod 703 = 9 \\ 3^4 \bmod 703 &= 9 \cdot 9 \bmod 703 = 81 \\ 3^8 \bmod 703 &= 81 \cdot 81 \bmod 703 = 234. \end{aligned}$$

It holds that $11 = 8 + 2 + 1$. Hence

$$\begin{aligned} 3^{11} \bmod 703 = 3^{1+2+8} \bmod 703 &= 3 \cdot 3^2 \cdot 3^8 \bmod 703 \\ &= 3 \cdot 9 \cdot 234 \bmod 703 = 694. \end{aligned}$$

No marks for this part if the answer is correct but there are no steps, because it is easy to obtain the answer using a calculator.

Supplementary Explanation

Answer. The ciphertext is 694.

Derivation/proof. The modulus is $n = 19 \cdot 37 = 703$ and $e = 11$. Compute $3^2 \equiv 9$, $3^4 \equiv 81$, $3^8 \equiv 81^2 = 6561 \equiv 234 \pmod{703}$. Since $11 = 8 + 2 + 1$, $3^{11} \equiv 234 \cdot 9 \cdot 3 = 6318 \equiv 694 \pmod{703}$. Direct powering agrees, and decrypting 694^{59} modulo 703 returns 3.

The final product satisfies $6318 = 8 \cdot 703 + 694$, and $694^{59} \bmod 703 = 3$ checks decryption.

N138

Cryptography

2017 Final | Question 4(a) | Source page 6 | Difficulty 3 | 5 marks
Q-2017-FINAL-P04-a | Answer provenance: official

> Recall that in RSA, we need to choose two primes p and q, and the public key is (n, e) where $n = pq$. Let $T = (p-1)(q-1)$. The private key is d, where $ed \equiv 1 \pmod{T}$. The encryption of x is $x^e \bmod n$. This means that d must be an integer in $\mathbb{Z}_T$ that is relatively prime with T. We can find d using the following algorithm:
>
> 1. Randomly pick $d \in \mathbb{Z}_T$.
> 2. If d and T are relatively prime, then return d. If they are not, then pick a new d and loop from step 1.
>
> Assume that we will never pick the same d twice.
>
> (a) Suppose that $p = 13, q = 61$. In the worst case, what is the number of times we will execute step 1 (that is, the number of times we have to pick a random d)?

Official Answer

Source: physical PDF page 6 (2017-final.pdf); physical PDF page 7 (2017-final.pdf).

$p - 1 = 2^2 \cdot 3, q - 1 = 2^2 \cdot 3 \cdot 5$. So this is equivalent to finding the number of integers from $\{0, 1, \cdots, (p-1)(q-1) - 1 = 719\}$ that are relatively prime to 2, 3, and 5. Using the inclusion exclusion principle, we first find the number of integers that are multiples of $2 : 360, 3 : 240, 5 : 144, 6 : 120, 10 : 72, 15 : 48, 30 : 24$. The number of integers that are not relatively prime to 2, 3, and 5 would then be $360 + 240 + 144 - 120 - 72 - 48 + 24 = 528$. By the pigeonhole principle, in the worst case it will take 529 tries to find a proper d.

Supplementary Explanation

Answer. 529 picks.

Derivation/proof. Here $T = 12 \cdot 60 = 720 = 2^4 3^2 5$. Bad candidates in $\{0, \ldots, 719\}$ are multiples of 2, 3 or 5. Their union has size $360 + 240 + 144 - 120 - 72 - 48 + 24 = 528$ by inclusion-exclusion (zero is included in each set). All 528 may be picked first without replacement; the next pick is then good. This ordering also proves the bound is attainable.

N139

Cryptography

2015 Final | Question 6(b) | Source page 11 | Difficulty 3 | 5 marks
Q-2015-FINAL-P06-b | Answer provenance: official

> Assume for the RSA scheme, you are given $p = 17$ and $q = 31$.
>
> (b) Find the private key d using the extended GCD algorithm for the public key e picked at Part (a). Show all steps of your calculations.

Official Answer

Source: PDF physical page 11 (2015-final.pdf).

(b) We use the extended GCD algorithm to find the d which is the inverse of e in modulo T, $T = (p - 1)(q - 1) = 480$. First we find the greatest common divisor of T and e

$$\begin{aligned} 480 &= 7 \times 68 + 4 \\ 7 &= 4 \times 1 + 3 \\ 4 &= 3 \times 1 + 1 \\ 3 &= 1 \times 3 + 0 \end{aligned}$$

Therefore gcd(480,7)=1. Now we roll back to get the inverse of 7

$$\begin{aligned} 1 &= 1 \times 1 + 3 \times 0 \\ &= 4 - 3 \\ &= 4 - (7 - 4 \times 1) \\ &= 4 \times 2 - 7 \\ &= (480 - 7 \times 68) \times 2 - 7 \\ &= 480 \times 2 + 7 \times (-137) \end{aligned}$$

Therefore $d = -137 \bmod 480 = 343$.

Supplementary Explanation

Answer. $d = 343$.

Derivation/proof. Euclid gives $480 = 68 \cdot 7 + 4$, $7 = 4 + 3$, $4 = 3 + 1$. Back-substitute: $1 = 4 - 3 = 4 - (7 - 4) = 2 \cdot 4 - 7 = 2(480 - 68 \cdot 7) - 7 = 2 \cdot 480 - 137 \cdot 7$. Thus $7^{-1} \equiv -137 \equiv 343 \pmod{480}$. Check: $7 \cdot 343 = 2401 = 5 \cdot 480 + 1$.

N140

Cryptography

2015 Final | Question 6(c) | Source page 11 | Difficulty 3 | 6 marks
Q-2015-FINAL-P06-c | Answer provenance: official

> Assume for the RSA scheme, you are given $p = 17$ and $q = 31$.
>
> (c) Suppose the ciphertext is 2. Decrypt it to find the plaintext. Show all steps of your calculations.
>
> You may find the following equalities helpful: $171^2 \bmod 527 = 256$, $256^2 \bmod 527 = 188$, $35 \times 171 \bmod 527 = 188$, and $188^2 \bmod 527 = 35$.

Official Answer

Source: PDF physical page 12 (2015-final.pdf).

(c) To decrypt the ciphertext 2, we need to do the following:

$$2^{343} \bmod (17 \times 31)$$

$$2^{343} \bmod 527$$

Since $343 = 101010111_{(2)}$ (i.e., $343 = 2^8 + 2^6 + 2^4 + 2^2 + 2 + 1$), we need to get I_0, I_1, I_2, I_4, I_6 and I_8:

I_0	$2 \bmod 527 = 2$
I_1	$I_0 \times I_0 \bmod 527 = 4$
I_2	$I_1 \times I_1 \bmod 527 = 16$
I_3	$I_2 \times I_2 \bmod 527 = 256$
I_4	$I_3 \times I_3 \bmod 527 = 188$ (given)
I_5	$I_4 \times I_4 \bmod 527 = 35$ (given)
I_6	$I_5 \times I_5 \bmod 527 = 171$
I_7	$I_6 \times I_6 \bmod 527 = 256$ (given)
I_8	$I_7 \times I_7 \bmod 527 = 188$ (given)

Now we have:

$$\begin{aligned} 2^{343} \bmod 527 &= (I_8 \times I_6 \times I_4 \times I_2 \times I_1 \times I_0) \bmod 527) \\ &= (188 \times 171 \times 188 \times 16 \times 4 \times 2) \bmod 527 \\ &= ((188^2 \bmod 527) \times 171 \times 16 \times 4 \times 2) \bmod 527 \\ &= ((35 \times 171 \bmod 527) \times 16 \times 4 \times 2) \bmod 527 \quad \text{(given)} \\ &= (188 \times 16 \times 4 \times 2) \bmod 527 \quad \text{(given)} \\ &= 24064 \bmod 527 \\ &= 349 \end{aligned}$$

Therefore the plaintext is 349.

Supplementary Explanation

Answer. The plaintext is 349.

Derivation/proof. Write $343 = 256 + 64 + 16 + 4 + 2 + 1$. The residues of 2^{2^j} for $j = 0, \ldots, 8$ are $2, 4, 16, 256, 188, 35, 171, 256, 188$. Hence $2^{343} \equiv 188 \cdot 171 \cdot 188 \cdot 16 \cdot 4 \cdot 2$. Use $188^2 \equiv 35$ and $35 \cdot 171 \equiv 188$ to reduce this to $188 \cdot 16 \cdot 4 \cdot 2 = 24064 \equiv 349 \pmod{527}$. Re-encryption gives $349^7 \bmod 527 = 2$, independently checking the plaintext.

N141

Cryptography

2014 Final | Question 3(a) | Source page 5 | Difficulty 3
Q-2014-FINAL-P03-a | Answer provenance: official

Consider the following simplified version of the RSA algorithm for public cryptography:

- (i) Bob's public key is a pair (n, e), where n is a prime number and e is a positive integer that is smaller than n and is relatively prime with $n-1$.
- (ii) Bob's private key is $d = e^{-1} \bmod (n-1)$.
- (iii) Alice encrypts a message m ($0 < m < n-1$) by calculating $c = m^e \bmod n$, and sends the ciphertext c to Bob.
- (iv) Bob decrypts the ciphertext c by calculating $c^d \bmod n$.

Suppose $n = 251$ and $e = 137$.

(a) Calculate d using the extended GCD algorithm. Show the computational steps.

Official Answer

Source: PDF physical page 5 (2014-final.pdf).

(a) Repeatedly using Euclid's Division Theorem, we get

$$\begin{aligned} 250 &= 1 \cdot 137 + 113 \\ 137 &= 1 \cdot 113 + 24 \\ 113 &= 4 \cdot 24 + 17 \\ 24 &= 1 \cdot 17 + 7 \\ 17 &= 2 \cdot 7 + 3 \\ 7 &= 2 \cdot 3 + 1 \end{aligned}$$

Rolling back, we get

$$\begin{aligned} 1 &= 7 - 2 \cdot 3 \\ &= 7 - 2 \cdot (17 - 2 \cdot 7) = 5 \cdot 7 - 2 \cdot 17 \\ &= 5 \cdot (24 - 17) - 2 \cdot 17 = 5 \cdot 24 - 7 \cdot 17 \\ &= 5 \cdot 24 - 7 \cdot (113 - 4 \cdot 24) = 33 \cdot 24 - 7 \cdot 113 \\ &= 33 \cdot (137 - 113) - 7 \cdot 113 = 33 \cdot 137 - 40 \cdot 113 \\ &= 33 \cdot 137 - 40 \cdot (250 - 137) = 73 \cdot 137 - 40 \cdot 250. \end{aligned}$$

Therefore, $d = 73$.

Supplementary Explanation

Answer. $d = 73$.

Derivation/proof. Euclid gives $250 = 137 + 113$, $137 = 113 + 24$, $113 = 4 \cdot 24 + 17$, $24 = 17 + 7$, $17 = 2 \cdot 7 + 3$, $7 = 2 \cdot 3 + 1$. Back substitution gives $1 = 73 \cdot 137 - 40 \cdot 250$, so $137^{-1} \equiv 73 \pmod{250}$.

Check: $137 \cdot 73 = 10001 = 40 \cdot 250 + 1$.

The intermediate back-substitution stage is $1 = 33 \cdot 24 - 7 \cdot 113$, before replacing 24 and 113.

N142

Cryptography

2014 Final | Question 3(b) | Source page 5 | Difficulty 3
Q-2014-FINAL-P03-b | Answer provenance: official

Consider the following simplified version of the RSA algorithm for public cryptography:

- (i) Bob's public key is a pair (n, e), where n is a prime number and e is a positive integer that is smaller than n and is relatively prime with $n-1$.
- (ii) Bob's private key is $d = e^{-1} \bmod (n-1)$.
- (iii) Alice encrypts a message m ($0 < m < n-1$) by calculating $c = m^e \bmod n$, and sends the ciphertext c to Bob.
- (iv) Bob decrypts the ciphertext c by calculating $c^d \bmod n$.

Suppose $n = 251$ and $e = 137$.

(b) Suppose $m = 200$. Calculate $c = m^e \bmod n$ using repeated squaring. Show the computational steps.

Official Answer

Source: PDF physical page 6 (2014-final.pdf).

(b) Observe that $137 = 2^7 + 2^3 + 2^0$. Using repeated squaring, we get

$$\begin{aligned} I_0 &= 200 \\ I_1 &= 200 \cdot 200 \bmod 251 = 91 \\ I_2 &= 91 \cdot 91 \bmod 251 = 249 \\ I_3 &= 249 \cdot 249 \bmod 251 = 4 \\ I_4 &= 4 \cdot 4 \bmod 251 = 16 \\ I_5 &= 16 \cdot 16 \bmod 251 = 5 \\ I_6 &= 5 \cdot 5 \bmod 251 = 25 \\ I_7 &= 25 \cdot 25 \bmod 251 = 123 \end{aligned}$$

So,

$$c = I_7 \cdot I_3 \cdot I_0 \bmod 251 = 123 \cdot 4 \cdot 200 \bmod 251 = 8.$$

Supplementary Explanation

Answer. $c = 8$.

Derivation/proof. $137 = 128 + 8 + 1$. Repeated squares modulo 251 give $200^2 \equiv 91$, $200^4 \equiv 249$, $200^8 \equiv 4$, $200^{16} \equiv 16$, $200^{32} \equiv 5$, $200^{64} \equiv 25$, $200^{128} \equiv 123$. Hence $200^{137} \equiv 123 \cdot 4 \cdot 200 \equiv 8$.

The final product is $123 \cdot 4 \cdot 200 = 98400 \equiv 8 \pmod{251}$.

In exponent order $2, 4, 8, 16, 32, 64, 128$, the residues are $91, 249, 4, 16, 5, 25, 123$.

N143

Cryptography

2013 Final | Question 2(c) | Source page 4 | Difficulty 3
Q-2013-FINAL-P02-c | Answer provenance: official

> Suppose for applying RSA, $p = 11$, $q = 19$, and $e = 7$.
>
> (c) Encrypt the message 100 using repeated squaring. Show the computational steps.

Official Answer

Source: PDF physical page 4 (2013-final.pdf).

(c) 100 is encrypted to $100^7 \bmod n$, where $n = 11 \times 19 = 209$.

$$100^7 \bmod 209 = 100^{4+2+1} \bmod 209 = 100 \times 100^2 \times 100^4 \bmod 209.$$

$$\begin{aligned} 100 \bmod 209 &= 100. \\ 100^2 \bmod 209 &= 10000 \bmod 209 = 177. \\ 100^4 \bmod 209 &= 177 * 177 \bmod 209 = 188. \end{aligned}$$

Thus $100^7 \bmod 209 = (100 \times 177 \times 188) \bmod 209 = 111$.

Supplementary Explanation

Answer. $100^7 \bmod 209 = 111$.

Derivation/proof. $7 = 4 + 2 + 1$. Compute $100^2 \equiv 177 \pmod{209}$ and $100^4 \equiv 177^2 \equiv 188 \pmod{209}$. Hence $100^7 \equiv 100 \cdot 177 \cdot 188 \equiv 111 \pmod{209}$.

The intermediate multiplication gives $100 \cdot 177 \equiv 144 \pmod{209}$, then $144 \cdot 188 = 27072 \equiv 111 \pmod{209}$.

REFERENCE

Source index

Exercise	Knowledge-base question ID	Source page
N001	Q-2019-MID-P06-b	8
N002	Q-2019-MID-P09-a	11
N003	Q-2019-MID-P09-b	11
N004	Q-2019-MID-P09-c	11
N005	Q-2017-MID-P04-a	6
N006	Q-2017-MID-P04-b	6
N007	Q-2015-FINAL-P05-b	10
N008	Q-2025S-FINAL-P02-a	3
N009	Q-2025S-FINAL-P03-a	4
N010	Q-2025S-MID-P07	11
N011	Q-2024F-FINAL-P05-a	15
N012	Q-2024F-FINAL-P05-b	15
N013	Q-2023F-FINAL-P06-b	10
N014	Q-2022-FINAL1-P04	3
N015	Q-2022-MID-P06-a	3
N016	Q-2019-MID-P06-a	8
N017	Q-2018-FINAL-P04-a	6
N018	Q-2016-FINAL-P04-b-ii	8
N019	Q-2015-FINAL-P05-a	10
N020	Q-2015-FINAL-P05-c	10
N021	Q-2025S-MID-P06-a	10
N022	Q-2025S-MID-P06-b	10
N023	Q-2024F-FINAL-P05-c	15
N024	Q-2024F-FINAL-P11-c	27
N025	Q-2024S-MID-P07	9

Exercise	Knowledge-base question ID	Source page
N026	Q-2023F-FINAL-P06-a	10
N027	Q-2023S-MID-P04	7
N028	Q-2022-MID-P07-b	3
N029	Q-2020-FINAL1-P04-a	3
N030	Q-2020-FINAL1-P04-c	3
N031	Q-2019-FINAL-P07	9
N032	Q-2019-MID-P12	14
N033	Q-2016-FINAL-P04-a	8
N034	Q-2016-FINAL-P04-b-i	8
N035	Q-2015-FINAL-P10	17
N036	Q-2024S-MID-P08	11
N037	Q-2020-FINAL1-P02	1
N038	Q-2025S-FINAL-P04-a	5
N039	Q-2019-MID-P10-a	12
N040	Q-2014-FINAL-P02-b	4
N041	Q-2014-FINAL-P02-d	4
N042	Q-2025S-FINAL-P02-b	3
N043	Q-2025S-MID-P08-a	12
N044	Q-2024F-FINAL-P06-a	17
N045	Q-2024F-FINAL-P07-1	19
N046	Q-2024S-MID-P12-a	18
N047	Q-2024S-MID-P12-b	18
N048	Q-2023F-FINAL-P01-a-i	3
N049	Q-2023F-FINAL-P01-a-iii	3
N050	Q-2022-MID-P07-a	3
N051	Q-2020-MID-P07	4
N052	Q-2020-MID-P08	4

Exercise	Knowledge-base question ID	Source page
N053	Q-2019-MID-P10-b	12
N054	Q-2019-MID-P10-c	12
N055	Q-2018-FINAL-P01-b	3
N056	Q-2018-MID-P03-a	5
N057	Q-2018-MID-P03-b	5
N058	Q-2018-MID-P05-b	7
N059	Q-2017-MID-P05-a	7
N060	Q-2015-FINAL-P07-b	13
N061	Q-2014-FINAL-P02-a	4
N062	Q-2014-FINAL-P02-c	4
N063	Q-2014-FINAL-P04-b	7
N064	Q-2013-FINAL-P01-b	3
N065	Q-2025F-FINAL-P06	10
N066	Q-2025S-FINAL-P03-b	4
N067	Q-2025S-FINAL-P04-b	5
N068	Q-2025S-MID-P08-b	12
N069	Q-2024F-FINAL-P07-2	19
N070	Q-2024S-FINAL-P08	9
N071	Q-2024S-FINAL-P09-a	11
N072	Q-2024S-FINAL-P09-b	11
N073	Q-2024S-MID-P10	14
N074	Q-2024S-MID-P11	16
N075	Q-2023F-FINAL-P01-a-ii	3
N076	Q-2023F-FINAL-P01-b	3
N077	Q-2023F-FINAL-P07	11
N078	Q-2023S-MID-P05	8
N079	Q-2023S-MID-P06	9

Exercise	Knowledge-base question ID	Source page
N080	Q-2022-FINAL2-P03	2
N081	Q-2022-MID-P06-b	3
N082	Q-2021-MID-P03	2
N083	Q-2021-MID-P07	3
N084	Q-2020-FINAL1-P04-b	3
N085	Q-2020-MID-BONUS	5
N086	Q-2020-MID-P06	3
N087	Q-2020-MID-P09	4
N088	Q-2019-FINAL-P02	4
N089	Q-2019-MID-P10-d	12
N090	Q-2018-FINAL-P09-a	13
N091	Q-2018-MID-P03-c	5
N092	Q-2018-MID-P05-a	7
N093	Q-2017-FINAL-P04-b-time	6
N094	Q-2017-FINAL-P04-c	6
N095	Q-2017-MID-P05-b	7
N096	Q-2016-FINAL-P06-a	11
N097	Q-2016-FINAL-P06-b	11
N098	Q-2015-FINAL-P07-a	13
N099	Q-2014-FINAL-P04-a	7
N100	Q-2013-FINAL-P01-a	3
N101	Q-2013-FINAL-P03	6
N102	Q-2013-FINAL-P04	7
N103	Q-2024F-FINAL-P06-b	17
N104	Q-2022-MID-P06-c	3
N105	Q-2021-FINAL1-P01	1
N106	Q-2018-FINAL-P03-b	5

Exercise	Knowledge-base question ID	Source page
N107	Q-2019-MID-P11-c	13
N108	Q-2013-FINAL-P02-a	4
N109	Q-2025F-FINAL-P07-i	12
N110	Q-2025F-FINAL-P07-iii	12
N111	Q-2024F-FINAL-P08-a	21
N112	Q-2023F-FINAL-P05-b	8
N113	Q-2019-MID-P11-a	13
N114	Q-2018-MID-P06-a	8
N115	Q-2018-MID-P06-b	8
N116	Q-2017-FINAL-P04-b-algorithm	6
N117	Q-2016-FINAL-P05-a-kenneth	10
N118	Q-2016-FINAL-P05-a-raymond	10
N119	Q-2016-FINAL-P05-b	10
N120	Q-2015-FINAL-P06-a	11
N121	Q-2014-FINAL-P03-c	5
N122	Q-2013-FINAL-P02-b	4
N123	Q-2025F-FINAL-P07-ii	12
N124	Q-2024F-FINAL-P08-b	21
N125	Q-2024F-FINAL-P08-c	21
N126	Q-2024S-MID-P09	13
N127	Q-2023F-FINAL-P05-a	8
N128	Q-2023S-MID-P07-a	10
N129	Q-2023S-MID-P07-b	10
N130	Q-2022-MID-P08-a	3
N131	Q-2022-MID-P08-b	3
N132	Q-2021-MID-P05	2
N133	Q-2021-MID-P06-a	3

Exercise	Knowledge-base question ID	Source page
N134	Q-2021-MID-P06-b	3
N135	Q-2020-MID-P10	5
N136	Q-2019-MID-P11-b	13
N137	Q-2018-MID-P06-c	8
N138	Q-2017-FINAL-P04-a	6
N139	Q-2015-FINAL-P06-b	11
N140	Q-2015-FINAL-P06-c	11
N141	Q-2014-FINAL-P03-a	5
N142	Q-2014-FINAL-P03-b	5
N143	Q-2013-FINAL-P02-c	4